

# Algebraicity of Weil classes on split abelian eightfolds

OpenAI

September 18, 2026

## Abstract

We prove that every rational Weil class on a split abelian eightfold of Weil type is algebraic. The result holds for every imaginary quadratic field, every compatible polarization type, and every member of the split family, including those with additional endomorphisms. Thus the full two-dimensional rational Weil space in codimension four is generated by algebraic cycle classes.

## Contents

|           |                                                                           |           |
|-----------|---------------------------------------------------------------------------|-----------|
| <b>1</b>  | <b>Introduction</b>                                                       | <b>1</b>  |
| <b>2</b>  | <b>The marked family and a middle-dimensional class</b>                   | <b>3</b>  |
| <b>3</b>  | <b>A Lagrangian representative with controlled topology</b>               | <b>7</b>  |
| <b>4</b>  | <b>Disk equations, compactness, and local gluing</b>                      | <b>15</b> |
| <b>5</b>  | <b>Finite integration with exact boundary products</b>                    | <b>28</b> |
| <b>6</b>  | <b>Signs and removal of the disk obstruction</b>                          | <b>37</b> |
| <b>7</b>  | <b>Exact coefficients and comparison with ordinary graph Floer theory</b> | <b>45</b> |
| <b>8</b>  | <b>The theta algebra and its projective family</b>                        | <b>54</b> |
| <b>9</b>  | <b>Ordinary graph branes and a simultaneous algebraic mirror</b>          | <b>61</b> |
| <b>10</b> | <b>Periodic modules and a nonpolarization class</b>                       | <b>68</b> |
| <b>11</b> | <b>Lifting the alternating Chern character</b>                            | <b>74</b> |
| <b>12</b> | <b>A general period and all special fibers</b>                            | <b>76</b> |

## 1 Introduction

The Weil classes are a particularly concrete test of the Hodge conjecture: they form a rational two-dimensional space defined by an imaginary quadratic action, while their algebraic representatives generally do not come from products of divisor classes. This paper establishes their algebraicity in the split eight-dimensional case.

## 1.1 Statement and scope

Let  $K = \mathbb{Q}(\sqrt{-d})$ , where  $d$  is a positive squarefree integer, and let  $A$  be a complex abelian variety of dimension eight. Suppose that

$$\iota : K \hookrightarrow \text{End}(A) \otimes_{\mathbb{Z}} \mathbb{Q}$$

is a  $\mathbb{Q}$ -algebra embedding and that a polarization  $\lambda$  has Rosati involution restricting to complex conjugation on  $\iota(K)$ . Write  $\sigma, \bar{\sigma} : K \hookrightarrow \mathbb{C}$  for its two embeddings. The *Weil-type condition* is

$$\dim_{\mathbb{C}} H^{1,0}(A)_{\sigma} = \dim_{\mathbb{C}} H^{1,0}(A)_{\bar{\sigma}} = 4.$$

On the  $K$ -vector space  $V = H_1(A, \mathbb{Q})$ , of dimension eight, let  $\psi$  be the rational Riemann form of  $\lambda$  and let  $J$  denote the action of  $\iota(\sqrt{-d})$ . The associated Hermitian form, linear in its first variable, is

$$h(x, y) = \psi(Jx, y) + \sqrt{-d} \psi(x, y). \quad (1.1)$$

We say that the tuple is *split* if  $V$  contains a four-dimensional  $K$ -subspace on which  $h$  vanishes identically. This definition fixes the convention without referring to a choice of sign for the discriminant.

The rational Weil space is

$$W_K(A) = \bigwedge_K^8 H^1(A, \mathbb{Q}) \subset H^8(A, \mathbb{Q}).$$

Its embedding is specified by the decomposition

$$W_K(A) \otimes_{\mathbb{Q}} \mathbb{C} = \bigwedge_{\mathbb{C}}^8 H^1(A, \mathbb{C})_{\sigma} \oplus \bigwedge_{\mathbb{C}}^8 H^1(A, \mathbb{C})_{\bar{\sigma}}.$$

Thus  $\dim_{\mathbb{Q}} W_K(A) = 2$ , and the Weil-type condition places it in  $H^{4,4}(A)$ .

**Theorem 1.1.** *For every split Weil-type tuple  $(K, A, \iota, \lambda)$  as above,*

$$W_K(A) \subset \text{im}(\text{CH}^4(A) \otimes_{\mathbb{Z}} \mathbb{Q} \xrightarrow{\text{cl}} H^8(A, \mathbb{Q})).$$

*The assertion holds for every imaginary quadratic field, every compatible polarization type, and every member of the split Weil-type families, including members with additional endomorphisms.*

The conclusion concerns rational linear combinations of algebraic cycles. It resolves the Weil-class algebraicity conjecture for these split eightfolds. It makes no assertion about all Hodge classes on arbitrary abelian eightfolds or about the Hodge conjecture for arbitrary smooth projective varieties.

## 1.2 Historical setting

Weil introduced the exceptional determinant classes in 1977 [28]. Markman's work proves their algebraicity on all abelian fourfolds of Weil type and on split abelian sixfolds of Weil type; it also yields the rational Hodge conjecture for abelian varieties of dimension at most five [18, 19]. The split eightfold case is a further instance beyond those results. The February 2026 version of [19, Question 11.4 and Section 12] proposes an eightfold construction using a weaker semiregularity condition. Both verification of that condition for the example and its sufficiency for the required deformation theorem remain open there.

Our proof proceeds through a symplectic construction and modules over theta section rings. Its main additional ingredients are a Lagrangian with prescribed homology and cohomological detection properties, a finite construction of weighted Floer operations and their obstruction identity, and an alternating specialization argument for periodic module cohomology. The last argument transports algebraic Chern characters without requiring a deformation theorem for a mirror object.

### 1.3 Proof architecture

The proof has four stages. First, Section 2 identifies all rational split Hermitian spaces with a fixed model and constructs a middle-dimensional class  $\alpha$ . Scalar graph tori pair trivially with the exceptional summand of  $\alpha$ , while one further graph detects it. Section 3 realizes a positive multiple of  $\alpha$  by an embedded spin Lagrangian  $L$ , graded modulo four. The maps  $H_2(L) \rightarrow H_2(T)$  and  $H_6(L) \rightarrow H_6(T)$  are injective, and a specified space of ambient two-forms restricts trivially in cohomology.

Second, Sections 4–7 construct the weighted Floer module. The analytic construction works with finite energy and arity cutoffs. It includes boundary gluing, compatible local thickenings, integration on overlaps, and the disk identity with one interior marked point. The two homology injections force the cohomological obstructions to vanish. A residue field of an ultraproduct then turns the finite constructions into exact algebraic identities over a power-series ring  $R = \kappa[[t]]$ .

Third, Section 8 computes the directed graph algebra as a theta section ring  $\mathcal{S}$ . Its relative projective spectrum  $\mathcal{X} = \text{Proj } \mathcal{S}$  is smooth and projective over  $R$ . At  $t = 0$ , Sections 9 and 10 apply ordinary mirror symmetry to finite lists of graph branes and recover the new module as a perfect object in a category with period-four grading. Its coherent cohomology sheaves give a finitely generated graded module in each residue degree. Euler pairings with the scalar graphs and the extra graph show that its alternating Chern character lies outside the span of the powers of the polarization.

Finally, Section 11 lifts finite generation and the alternating character across  $R$ . The  $t$ -torsion terms cancel in the alternating sum of derived restrictions. The theta ring can be realized at complex periods avoiding specified countably many analytic loci. Section 12 chooses such a period, isolates a nonzero Weil component, and spreads its marked class to the entire family using proper Hilbert parameter spaces. The quadratic-field action supplies a second independent Weil class; the initial isogeny reduction then proves Theorem 1.1.

### 1.4 Conventions and inputs

The complex abelian eightfolds  $A_\Pi$  and the real symplectic sixteen-torus  $T$  play different roles. The symplectic form on  $T$  is fixed throughout. Variation of  $\Pi$  enters the Floer operations through an additional formal relative two-class. Only the theta multiplication constants are prescribed under the field embeddings used to realize a complex period; there is no assertion that the new Lagrangian’s Floer series converge at that period.

We use established immersion and embedding results, the ordinary family Floer theorems, and standard algebraic and analytic geometry with their hypotheses stated at the point of application. The finite weighted virtual construction and its comparison identities are proved in Sections 4–7. In particular, the ordinary mirror theorems are applied only to tautologically unobstructed graphs. The new brane enters through its module and the proof does not assume a mirror equivalence for the deformed bounding-cochain category.

Homology and cohomology are rational unless another coefficient field is displayed. Periodic degrees are indexed by  $\mathbb{Z}/4$ ; the function  $r \mapsto (-1)^r$  is well defined on this group. Chern characters are rational algebraic Chern characters, with their Betti or  $\ell$ -adic cycle classes specified where needed.

## 2 The marked family and a middle-dimensional class

We first put all split Weil-type eightfolds for a fixed imaginary quadratic field into one rational model. The same model supplies the homology class and the space of two-forms used in the symplectic construction.

## 2.1 A model for every split Hermitian space

Fix a positive squarefree integer  $d$  and put  $K = \mathbb{Q}(\sqrt{-d})$ . Choose an integral quaternion of norm  $d$ , which exists by the four-squares theorem. Its left-multiplication matrix  $Q$  satisfies  $Q^t Q = Q Q^t = d1_4$ . On  $F = \mathbb{Q}^8$  set

$$D = \begin{pmatrix} 0 & -Q^t \\ Q & 0 \end{pmatrix}, \quad S_- = \text{diag}(1_4, -1_4). \quad (2.1)$$

Thus  $D^t = -D$ ,  $D^2 = -d1_8$ , and  $DS_- = -S_-D$ . Let

$$H = F_q \oplus F_p = F \otimes S, \quad S = \mathbb{Q}e \oplus \mathbb{Q}f, \quad \psi_0 = dq^t \wedge dp.$$

The action of  $\sqrt{-d}$  is  $D$  on both copies of  $F$ . Our convention is  $\psi_0(x, Ix) > 0$  for a positive compatible complex structure  $I$ . For any matrix  $M$  we write  $dq^t M \wedge dp = \sum_{a,b} M_{ab} dq_a \wedge dp_b$ .

Consider the marked family

$$\begin{aligned} \mathcal{U} &= \{\Pi \in \text{Mat}_8(\mathbb{C}) : \Pi = \Pi^t, \Im \Pi > 0, [\Pi, D] = 0\}, \\ A_\Pi &= \mathbb{C}^8 / (\mathbb{Z}^8 + \Pi \mathbb{Z}^8), \quad z = q + \Pi p. \end{aligned} \quad (2.2)$$

The principal polarization has class  $\theta$ , identified with  $\psi_0$  in the fixed torus marking. The domain  $\mathcal{U}$  is a nonempty convex open subset of the complex vector space of symmetric matrices commuting with  $D$ : it contains  $i1_8$ .

**Proposition 2.1** (Reduction to the marked family). *Every split Weil-type polarized abelian eightfold with  $K$ -action is  $K$ -compatibly isogenous to a member of (2.2). Consequently, algebraicity of the full rational Weil space on every  $A_\Pi$  implies the assertion for every compatible polarization type.*

*Proof.* First, (2.2) describes exactly the positive compatible complex structures on  $(H_{\mathbb{R}}, \psi_0)$  commuting with  $D$ . Indeed the real Lagrangian  $F_q$  is transverse to  $IF_q$ : if  $Iy \in F_q$  with  $y \in F_q$ , then  $\psi_0(y, Iy) = 0$ , so  $y = 0$ . A basis of  $F_q$  is therefore a complex basis. In that basis the other lattice generators give a symmetric period matrix with positive imaginary part, by the Riemann bilinear relations. The action of  $D$  on the first basis is its fixed matrix  $D$ , so complex linearity on the second basis says  $D\Pi = \Pi D$ . The converse follows from the same relations.

The  $K$ -Hermitian form

$$h_0(x, y) = \psi_0(Dx, y) + \sqrt{-d} \psi_0(x, y)$$

is nonsingular and has the  $K$ -isotropic half-space  $F_q$ . At  $\Pi = i1_8$  the two eigenvalues of  $D$  on  $H^{1,0}$  have multiplicities four and four. Those multiplicities are locally constant and  $\mathcal{U}$  is connected, so every member has the required Weil type.

For an arbitrary tuple in the theorem, write  $(V, h)$  for its rational  $K$ -Hermitian space. The prescribed isotropic  $K$ -subspace has dimension four. Choose a basis  $e_1, \dots, e_4$  of it and vectors  $f_1, \dots, f_4$  with  $h(e_i, f_j) = \delta_{ij}$ . If  $B_{ij} = h(f_i, f_j)$ , replace

$$f_i \quad \text{by} \quad f_i - \frac{1}{2} \sum_j B_{ij} e_j.$$

Using that  $h$  is linear in its first variable and  $B_{ij} = \overline{B_{ji}}$ , the new  $f_i$  span an isotropic complement. Thus  $(V, h)$  is hyperbolic. Applying this construction to  $h_0$  gives a  $K$ -linear isometry  $(H, h_0) \simeq (V, h)$ . Taking the coefficient of  $\sqrt{-d}$  shows that it also identifies the alternating forms. The positive complex structure on  $V_{\mathbb{R}}$  therefore pulls back to a point of  $\mathcal{U}$ .

The transported integral homology lattice and  $\mathbb{Z}^{16}$  are full lattices in the same rational space, hence commensurable. Multiplying the rational identification by a suitable positive integer yields a  $K$ -compatible isogeny of the corresponding complex tori. Both are abelian varieties, so the isogeny is algebraic. If  $f : B \rightarrow A$  is such an isogeny, then  $f^* : H^*(A, \mathbb{Q}) \rightarrow H^*(B, \mathbb{Q})$  is an isomorphism, its inverse is  $f_*/\deg f$ , and both maps preserve algebraic rational cycle classes. They identify the two determinant lines defining the Weil spaces because  $f$  commutes with  $K$ . This proves the reduction without a restriction on integral polarization type.  $\square$

## 2.2 Hodge classes at a general period

We regard all rational cohomology groups as subspaces of  $\bigwedge^* H^*$  by the marking. In particular the Weil space is the same rational two-plane  $W_K$  throughout  $\mathcal{U}$ .

**Proposition 2.2.** *There is a countable collection of proper closed analytic subsets of  $\mathcal{U}$  outside which*

$$H^{2j}(A_\Pi, \mathbb{Q}) \cap H^{j,j}(A_\Pi) = \begin{cases} \mathbb{Q}\theta^j, & j \neq 4, \\ \mathbb{Q}\theta^4 \oplus W_K, & j = 4. \end{cases} \quad (2.3)$$

Here  $0 \leq j \leq 8$ .

*Proof.* For each rational vector  $v \in \bigwedge^{2j} H^*$ , the condition  $v \in F_{\text{Hdg}}^j$  is closed analytic, since the Hodge filtration is a holomorphic subbundle of the constant cohomology bundle. As  $v$  is rational, this is equivalent to type  $(j, j)$ . There are only countably many such vectors. It remains to identify those which have type  $(j, j)$  at every period.

The centralizer of  $K$  in  $\text{Sp}(H_{\mathbb{R}}, \psi_0)$  is the unitary group of  $h_0$ , of signature  $(4, 4)$ . This follows directly from the formula for  $h_0$  and its recovery of  $\psi_0$ . It acts on the compatible complex structures in the family. Their Hodge circles lie in  $\text{SU}(4, 4)$ : on the  $K$ -complex representation their two circle weights each occur four times, so their determinants are one. A circle is noncentral, since both weights occur. The Lie algebra generated by all its  $\text{SU}(4, 4)$ -conjugates is a nonzero ideal in the simple real Lie algebra  $\mathfrak{su}(4, 4)$ , hence is the whole algebra. Thus a class fixed by every Hodge circle is fixed by the connected group  $\text{SU}(4, 4)$ .

After complexification this is the  $\text{SL}_8$ -representation on a standard space  $E$  and its dual. Its exterior algebra decomposes as

$$\bigwedge^k (E \oplus E^*) = \bigoplus_{a+b=k} \bigwedge^a E \otimes \bigwedge^b E^*.$$

An invariant in the  $(a, b)$  summand is an intertwiner  $\bigwedge^b E \rightarrow \bigwedge^a E$ . The exterior powers are irreducible and mutually inequivalent for  $1 \leq a \leq 7$ , whereas the zeroth and eighth powers are both trivial. The resulting invariant lines are the matched summands  $a = b$ , together with  $(a, b) = (8, 0), (0, 8)$ . The matched invariants are the powers of the nondegenerate pairing, which is  $\theta$  up to a nonzero scalar. The two remaining lines are exactly  $W_K \otimes \mathbb{C}$ . Conversely  $\theta$  and these determinant lines have the asserted Hodge types at every period. Descent to rational cohomology gives the common spaces in (2.3). Exclude the proper Hodge locus of each remaining rational vector.  $\square$

### 2.3 The symplectic model and its exceptional class

The construction takes place on the separate symplectic torus

$$\begin{aligned} T &= (\mathbb{R}_q^8/\mathbb{Z}^8) \times (\mathbb{R}_p^8/\mathbb{Z}^8), \quad \omega = dq^t \wedge dp, \\ \Omega &= \{dq^t M \wedge dp : M \in \text{Mat}_8(\mathbb{Q}), M = M^t, [D, M] = 0\} \subset \bigwedge^2 H^*. \end{aligned} \quad (2.4)$$

Use its symplectic orientation, the constant complex volume for  $q + ip$ , and spin background zero. Fix an integer  $h_0 \geq 3$  and orient the graphs

$$G_k = \{p = h_0 k q\}, \quad k \geq 1, \quad C = \{p = S_- q\} \quad (2.5)$$

by their  $q$ -coordinates. Write  $g_k, c \in \bigwedge^8 H = H_8(T, \mathbb{Q})$  for their fundamental classes. Their slope matrices are symmetric, so the graphs are Lagrangian.

Let  $F_{\mathbb{C}} = U_+ \oplus U_-$  be the eigenspace decomposition of  $D$ , with eigenvalues  $\pm i\sqrt{d}$ , and set  $H_{\pm} = U_{\pm} \otimes S_{\mathbb{C}}$ . Each  $U_{\pm}$  has dimension four and each  $H_{\pm}$  has dimension eight.

**Lemma 2.3.** *There exists a rational class  $\alpha_{\text{ex}} \in \bigwedge^8 H$  whose complexification belongs to  $\bigwedge^8 H_+ \oplus \bigwedge^8 H_-$  and satisfies*

$$\alpha_{\text{ex}} \wedge c \neq 0, \quad \alpha_{\text{ex}} \wedge g_k = 0 \quad (k \geq 1). \quad (2.6)$$

*Both of its pure eigenspace components are nonzero.*

*Proof.* The sum of the two determinant lines descends to a rational two-plane: the eigenspace splitting is defined over  $K$ , and its Galois involution exchanges the two lines. Projection of the graph vector space of  $C$  onto  $H_-$  is an isomorphism. Indeed, writing  $x = x_+ + x_-$ , it is

$$x \mapsto x_- \otimes e + (S_- x_+) \otimes f,$$

and  $S_- : U_+ \rightarrow U_-$  is invertible. Thus wedging with  $c$  is nonzero on  $\bigwedge^8 H_+$  and hence on the rational two-plane. Choose a rational vector where it is nonzero. Its two components are conjugate, so neither vanishes. Finally  $g_k$  has eigenspace bidegree  $(4, 4)$ , whereas each component of  $\alpha_{\text{ex}}$  is pure of degree eight; their wedge is zero by dimension.  $\square$

Set

$$\alpha = \alpha_{\text{ex}} + g_1 + g_2 + g_3. \quad (2.7)$$

The three scalar slopes have a specific purpose: they detect the three coefficients of a binary quadratic form. This gives the two identities which control the topology of the eventual Lagrangian.

**Lemma 2.4.** *The space  $\Omega$  has dimension sixteen, and*

$$\ker \left( \bigwedge^2 H^* \longrightarrow \bigwedge^6 H, \eta \mapsto \iota_{\eta} \alpha \right) = \Omega, \quad H^* \otimes \Omega \xrightarrow{\wedge} \bigwedge^3 H^* \text{ is injective.} \quad (2.8)$$

*Proof.* We may extend scalars to  $\mathbb{C}$ . Skewness of  $D$  makes  $U_+$  and  $U_-$  isotropic for the dot product and pairs them perfectly. A symmetric matrix commuting with  $D$  therefore determines, and is determined by, an arbitrary pairing between  $U_+$  and  $U_-$ . In the two  $S$ -coordinates the corresponding two-form has alternating  $S$ -factor. Consequently

$$\Omega_{\mathbb{C}} = U_+^* \otimes U_-^* \otimes \bigwedge^2 S_{\mathbb{C}}^* \subset \bigwedge^2 H_{\mathbb{C}}^*, \quad (2.9)$$

which has dimension sixteen. These forms contract to zero against the pure determinant lines and against every scalar graph volume, so  $\Omega_{\mathbb{C}}$  lies in the stated kernel.

Conversely, suppose  $\iota_{\eta}\alpha = 0$ . The components of its value in eigenspace bidegrees  $(6, 0)$  and  $(0, 6)$  come only from contracting the two nonzero pure volumes of  $\alpha_{\text{ex}}$  by the pure parts of  $\eta$ . Contraction into a nonzero top exterior power is an isomorphism from two-forms to the complementary exterior power. Hence both pure parts of  $\eta$  vanish, and  $\eta$  is mixed.

Put  $s_j = e + h_0 j f$  for  $j = 1, 2, 3$ . Project the  $(3, 3)$  component of the contraction to

$$\bigwedge^3 U_+ \otimes \bigwedge^3 U_- \otimes \text{Sym}^6 S_{\mathbb{C}}$$

by alternating in each  $U$ -factor and multiplying the six  $S$ -factors symmetrically. Contraction of  $g_j$  projects, up to one fixed nonzero factor and the fixed volume isomorphisms, to

$$\eta(- \otimes s_j, - \otimes s_j) \otimes s_j^6.$$

The vectors  $s_1^6, s_2^6, s_3^6$  are independent: their coefficients of  $e^6, e^5 f, e^4 f^2$  give a nonzero Vandermonde determinant. Thus all three pairing-valued evaluations vanish. Every coefficient of  $\eta(- \otimes s, - \otimes s)$  is a binary quadratic in  $s$ , and a binary quadratic vanishing on these three distinct lines is zero. The symmetric  $S$ -part of  $\eta$  therefore vanishes, leaving exactly (2.9). This proves the kernel assertion.

For wedge injectivity, use bases  $u_i$  of  $U_+^*$  and  $v_j$  of  $U_-^*$ . The subscripts  $e, f$  denote their four copies in  $H_{\mathbb{C}}^*$ , and a basis of  $\Omega_{\mathbb{C}}$  is

$$z_{ij} = u_{i,e} \wedge v_{j,f} - u_{i,f} \wedge v_{j,e}.$$

The following projections of a wedge product detect the four possible types of first factor independently:

| first factor | three-letter block                            |
|--------------|-----------------------------------------------|
| $U_{+,e}^*$  | $U_{+,e}^* \wedge U_{+,f}^* \wedge U_{-,e}^*$ |
| $U_{+,f}^*$  | $U_{+,e}^* \wedge U_{+,f}^* \wedge U_{-,f}^*$ |
| $U_{-,e}^*$  | $U_{+,e}^* \wedge U_{-,e}^* \wedge U_{-,f}^*$ |
| $U_{-,f}^*$  | $U_{+,f}^* \wedge U_{-,e}^* \wedge U_{-,f}^*$ |

For example, the first projection of  $u_{a,e} \wedge z_{ij}$  is  $-u_{a,e} \wedge u_{i,f} \wedge v_{j,e}$ , and no other first-factor type contributes to that block. The three indices freely recover every coefficient. The other rows give the same argument. Hence all coefficients of an element in the wedge kernel vanish.  $\square$

### 3 A Lagrangian representative with controlled topology

The obstruction calculation will use two properties of a Lagrangian representative: ambient forms must detect its cohomology in degrees two and six, and its Maslov grading must be defined modulo four. We construct these properties together. All homology and cohomology in this section have rational coefficients unless another coefficient ring is displayed. The symplectic torus  $T$ , the class  $\alpha$ , and the space  $\Omega$  are those of Section 2.

**Proposition 3.1** (The auxiliary brane). *There are a positive integer  $\nu$  and a closed connected oriented spin Lagrangian embedding  $i_L : L \hookrightarrow T$  such that*

$$(i_L)_*[L] = \nu\alpha. \tag{3.1}$$

*The embedding has a grading in the fourfold Maslov cover determined by the standard complex volume form on  $T$ . Moreover,*

$$i_L^* : H^2(T) \longrightarrow H^2(L) \text{ is surjective,} \quad \ker i_L^* = \Omega, \tag{3.2}$$

and

$$(i_L)_* : H_j(L) \longrightarrow H_j(T) \text{ is injective for } j = 2, 6. \quad (3.3)$$

In particular, every form in  $\Omega_{\mathbb{C}}$  restricts to an exact complex two-form on  $L$ .

We first realize  $\alpha$  by a stably framed manifold with the desired cohomology, then use a local exact domain to control the signs of a Lagrangian immersion. The final surgeries preserve precisely the homology groups in (3.3).

### 3.1 A nilmanifold and a framed representative

Choose an integral basis  $\eta_1, \dots, \eta_{16}$  of a full lattice in  $\Omega$ . Let  $b : H_{\mathbb{R}} \times H_{\mathbb{R}} \rightarrow \mathbb{R}^{16}$  be bilinear, integral on the coordinate lattice, with

$$b(x, y) - b(y, x) = (\eta_1(x, y), \dots, \eta_{16}(x, y)).$$

For example, take the upper triangular part of each integral alternating matrix. The multiplication

$$(x, z)(y, w) = (x + y, z + w + b(x, y))$$

makes  $H_{\mathbb{R}} \oplus \mathbb{R}^{16}$  a simply connected two-step nilpotent Lie group  $N$ . Its integral points form a cocompact subgroup  $\Gamma$ . Projection onto the first coordinate descends to

$$\pi : \mathcal{N} = \Gamma \backslash N \longrightarrow T.$$

The universal cover of  $\mathcal{N}$  is contractible; thus  $\mathcal{N}$  is a classifying space for  $\Gamma$ , and  $\Gamma \rightarrow \pi_1(T)$  is surjective.

**Lemma 3.2.** *Pullback induces an isomorphism*

$$H^2(\mathcal{N}) \cong \bigwedge^2 H^*/\Omega.$$

There is a class  $\tilde{\alpha} \in H_8(\mathcal{N})$  with  $\pi_* \tilde{\alpha} = \alpha$ .

*Proof.* Nomizu's Theorem [21, Theorem 1] computes real cohomology by invariant forms. The invariant complex is generated in degree one by  $H_{\mathbb{R}}^*$  and  $z_1^*, \dots, z_{16}^*$ , with  $dH_{\mathbb{R}}^* = 0$  and  $dz_i^* = \eta_i$ , after changing signs if necessary. On  $H_{\mathbb{R}}^* \otimes \langle z_i^* \rangle$ , the differential is the wedge map in Lemma 2.4, hence is injective. It is also injective on  $\bigwedge^2 \langle z_i^* \rangle$ : in

$$d(z_i^* \wedge z_j^*) = \eta_i \wedge z_j^* - \eta_j \wedge z_i^*,$$

the independence of the  $\eta_i$  detects every coefficient. These two images have different central degrees and cannot cancel. The closed two-forms are therefore the horizontal ones, modulo the boundaries  $\Omega_{\mathbb{R}}$ .

Dually, the horizontal Lie chain  $\alpha \in \bigwedge^8 H$  has boundary

$$\partial \alpha = \pm \sum_i z_i \wedge \iota_{\eta_i} \alpha = 0$$

by Lemma 2.4. Its image under projection is  $\alpha$ . Naturality of the invariant complex proves the lifting statement over  $\mathbb{R}$ . Both the cohomology pullback and the homology pushforward are rational linear maps. Their asserted kernel and image statements consequently descend from  $\mathbb{R}$  to  $\mathbb{Q}$ .  $\square$

**Lemma 3.3.** *There are a closed connected stably framed eight-manifold  $L_0$ , a map  $f_0 : L_0 \rightarrow T$ , and a positive integer  $\nu$  such that*

$$f_{0*}[L_0] = \nu\alpha, \quad \chi(L_0) > (\nu\alpha) \cdot (\nu\alpha).$$

*The map is surjective on fundamental groups, has an additional free circle generator mapping trivially to  $T$ , and satisfies (3.2) and (3.3) with  $f_0$  in place of  $i_L$ .*

*Proof.* Rational framed bordism maps isomorphically to rational homology: the Pontryagin–Thom identification and the framed-bordism spectral sequence give this because the positive stable stems of the sphere are torsion [29, proof of Proposition 3.1]. Clearing denominators therefore represents  $\nu\tilde{\alpha}$ , for some positive integer  $\nu$ , by a closed stably framed eight-manifold mapped to  $\mathcal{N}$ .

Perform the following surgeries over  $\mathcal{N}$ . First connect the components by framed one-handles and add loops mapping to a finite generating set of  $\Gamma$ , making the fundamental-group map surjective. Its kernel is normally finitely generated, since both groups are finitely presented. Represent normal generators by disjoint embedded circles and perform framed surgery on them, using their null homotopies in  $\mathcal{N}$ . The normal rank is seven; twisting a normal framing by  $\pi_1\mathrm{SO}(7)$  makes it compatible with the given stable framing. The resulting fundamental group is  $\Gamma$ .

Next,  $\pi_2$  of the source is finitely generated as a  $\mathbb{Z}\Gamma$ -module. Indeed, Hurewicz identifies it with the second homology of the universal cover. The cellular chain modules of that cover are finite free over  $\mathbb{Z}\Gamma$ , and this ring is Noetherian because  $\Gamma$  is finitely generated nilpotent, hence polycyclic [17]. Represent module generators by disjoint embedded two-spheres, possible by general position in dimension eight. Their normal bundles have rank six and are stably trivial, hence trivial; the stable framing extends over surgery since  $\pi_2\mathrm{SO} = 0$ . The maps extend because  $\pi_2(\mathcal{N}) = 0$ . The reverse handles have indices at least six, so these surgeries kill  $\pi_2$  without altering  $\pi_1$ . All these modifications are bordisms over  $\mathcal{N}$  and preserve the pushed fundamental class.

The resulting source has two-connected universal cover. Its classifying map to  $\mathcal{N}$  induces an isomorphism on  $H^2$ , as follows, for example, from the low-degree homology sequence of the universal-cover fibration. Now take a framed connected sum with  $S^1 \times S^7$  and with sufficiently many copies of  $S^4 \times S^4$ , mapping the added summands constantly. These products are stably parallelizable. The first summand supplies the extra null-image generator, while each of the latter summands raises the Euler characteristic by two. None changes  $H^2$ . Call the resulting map to  $T$   $f_0 : L_0 \rightarrow T$ .

Lemma 3.2 now gives the surjectivity and kernel in (3.2). Duality gives injectivity on  $H_2$ . For  $u \in H_6(L_0)$ , Poincaré duality and that same surjectivity write  $u = f_0^*\eta \cap [L_0]$  for some  $\eta \in H^2(T)$ . Then

$$f_{0*}u = \eta \cap (\nu\alpha) = \nu\iota_\eta\alpha.$$

If this vanishes, Lemma 2.4 puts  $\eta$  in  $\Omega$ , so  $u = 0$ . This proves the  $H_6$  injection.  $\square$

### 3.2 A graded immersion and its branch actions

Since  $\omega \in \Omega$ , we have  $f_0^*[\omega] = 0$  over  $\mathbb{R}$ . The stable real framing gives a stable complex trivialization of  $TL_0 \otimes \mathbb{C}$  with trivial determinant phase on the oriented real tangent. Rank eight is in the unitary stable range over an eight-dimensional manifold, so this trivialization can be destabilized, preserving its stable homotopy class. Its restriction to  $TL_0$  is a Lagrangian bundle monomorphism into the trivial symplectic bundle  $f_0^*TT$ , of Maslov class zero. The Gromov–Lees immersion theorem, with its cohomological hypothesis  $f_0^*[\omega] = 0$ , realizes it by a Lagrangian immersion homotopic to  $f_0$  [23, Theorem 2.3]. Choose its integral grading with the parity corresponding to the given orientation.

For such an immersion  $f$ , let  $K_f$  be the kernel of  $\pi_1(L_0) \rightarrow \pi_1(T)$ . A loop representing  $\gamma \in K_f$  has a filling in  $T$ , and its symplectic area defines  $\mathcal{A}_f(\gamma) \in \mathbb{R}$ . This is independent of the filling because  $\pi_2(T) = 0$ , and of the representative because  $f$  is Lagrangian. Concatenation makes  $\mathcal{A}_f$  additive on  $K_f$ ; transport and conjugation do not change it.

**Lemma 3.4.** *After Lagrangian perturbations of  $f$  there is a self-transverse immersion  $f_1$  without triple points, still integrally graded and homotopic to  $f_0$ , with the following property. Its double points can be written  $f_1(a_j) = f_1(b_j)$ , and there are pairwise disjoint embedded arcs from  $a_j$  to  $b_j$  whose image loops are contractible in  $T$  and have area zero.*

*Proof.* The action subgroup contains 1. To see this, lift based loops mapping to lattice vectors  $v, w$  with  $\omega(v, w) = 1$ ; this is possible by fundamental-group surjectivity. Their commutator lies in  $K_f$ , and integration of  $-p^t dq$  on the universal cover gives its area as  $\pm\omega(v, w)$ : the translated opposite sides cancel except for this lattice pairing.

The extra circle of Lemma 3.3 admits an independently adjustable action. In an immersed cotangent neighborhood of  $f$ , replacing the zero section by a small closed one-form changes its action by that form's period, up to sign. Choose this period so that the resulting action is irrational. The action subgroup then contains  $\mathbb{Z} + \rho\mathbb{Z}$  with  $\rho \notin \mathbb{Q}$ , and is dense in  $\mathbb{R}$ . Small exact perturbations on source patches make the immersion self-transverse without triple points; they do not change these loop periods.

For each double point choose a path  $a_j \rightarrow b_j$ . By inserting a loop at  $a_j$ , fundamental-group surjectivity first makes its image contractible. Inserting an element of  $K_f$  then makes its filling area arbitrarily small. Choose mutually disjoint small source patches at the  $a_j$ , avoiding the other double-point preimages. An exact perturbation by  $dF_j$ , where  $F_j$  is constant near  $a_j$  and zero outside its patch, fixes the sheet near the double point but changes this path action by  $F_j(b_j) - F_j(a_j)$ , up to the fixed sign convention. Thus it makes the action exactly zero. This endpoint formula remains valid if a chosen path revisits the patch. For the fixed patches there is a uniform smallness bound preserving all transverse double points and introducing no new ones; choose the preceding residual areas within that bound.

Finally, relative homotopies of these paths preserve their actions. General position in the eight-dimensional source makes them embedded, pairwise disjoint, and disjoint from every other double-point preimage.  $\square$

The algebraic number of double points of  $f_1$  is

$$I(f_1) = \frac{(\nu\alpha) \cdot (\nu\alpha) - \chi(L_0)}{2} < 0. \quad (3.4)$$

Indeed, the normal Euler number of an oriented Lagrangian eight-manifold is  $(-1)^{8 \cdot 7/2} \chi(L_0) = \chi(L_0)$ , and the self-intersection formula is the normal Euler number plus twice the signed double-point count.

### 3.3 An embedded exact domain containing the double points

The next lemma makes the simply connected flexibility theorem applicable inside a torus. Its exactness and its embedding in  $T$  are both part of the construction.

**Lemma 3.5** (The local exact domain). *There is a compact contractible Weinstein domain  $(P, \lambda) \subset (T, \omega)$ , with  $d\lambda = \omega|_P$ , such that  $W = f_1^{-1}(P)$  is a disk containing all double-point preimages. The Liouville vector field is tangent to its immersed sheets,  $f_1^* \lambda|_W = 0$ , and the immersion is cylindrical in a collar of  $\partial W$ . At one interior point of  $W$  its image is a single linear Lagrangian sheet in a radial Darboux ball.*

*Proof.* Extend the arcs of Lemma 3.4 to a finite embedded tree  $\Theta \subset L_0$ , using connector edges avoiding all double-point preimages. Set  $G = f_1(\Theta)$ . The only identifications on this tree are the paired endpoints  $a_j, b_j$ , so  $f_1^{-1}(G) = \Theta$ . Its graph cycles are freely generated by the chosen arcs, with suitable connecting paths. They are contractible in  $T$  and have filling area zero. Subdivide an ordinary edge to provide a vertex with just one preimage.

**The zero- and one-handles.** At every vertex take a small Darboux ball with radial Liouville field. A single sheet, or the two transverse sheets at a double point, can be made linear simultaneously. For the latter claim, first make one sheet horizontal; the other is a gradient graph over the vertical plane, and a symplectic shear straightens it while fixing the first. By homotoping the tree near its vertices, make its incident edges distinct radial rays. Join these balls by thin Weinstein one-handles along the remaining arcs, carrying Lagrangian strips in the corresponding sheets.

We describe the attachment and its relative neighborhood data, also for later use with two-handles. For index  $k = 1, 2$ , use symplectic coordinates  $(u, v, z)$  with  $u, v \in \mathbb{R}^k$  and  $z \in \mathbb{R}^{2(8-k)}$ . The vector field

$$Z_k = (-u, 2v, z/2) \quad (3.5)$$

is Liouville. A handle box has  $|u| \leq 1$ ,  $|(v, z)| \leq \epsilon$ , with negative face near  $|u| = 1$  and core  $v = z = 0$ . It has a quadratic Lyapunov function with one critical point of index  $k$ . For  $k = 1$ , the set  $v = 0$  with  $z$  in a fixed Lagrangian plane is an invariant Lagrangian strip.

The contact neighborhood identifications at the attaching locus may be made strict for the specified contact forms. Here is the needed refinement of the neighborhood theorem. Choose, on each side, a hypersurface containing the isotropic locus and tangent to the contact plane along it. The Reeb field is transverse on a sufficiently small neighborhood; its flow writes the contact form as  $da + \beta$  on that neighborhood, with  $d\beta$  symplectic on the slice. Match the slices by the symplectic isotropic neighborhood theorem using the prescribed tangent and symplectic normal data. The two slice primitives differ by a closed form whose restriction to the attaching locus is zero. Since the neighborhood retracts onto that locus, this difference is  $dg$ , with  $g = 0$  there. Shifting the Reeb coordinate by  $g$ , with the appropriate sign, makes the identification strict. This argument also applies to an entire Legendrian patch. Extension along the Liouville flows gives a symplectic collar identification preserving the primitive.

For the one-handles, choose the Legendrian patch identifications using a transverse strip framing along each source arc. Either orientation of such a patch can be chosen. Extend the prescribed end germs over the strip by tubular coordinates. The relative Lagrangian neighborhood theorem extends this strip map symplectically, keeping the collar germs: after the tangent map is specified, the remaining linear lift choices are contractible, and the tubular construction and Moser argument are relative to the collars. On shrinking the handles, they are embedded, mutually disjoint away from their prescribed attachments, and meet no other sheets. This follows from compactness and  $f_1^{-1}(G) = \Theta$ . Transport the model primitives through these embeddings and glue them by the strict collars.

Rounding the positive corners transversely to the Liouville field gives an embedded Weinstein zero- and one-handlebody  $P_1$  with spine  $G$ . Its preimage thickens the original tree  $\Theta$ , hence is a disk. The field is tangent to every sheet there; equivalently, the primitive restricts to zero. Its boundary is therefore Legendrian and has the required positive cylindrical collar. The chosen single-sheet vertex retains a smaller radial Darboux ball.

**Exterior isotropic disks.** The inclusions

$$\partial P_1 \longrightarrow P_1, \quad T \setminus \text{int } P_1 \longrightarrow T$$

induce isomorphisms on fundamental groups. For the first this is the dual handle decomposition of a sixteen-dimensional zero- and one-handlebody; for the second, attaching the removed handlebody from its boundary involves only indices fifteen and sixteen. Choose loops giving a free basis of  $\pi_1(P_1)$  in  $\partial P_1$ . The relative subcritical isotropic embedding  $h$ -principle in contact manifolds [8, Theorem 12.4.1] makes them disjoint isotropic circles, avoiding the Legendrian boundary of the immersed disk. The required formal tangent lines can be deformed into the contact hyperplanes, and all the avoidance conditions are in the general-position range.

Each circle bounds a smooth disk outside  $P_1$ : its class maps trivially to  $\pi_1(T)$ , hence to the fundamental group of the exterior. Prescribe as its initial collar the outward Liouville-flow cylinder, which is isotropic. General position relative to these collars gives embedded disks with mutually disjoint interiors, avoiding the immersed eight-manifold; here  $2 + 2 < 16$  and  $2 + 8 < 16$ .

Each disk has symplectic area zero. Indeed an annulus in  $P_1$  joins its boundary to the corresponding graph cycle. The  $\lambda$ -integral on both boundary curves is zero: one is isotropic in the contact boundary and the other lies in the Lagrangian sheets where  $f_1^*\lambda = 0$ . The annulus therefore has zero area. The graph cycle has a zero-area filling by its construction, and  $\pi_2(T) = 0$  makes the area independent of the chosen filling.

We need isotropic disks, with the already prescribed collars. The following formal check ensures that the subcritical  $h$ -principle supplies them. Take a prequantization circle bundle of the integral symplectic form  $\omega$ , with its contact connection. The isotropic disk collars lift horizontally. The zero filling area makes their holonomy trivial and, in a trivialization over each disk, gives winding number zero for the boundary lift. Thus these horizontal lifts extend to smooth lifts of the disks. This uses area zero, rather than merely an integral area.

Trivialize the contact bundle over each lifted disk. The space of isotropic ordered two-frames retracts to  $U(8)/U(6)$ , whose fundamental group is zero. The prescribed isotropic tangent frames on the collar consequently extend over the disk. The ordinary real two-frame space in rank seventeen is two-connected, so the lifted disk derivative is homotopic, relative to that collar, to this isotropic monomorphism. The relative,  $C^0$ -dense subcritical contact embedding  $h$ -principle now realizes the formal data [8, Theorem 12.4.1]. Fix a slightly narrower collar throughout. The approximations preserve avoidance by  $C^0$ -smallness away from it.

Projection to  $T$  is an isotropic immersion. It may be chosen embedded, with all the disks disjoint. In fact contact Hamiltonian perturbations supported near distinct lift points give independently arbitrary projected variations, so projected double points and intersections are removed by transversality in dimension sixteen. The same argument preserves avoidance of  $f_1(L_0)$ , and the fixed narrower collars protect the attachments. We have therefore obtained actual disjoint isotropic core disks in the exterior, relative to the required germs.

**The two-handles.** Attach the model (3.5) with  $k = 2$  along these cores. Its symplectic normal quotient along the attaching circle is the contact conformal symplectic normal bundle, with scale fixed by the contact form. Choose its attaching framing from a symplectic normal framing over the core disk, and its parameterization from one extending over that disk. Thus both the tangent and normal data of the prescribed collar identification extend. The relative isotropic neighborhood theorem, followed by relative Moser, embeds a sufficiently thin actual handle with that collar germ. These handles miss the immersion; their primitives glue by the strict identifications above. Rounding corners gives the desired domain  $P$ .

Topologically,  $P$  is the graph  $G$  with a two-cell attached along each free generator, hence is contractible. More explicitly, after collapsing a maximal tree its attaching maps are a free basis of a bouquet of circles, so this two-complex is homotopy equivalent to a point. The two-handles are disjoint from the immersion, so its preimage remains the disk already

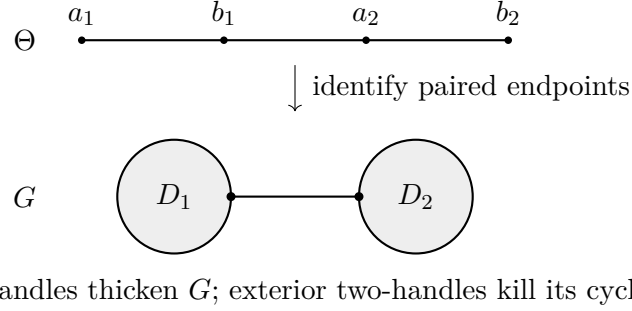


Figure 1: The topology of the local domain, illustrated with two double points. The preimage tree thickens to an eight-disk. The shaded disks indicate the two-handle attachments only: their actual isotropic cores attach to circles in  $\partial P_1$  and lie off the immersed sheets.

constructed in  $P_1$ . The primitive, collar, and radial single-sheet ball remain unchanged.  $\square$

### 3.4 Removing positive double points and extending the grading

**Lemma 3.6.** *The immersion can be replaced on the disk  $W$  by an oriented spin Lagrangian immersion with only negative transverse double points. It retains the boundary collar, its fundamental class, its integral Maslov grading, and the homology-map properties in Lemma 3.3.*

*Proof.* Complete  $P$  by its positive cylindrical end and extend the immersed disk by its cylinder. Remove the center of the single-sheet radial ball supplied by Lemma 3.5. The punctured ball is a negative Liouville end. The source becomes an annulus  $S^7 \times \mathbb{R}$ , with standard real-plane Legendrian link  $\phi_0 \subset S^{15}$  at the negative end. The punctured completion is simply connected, since removing a point from a sixteen-manifold does not affect  $\pi_1$ .

We use the stabilization traces of [6, Lemma 3.2 and proof of Lemma 3.4]. In source dimension  $2k$ , a trace going outward from a Legendrian link  $\phi$  to its loose stabilization  $\phi_1$  has one double point of sign  $(-1)^{k-1}$ ; the reverse trace has sign  $(-1)^k$ . For  $k = 4$  the signs are therefore

$$\phi \longrightarrow \phi_1 : -1, \quad \phi_1 \longrightarrow \phi : +1. \quad (3.6)$$

Put  $m = -I(f_1) > 0$ . Insert  $m$  reverse traces in separated radial collars near the puncture, ordered from the  $m$ -fold stabilization  $\phi_m$  at the innermost end to  $\phi_0$  outward. The negative link is now loose and the total signed double count is  $I(f_1) + m = 0$ .

Apply Eliashberg–Murphy’s exact cancellation theorem [9, Theorem 2.3]. Its hypotheses here are as follows: the target is a simply connected Liouville manifold with a negative end; the immersion is proper, self-transverse, and has finitely many double points; its connected source has dimension eight; the negative asymptote has one loose component; and its signed double count is zero. The immersion is exact because  $H^1(S^7 \times \mathbb{R}; \mathbb{R}) = 0$ . The additional width assumption of that theorem applies only in source dimension three. Its conclusion is a compactly supported Hamiltonian regular homotopy to an embedding. Compact support preserves both cylindrical tails, including the prescribed positive tail.

In the unchanged negative region, insert  $m$  forward traces from  $\phi_0$  to  $\phi_m$  and fill the center by the real plane. Equation (3.6) shows that the filled immersion has exactly  $m$  double points, all negative. Its noncylindrical part is compact. Apply a sufficiently long negative Liouville flow in the filled positive completion to bring this compact part inside  $P$ . Backward flow exists for all time; points inside  $P$  cannot leave it backward, and on the positive end it translates the cylindrical coordinate. The flow rescales the symplectic form, so preserves

Lagrangianity. The positive cylinder remains the same cylinder as a subset and crosses  $\partial P$  in the original collar. Trim there and glue to the unmodified immersion outside  $W$ .

The replaced source piece is a disk with collars. A boundary parameterization extends to a homeomorphism of disks, so the old and new maps can be compared continuously relative to the common complement. As  $T$  is aspherical, this disk replacement preserves the homotopy class and the fundamental class. It also preserves the degree-two and degree-six homology maps through the common complement. Orient the replacement compatibly. Spin extends because the boundary  $S^7$  has a unique spin structure. Finally every first homology class is represented off the replaced disk; the Maslov class therefore remains zero. Choose the integral grading with the original orientation parity.  $\square$

*Proof of Proposition 3.1.* Apply local Lagrangian surgery at the negative double points of Lemma 3.6. We spell out the grading choice, since it is the reason for using period four. In symplectic coordinates the two sheets are  $\mathbb{R}^8$  and  $i\mathbb{R}^8$ . The handle is

$$(s, x) \mapsto \gamma(s)x, \quad x \in S^7,$$

where  $\gamma$  rounds a corner from the positive real ray to the positive imaginary ray through the first quadrant, or to the negative imaginary ray through the fourth quadrant. The first ray is traversed inward and the second outward. With the usual monotone corner shapes these are embedded Lagrangian handles: tangent vectors in the sphere directions pair trivially, and their symplectic pairing with  $\gamma'x$  vanishes because  $x \perp T_x S^7$ .

These handles glue orientably exactly at a negative double point. For the coordinate orientations the two planes have positive intersection sign in dimension eight, whereas the handle gives opposite relative orientation marks at its two ends. Thus the negative intersection condition is precisely the required orientation agreement.

Identify the ambient Maslov cover over each surgery ball with the standard one. For the phase lift  $\arg \det / \pi$ , a handle tangent frame has determinant proportional to  $\gamma'\gamma^7$ . Its total phase changes along the two corners are

$$+3 \quad \text{and} \quad -3, \tag{3.7}$$

respectively; see also [7, Lemma 2.4]. For example, along the first corner the initial argument is  $\pi$  and the final argument is  $4\pi$ . The integral lifts of the two original sheet phases differ by an odd integer, because their orientation marks are opposite. The two residues in (3.7) are the two odd residues modulo four. Consequently one of the two handles extends the grading modulo four at each double point. The choices are independent in disjoint surgery balls. The unique spin structures on their boundary seven-spheres also give spin extensions.

Let  $i_L : L \hookrightarrow T$  be the resulting embedded Lagrangian. Each oriented local surgery preserves its pushed fundamental class. Removing the pairs of eight-disks and inserting  $S^7 \times [0, 1]$  changes neither  $H_2$ ,  $H_6$ , nor  $H^2$ : the isomorphisms through the common complement follow directly from the relative homology sequence and Mayer–Vietoris. They intertwine the maps to  $T$ . Thus all the properties of Lemma 3.3 asserted in the proposition survive. Finally (3.2), extended to  $\mathbb{C}$ , and de Rham’s Theorem give exactness of the restricted forms in  $\Omega_{\mathbb{C}}$ .  $\square$

**Corollary 3.7** (Ambient detection). *Let  $r = 2$  or  $6$ , and let  $u$  be a closed complex  $r$ -form on  $L$ . If*

$$\int_L i_L^* \eta \wedge u = 0 \quad \text{for every closed complex } (8-r)\text{-form } \eta \text{ on } T,$$

*then  $u$  is exact.*

*Proof.* The displayed pairings say that  $(i_L)_*([u] \cap [L]) = 0$  in  $H_{8-r}(T; \mathbb{C})$ . The relevant injection in (3.3) and Poincaré duality give  $[u] = 0$ .  $\square$

## 4 Disk equations, compactness, and local gluing

We now fix the symplectic form on  $T$  once and for all. The deformation parameter will enter only through a relative closed two-form; it will not change the equation for holomorphic curves. This section constructs the local analytic objects needed for the finite integrations in the next section. Particular care is needed at an unmarked boundary, at a transverse boundary jump, and when several boundary nodes are smoothed simultaneously.

### 4.1 Boundary labels and the relative weight

Let  $L \subset T$  be the embedded spin Lagrangian of Proposition 3.1, with its grading modulo four. Set  $J_*$  to be the flat complex structure determined by  $q + ip$ . We use a fixed sufficiently small uniformly tame neighborhood of  $J_*$  in the space of  $\omega$ -compatible almost complex structures. All the structures below take values in this neighborhood.

There are countably many graph labels, arranged in layers indexed by  $\mathbb{N} \cup \{0\}$ . Each layer contains one copy of every scalar graph  $G_k$ ,  $k \geq 1$ , and of  $C$ . A label has the form

$$X = \{p = A_X q + \nabla f_X(q)\}, \quad A_X = h_0 k \, 1_8 \quad \text{or} \quad S_-, \quad (4.1)$$

where  $f_X$  is a smooth periodic real function. Layers are ordered increasingly; within each layer the scalar slopes are ordered decreasingly, followed by  $C$ . Write  $X \prec Y$  for this order, and  $X^+$  for the copy of the same slope in the next layer. The order need not be a well-order; all analytic constructions first use a finite sublist. Each graph has the orientation and spin structure from its base frame, trivial local system, and the integral grading continued from the horizontal plane through symmetric matrices. In particular

$$\pi_2(T, X) = 0,$$

by the relative homotopy sequence and injectivity of  $\pi_1(X) \rightarrow \pi_1(T)$ . Thus a graph supports no nonconstant holomorphic disk.

Choose the potentials so that all labels, including  $L$ , are pairwise transverse and have no triple intersections. In layer zero their Hessians are as small as needed. Require  $f_{X^+} - f_X$  to be a sufficiently small perfect Morse function; it may be chosen in the open set containing a small split sum of cosines. These choices are simultaneous: enumerate the countably many required conditions, and use transversality of the first-jet evaluation of a periodic function. At each step the conditions already imposed, as well as the perfect Morse condition, are open, and the remaining transversality and avoidance conditions are residual in an arbitrarily small open set. Equivalently one may use the Baire theorem on the product of these open sets of successive potential differences.

Our strip convention places the earlier boundary condition at height zero and the later one at height one, with inputs at positive ends. For graded planes with positive rotation angles  $\alpha_j \in (0, \pi)$ , the degree is

$$|x| = 8 + \vartheta_2 - \vartheta_1 - \sum_{j=1}^8 \alpha_j / \pi. \quad (4.2)$$

For two graph tangent planes this is the number of positive eigenvalues of the second Hessian matrix minus the first. To check this, keep their difference fixed, move the first matrix to zero, and diagonalize the difference. Consequently the maximum of  $f_{X^+} - f_X$  gives a unique degree-zero generator

$$w_X: X \longrightarrow X^+,$$

and all generators of  $G_i \rightarrow G_j$ ,  $i > j$ , in layer zero have degree zero. For arrows incident to  $L$ , degrees are taken modulo four.

Fix a complex symmetric matrix  $M$  commuting with  $D$ , and put  $B = dq^t M \wedge dp$ . For  $L$  and for scalar-slope labels choose forms  $\xi_A$  with  $d\xi_A = i_A^* B$ . On  $L$  these exist by the pullback-kernel property proved above. On a scalar graph we take

$$\xi_X = -(\nabla f_X)^t M dq. \quad (4.3)$$

Indeed symmetry of  $M$  makes the scalar linear part of  $i_X^* B$  vanish, and differentiation of (4.3) gives its remaining part. For a polygon with fixed jump limits set

$$B_{\text{rel}}(u) = \int u^* B - \sum_{\text{boundary arcs } a} \int_a u^* \xi_{A(a)}, \quad \text{wt}(u) = \mathbf{q}^{\int u^* \omega} \exp(t B_{\text{rel}}(u)). \quad (4.4)$$

Here  $t$  is an ordinary formal variable and  $\mathbf{q}$  is the Novikov monomial; the coefficient field is specified in Section 7. Only the unweighted specialization  $t = 0$  is used on words containing a  $C$ -slope label.

**Lemma 4.1.** *With the jump limits fixed,  $B_{\text{rel}}$  is invariant under homotopy through maps with the prescribed boundary lifts. It is additive on every nodal splitting. Thus the weight in (4.4) is multiplicative on trees, coefficient by coefficient in  $t$ .*

*Proof.* Apply Stokes to a homotopy cylinder. Its boundary strips lie on the prescribed labels, where  $B = d\xi_A$ , and cancel the variation of the arc integrals. The endpoint terms vanish because the jump limits are fixed. On splitting a disk, integrals add over the components; the inserted constant boundary paths at a node contribute zero. This also proves the assertion for refinements that retain relative classes at every vertex.  $\square$

## 4.2 The area gap and the later-copy comparison

**Lemma 4.2** (Uniform gap for a finite list). *For a finite transverse list of the labels and  $L$ , with no triples, there is  $\delta > 0$  such that every nonconstant unforced disk or polygon with these boundary conditions has area at least  $\delta$ . The constant is independent of the number of marked points and of the domain-dependent compatible structure in the fixed neighborhood of  $J_*$ . In particular fix  $\delta_L > 0$  for nonconstant pure disks on  $L$  with structure  $J_*$ .*

*Proof.* Choose finitely many sufficiently small target charts. A chart containing a pair intersection meets precisely the two local exact sheets, with that single intersection; elsewhere a sufficiently small chart meets at most one sheet. If the image has sufficiently small diameter, it lies in one of slightly larger such charts. In the two-sheet case every jump is at the same point. Normalize the two primitives to agree there. Stokes then gives zero area, even with repeated jumps; the one-sheet case is the same. Positivity implies that the map is constant.

Otherwise the connected image has a fixed positive diameter. Remove small disjoint neighborhoods of the finitely many pair intersections. There is a regular image point outside them through which the image crosses a ball of a fixed radius, meeting at most one local sheet. Here is the monotonicity estimate needed for this assertion. Let  $a(r)$  be area in a subball and let  $\ell(r)$  be the length, with multiplicities, of its level contours. Coarea and uniform compatibility give  $a'(r) \geq c\ell(r)$  for almost every  $r$ . Each contour arc ending on the local sheet can be closed there by a path whose length is at most a fixed multiple of the arc length. The primitive is exact on that sheet, so this does not alter the Stokes computation. A local primitive centered at a point of each resulting loop gives the quadratic bound  $a(r) \leq C\ell(r)^2$ . Hence

$$a'(r) \geq cC^{-1/2} \sqrt{a(r)}, \quad a(r) \geq c_0 r^2$$

up to reducing the fixed radius. All constants use target charts and uniform compatibility only. They do not use derivatives in the domain of the almost complex structure. This supplies the asserted gap.  $\square$

We impose a further smallness requirement on  $X^+$ . It is the geometric reason its eventual action on the new module will be invertible.

**Lemma 4.3** (Comparison with vertical fibers). *For each  $X$ , its next copy  $Y = X^+$  may be chosen with a number  $0 < \epsilon_X < \delta_L$  and the following properties. There are symplectic coordinates in a tube about  $X$  in which  $X$  is horizontal,  $Y$  is  $p = \nabla f(q)$  with  $\text{osc } f < \epsilon_X$ , and each component of  $L$  in that tube meeting  $X$  is a vertical patch at a point  $q_j$ . Let  $F_j$  be the full vertical fiber there. Then:*

- (i) *Unforced  $X, Y, L$  triangles of area at most  $\epsilon_X$  coincide literally with  $X, Y, F_j$  triangles in the corresponding patch; no such triangle joins different patches.*
- (ii) *Every  $X, Y, F_j$  triangle, with  $X, Y$  corner  $r$ , has area  $f(r) - f(q_j) \leq \text{osc } f$ . In particular the corner  $w_X$  gives area  $\Delta_j = f(\max) - f(q_j) < \epsilon_X$ .*
- (iii) *There are no nonconstant  $X, L$  or  $Y, L$  strips of area at most  $\epsilon_X$ , and no nonconstant  $X, F_j$  or  $Y, F_j$  strips of any area.*

*The same choices work for any of the allowed almost complex data after pushing those data forward by the fixed symplectic coordinate change. If  $X \cap L = \emptyset$ , choose  $Y \cap L = \emptyset$  as well.*

*Proof.* The integral shear  $p \mapsto p - A_X q$  and the gradient translation flatten  $X$ . Near a transverse intersection, the other sheet has equation  $q = q_j + \nabla_p g(p)$ , with  $dg(0) = 0$ . A Hamiltonian shear, cut off in that chart and generated using  $g - g(0)$ , makes it vertical while fixing the horizontal sheet. There are finitely many disjoint such charts. By compactness a sufficiently thin tube about the horizontal torus meets  $L$  only in these vertical patches. A sufficiently close  $Y$  remains an exact graph throughout this transformation.

The crossing estimate of Lemma 4.2, applied away from the two nearby graph boundaries, gives a fixed positive area cost for leaving this tube. It is uniform as  $Y$  approaches  $X$ . Choose  $\epsilon_X$  below this cost and  $\delta_L$ , and then choose  $Y$  with  $\text{osc } f < \epsilon_X$ . A low-area curve stays in the tube and its third boundary arc stays in one vertical patch. Conversely, lift a triangle with third boundary  $F_j$  to the cover and use the primitive  $-p^t dq$ . The horizontal and vertical boundary actions vanish, while the graph action is  $-df$ . Stokes gives  $f(r) - f(q_j)$ . Thus every such triangle also stays in the tube and has the required correspondence. A strip in a single patch has equal endpoint actions and area zero. For the full vertical fiber the lifted graph and fiber have a unique intersection, so this last argument applies without an area bound.  $\square$

### 4.3 Harmonic domain data

An  $m$ -disk has a distinguished boundary output followed counterclockwise by its ordered inputs. We use four sectors:

- pure  $m_k$ , with all boundary on  $L$  and smooth input markings;
- directed polygons with successive labels  $X_0 \prec \cdots \prec X_k$ , and all inputs and output transverse jumps;
- mixed polygons with successive graph labels  $X_0 \prec \cdots \prec X_b$ , then  $L$ , with graph inputs followed by an  $X_b, L$  input and any number of smooth  $L$  inputs;

- pure  $p_k$  disks with one interior output and  $k$  cyclically ordered smooth boundary inputs. For nonconstant maps  $k = 0$  is allowed.

We count stable maps only. The pure unary differential also has its usual unstable term  $d$  on differential forms. Pure disks, including all  $p$  disks, use  $J_*$ . In a sector with jumps, ignore all smooth and auxiliary markings when choosing the equation.

For an ordered jump-arc list  $\mathbf{A} = (A_0, \dots, A_{m-1})$ , let  $h_j(z)$  denote harmonic measure of its  $j$ th arc as seen from  $z$ . Equivalently, send  $z$  to the center of the disk and divide the arc's angular length by  $2\pi$ . Put

$$J(z, x) = J_{\mathbf{A}}(h_0(z), \dots, h_{m-1}(z); x), \quad h(z) \in \Delta^{m-1}. \quad (4.5)$$

The functions  $J_{\mathbf{A}}$  are smooth on the closed simplices.

**Proposition 4.4.** *The functions in (4.5) can be chosen simultaneously with the following properties.*

- (i) *On a simplex face they agree with the function for the surviving subsequence. They equal  $J_*$  near each vertex.*
- (ii) *Near an edge joining labels  $A, B$ , they equal exactly  $J_{AB, h_B}$ , for a height-dependent strip datum constant  $J_*$  near heights zero and one. Take  $J_{X_L, \tau} = J_*$ .*
- (iii) *All nonconstant graph-pair strips and all directed polygons are regular, including domain variations. The comparison triangles of Lemma 4.3, with  $L$  replaced by any  $F_j$ , are regular for the pushforward of the same triangle data.*

*No regularity of general mixed or pure maps is required.*

*Proof.* First prescribe generic graph-pair data, supported in heights  $(0.1, 0.9)$ . Ordinary strip transversality applies to these  $s$ -independent data. Its injective-point condition can be expressed as injectivity of  $(s, \tau) \mapsto (\tau, u(s, \tau))$ : if an open immersed patch had a same-height partner on every point, a Baire argument among compact immersed patches would give overlapping branches  $u(s, \tau) = u(\phi(s, \tau), \tau)$ . The equation forces  $\phi(s, \tau) = s + c$ ; unique continuation would then contradict finite energy for a nonconstant strip. Critical images are negligible and immersive points are dense. Hence variations of  $J_\tau$  near an injective point kill any proposed cokernel element, also when the variation is restricted to the indicated open height interval.

Extend by word size. Require the pair formula where  $h_A + h_B > c_m$  and the vertex formula where  $h_A > d_m$ , using thresholds increasing slightly with  $m$ , with  $c_m$  close to one and  $d_m$  close to 0.94. Pair variations vanish near the endpoints, so these requirements agree on overlaps. In a local linear chart of compatible structures near  $J_*$ , first extend the required collar formulas and then extend the remaining boundary values over the simplex. Inclusion–exclusion in corner charts extends functions agreeing on intersecting faces; a partition of unity combines the extensions. A smooth retraction into the chosen small neighborhood retains all already prescribed values.

For  $m \geq 3$ , a jump configuration together with one interior point, modulo disk automorphisms, is identified with the open simplex by  $h$ : after sending the point to the center, the successive arc lengths recover all endpoints up to rotation. There is a nonempty free open set outside the prescribed edge and vertex collars. Each fixed domain has interior points mapping to this set under  $h$ : send any three of its jumps to equally spaced circle points. At the center every individual weight is at most  $1/3$  and every pair has total at most  $2/3$ , so a neighborhood of the center is free. There is an immersive point there for a nonconstant solution. A variation of  $J_{\mathbf{A}}$  supported near that simplex point isolates its domain point, even

if the map has multiple images. The usual pairing with a nonzero adjoint solution and unique continuation prove surjectivity of the universal linearization. A directed map is nonconstant because there are no triples. Sard–Smale therefore gives the required regularity relative to all prescribed faces and collars. Apply the same argument also to the finitely or countably many auxiliary fiber triangles using precisely the pushed-forward datum. There are only countably many words, classes, and such transversality requirements, so the successive residual choices can all be made.  $\square$

#### 4.4 Boundary compactification and finite cuts

**Lemma 4.5** (Only disk components occur). *Every stable limit in the four sectors consists of disks joined at boundary nodes. Smooth nodes in a mixed sector split off pure  $L$  disks. At jump nodes the two ends have opposite transitions, and each vertex has the corresponding subsequence of the original ordered labels. The same conclusion holds for positive-area  $p_0$  disks; there is no additional stratum in which the whole boundary collapses.*

*Proof.* There are no nonconstant spheres because  $\pi_2(T) = 0$ . The disk vertices in the doubled domain form a connected subtree: they are fixed vertices of its real involution, and the path between fixed vertices is fixed. Consequently other sphere trees could only hang from a disk at an interior node. A tree of  $v$  constant spheres with one attachment and at most one interior marking has at most  $2v$  special points counted with multiplicity, whereas stability requires at least  $3v$ ; hence no such tree occurs.

The issue when  $p_0$  has no boundary markings is the possible collapse of the entire real circle. In the doubled dual tree the collapsed boundary would be the midpoint of an edge whose branches are exchanged by the involution. This cannot coexist with a fixed disk vertex: the path from a fixed vertex to that midpoint would be fixed, including its final half-edge, contrary to exchanging the two half-edges. Thus all normalized components occur in conjugate sphere pairs. In the original half these are sphere components. Positive area would force one of them to be nonconstant, which is impossible. This rules out boundary collapse in the only sector where it is not excluded by a boundary marking.

Following labels around a boundary smoothing proves the subsequence assertion. A smooth node on a graph would cut off a single-graph subtree with no original inputs. An outermost nonconstant component would be a disk on that weakly exact graph, while a wholly constant subtree is unstable. Such a subtree is impossible. Thus smooth attachments in a mixed sector are pure  $L$  branches. On a limiting core the omitted arcs have harmonic measure zero, so Proposition 4.4(i) gives exactly the equation of that vertex.  $\square$

We will verify compactness, including absence of neck drift, after giving the explicit collar coefficients below. Its finite-cut consequence can already be stated. For a finite list and a positive integer  $N$  choose

$$0 < a < \min(1, \delta_L, \delta),$$

where  $\delta$  is the gap of Lemma 4.2. Define

$$c_m(\beta, k) = \omega(\beta) + a(k - 1), \quad c_p(\beta, k) = \omega(\beta) + ak. \quad (4.6)$$

Retain all stable types of cost at most  $4N$ , with their relative-class refinements and all nodal types whether or not they are smoothable.

**Proposition 4.6.** *This is a finite collection of compact moduli problems, closed under normalized boundary and corners. Costs add on every splitting, every stable vertex has positive cost, and only finitely many relative classes occur. No discreteness of the symplectic period group is needed.*

*Proof.* For an  $m$  splitting with child arity  $s$  and parent arity  $k - s + 1$ ,  $(s - 1) + (k - s) = k - 1$ . For a  $p$  root the analogous equality is  $(s - 1) + (k - s + 1) = k$ . This proves additivity, including  $s = 0$ . A stable constant pure  $m$  vertex has  $k \geq 2$ , and a constant  $p$  vertex has  $k \geq 1$ . A stable constant mixed vertex likewise has at least three boundary special points; directed constants are excluded by no triples. Such vertices cost at least  $a$ . Every nonconstant  $m$  vertex costs at least  $\delta - a > 0$ , and every nonconstant  $p$  vertex costs at least  $\delta$ . Thus the cut bounds the number of vertices, total area, and arity. Only finitely many combinatorial patterns and asymptotic generators occur.

For completeness the compactness argument, proved below in Lemma 4.9, is for the union over relative classes with these bounds. In a convergent sequence, maps on retained compact cores are homotopic to their limits, and the remaining thin parts have arbitrarily small diameter in the appropriate local one- or two-sheet charts. Contracting these parts and matching at the nodes preserves the total relative class. The class is therefore locally constant in the compact stable-map space. A finite subcover yields finitely many classes. The same reasoning retains any relative refinement used to record the additive weight. Positivity and additivity show that every component of a boundary type stays in the cut, proving boundary closure.  $\square$

#### 4.5 Real plumbing and quantitative collar coefficients

Since all nodes are boundary nodes, we can use whole-disk projective coordinates for every retained core. Root a disk tree at its boundary output, or at the disk carrying the interior output in the  $p$  sector. On a child use an upper-half-plane coordinate  $z$  with its output at infinity; on its parent put the attaching point at  $a_e$ . The insertion

$$\zeta = a_e + \rho_e z, \quad \rho_e = e^{-\pi T_e}, \quad T_e \gg 1, \quad (4.7)$$

is real plumbing, using  $-1/z$  as the child's nodal coordinate. Cutting out the corresponding half-disks and making this identification gives another upper half-plane. Iteration therefore identifies each core by the restriction of a whole-disk biholomorphism. This point will also make the intrinsic normalizations in the next section compatible with gluing.

Use node-specific projective coordinates varying smoothly in stabilized component configurations. They can be chosen equivariantly: a finite-order orientation-preserving disk automorphism fixing a boundary point is the identity, so a node stabilizer acts trivially on its incident disks. Choose the germs on representatives of the node orbits and transport them; invariant local slices for the finite stabilizer handle component configurations. Descendant arc endpoints in a parent coordinate have the form  $a_e + O(\rho_e)$ , and exterior endpoints in a child coordinate have reciprocals  $O(\rho_e)$ . Iterated insertions give smooth expressions in all plumbing parameters; apparent denominators stay separated by using the nearest common ancestor coordinate.

**Lemma 4.7** (Harmonic collar estimates). *On a jump neck  $e$ , write*

$$\zeta - a_e = -e^{-\pi(s+i\tau)}, \quad R \leq s \leq T_e - R, \quad 0 \leq \tau \leq 1, \quad r_l = e^{-\pi s}, \quad r_r = e^{-\pi(T_e - s)}.$$

*The transitions at the ends are an ordered pair  $A, B$ . For sufficiently large fixed  $R$ , the equation uses precisely its pair collar, and*

$$h_B = \tau + O(r_l + r_r), \quad h_A = 1 - \tau + O(r_l + r_r), \quad \sum_{C \neq A, B} h_C = O(r_l + r_r). \quad (4.8)$$

*On the parent half the difference from the partially broken parent coefficient is  $O(r_r)$ ; on the child half the corresponding difference is  $O(r_l)$ . These estimates hold after any prescribed*

finite number of strip, target, and compact component-parameter derivatives. Logarithmic length derivatives have the same estimates, and a derivative with respect to another length  $T_f$  carries an additional factor  $O(e^{-\pi T_f})$ , uniformly in the other large lengths. In particular all coefficients have uniformly bounded strip derivatives.

On a smooth matching neck the datum is exactly  $J_*$  throughout a long middle collar. The same one-sided estimates hold at external jump ends.

*Proof.* Harmonic measure in the upper half-plane is the angle subtended by an interval, divided by  $\pi$ . For the two intervals adjacent to  $a_e$ , the limiting angles in the displayed coordinate are  $\pi\tau$  and  $\pi(1-\tau)$ . A displacement of a clustered endpoint is  $O(\rho_e)$ ; dividing by  $|\zeta - a_e| = r_l$  gives  $O(r_l)$ . A separated exterior endpoint contributes an angle  $O(r_l)$ , with its reciprocal distance bounded. The smooth angle function and all of its derivatives on these separated configurations give (4.8) and the two one-sided comparisons. Differentiating  $\rho_e$  logarithmically only multiplies it by  $-\pi$ . If an endpoint shift comes from a descendant node  $f$ , its differentiated displacement is  $O(\rho_e \rho_f)$  before dividing by  $r_l$ ; the same argument in an ancestor or nearest-common-ancestor chart treats all other cases. This proves uniformity in every other length as well as all finite derivative versions.

For large  $R$  the total omitted weight is uniformly small, so the simplex point lies in the pair neighborhood of Proposition 4.4; the coefficient is exactly  $J_{AB, h_B}$ . At a smooth node only a pure  $L$  branch is inserted. No original jump lies down that branch. On the long collar the weight of  $L$  is in the vertex region, where the datum is exactly  $J_*$ .  $\square$

## 4.6 Decay and compactness

**Lemma 4.8** (Transverse ends and smooth-node decay). *At a jump limit  $x \in A \cap B$ , the operator*

$$\mathcal{A}\xi = J_{AB, \tau}(x) \partial_\tau \xi, \quad \xi(0) \in T_x A, \quad \xi(1) \in T_x B,$$

*on the interval is self-adjoint for the  $\tau$ -dependent compatible  $L^2$  metric, has compact resolvent, and has zero kernel. In particular it has a positive spectral gap.*

*On a sufficiently small-image long neck, solutions with no obstruction support satisfy uniform exponential estimates from its two ends for all positive derivatives. At a jump the limiting value is the fixed intersection; at a smooth node one separates a single movable value in  $L$ . On a strip  $[0, T] \times [0, 1]$ , after removing fixed end collars, one may take*

$$|\nabla^j du(s, \tau)| \leq C_j (e^{-\mu s} + e^{-\mu(T-s)}) \quad (j \leq j_0), \quad (4.9)$$

*for any preassigned finite  $j_0$ , with constants uniform near the given nodal map. The same estimates hold for small compact-core-forced solutions, since their neck equations are unforced.*

*Proof.* Put  $g_\tau(v, w) = \omega(v, J_{AB, \tau}(x)w)$ . Compatibility gives

$$\int_0^1 g_\tau(\mathcal{A}\xi, \eta) d\tau = \int_0^1 \omega(\partial_\tau \xi, \eta) d\tau = - \int_0^1 \omega(\xi, \partial_\tau \eta) d\tau = \int_0^1 g_\tau(\xi, \mathcal{A}\eta) d\tau.$$

The boundary term vanishes by Lagrangianity. The adjoint has the same boundary domain, which proves self-adjointness; the one-dimensional elliptic estimate and compact embedding give compact resolvent. Its kernel consists of constant paths in  $T_x A \cap T_x B$ , hence is zero. Notice that no constancy of  $J_{AB, \tau}$  in  $\tau$  was used.

The full-cylinder inverse of  $\partial_s + \mathcal{A}$  is obtained by integrating the positive spectral subspace forward and the negative subspace backward. Its kernels are bounded by  $Ce^{-\mu_0|s-s'|}$ , where  $\mu_0$  is the spectral gap. This gives the weighted estimate for any logarithmic weight slope less than  $\mu_0$ ; local elliptic estimates give its Sobolev versions. Straighten the two boundary

sheets at  $x$ . The nonlinear equation on a small-image neck differs from this model by a small first-order term, a zero-order term bounded by  $C|du|$ , and the controlled coefficients in Lemma 4.7. After a fixed shortening of the collar, absorb the small first-order term in the cylinder estimate. Cutting off at the two ends yields (4.9); differentiation and elliptic bootstrapping give the asserted finite range of derivatives.

For a smooth node both boundaries lie on the same sheet. Work first in its cotangent tube, whose primitive vanishes on the zero section. For a short path with endpoints on the sheet, its action is bounded by a fixed constant times the square of its length: close it by a short sheet path and use a centered primitive in a local chart. If  $E(R)$  is the energy in  $[R, T - R] \times [0, 1]$ , Stokes and this estimate at the two slices give

$$E(R) \leq C(e(R) + e(T - R)) = -CE'(R), \quad (4.10)$$

where  $e(s)$  is slice energy, up to a fixed uniform comparison of metrics. Therefore  $E(R) \leq C'e^{-R/C}$ . Small-energy regularity on unit strips, with the uniform coefficient bounds above, gives exponential decay of positive derivatives. Integrating the first derivative gives convergence to one value and the same two-sided estimate measured relative to a middle boundary value. This argument also shows that the constant asymptotic mode must be retained, rather than estimated by an invertible transverse model. The argument for two transverse sheets can alternatively use (4.10) by closing each slice to their intersection along the sheets.  $\square$

**Lemma 4.9.** *For a finite list, fixed arity bound, and bounded area, the unforced stable-map spaces of the four sectors are compact after adjoining the disk trees of Lemma 4.5. There is no lost energy or drift of the matching values. On the cores the convergence is smooth, and relative classes agree eventually with the total limiting class.*

*Proof.* Apply the usual disk and strip rescaling procedure, retaining marked-point clustering scales as well as nonconstant bubbling scales. The required coefficient compactness holds here directly: harmonic measures are bounded harmonic functions and have uniform derivatives on interior hyperbolic balls. Under any disk reparametrization, clustered endpoints converge to a limiting configuration; on a retained core their harmonic measures converge to the surviving ones. Along a retained boundary arc the datum is constant in a vertex neighborhood; on a jump collar use Lemma 4.7. An interior blowup at scale much smaller than the boundary distance sees constant-in-domain data. Consequently any new gradient concentration gives an ordinary disk, a transverse strip, or an interior sphere. Spheres are impossible, and each nonconstant level consumes the positive area gap. There are therefore only finitely many nonconstant levels, with precisely the equations prescribed above.

After all these levels have been extracted, derivatives on the remaining middle strips are small by small-energy regularity. At a jump, a short slice between two distinct compact sheets is close to their finite intersection set. Continuity makes it remain near one intersection throughout each remaining thin strip. For one sheet, short slices already lie in its cotangent tube, even before one knows that their longitudinal motion is small. Thus (4.10) applies in both cases. The constants in the single-sheet action estimate are uniform along the compact sheet; it does not assume the image lies near a fixed point in advance. Its exponential consequence and integration of  $|du|$  show that the central oscillation tends to zero. This excludes slow travel along  $L$ , as well as loss of energy, and identifies the two limiting values at a smooth node. Elliptic estimates give the claimed core convergence. Finally, thick-region homotopies and contractions of these small thin parts within their prescribed local sheets identify the total relative classes. Stability and Lemma 4.5 give the stated compactification.  $\square$

## 4.7 Fixed-domain augmentation

We specify precisely the local input from the established disk theory. For a compact target, a compact relatively spin boundary Lagrangian, and fixed tame  $J$ , compact-core obstruction data give finite-dimensional disk charts and smooth evaluations; augmented boundary-evaluation surjectivity gives an output submersion [10, Theorem 7.1, Condition 7.11, and Lemma 7.12]. The gluing theorem assumes surjectivity of the augmented component operators and of the difference of their matching evaluations. It gives local gluing and exponentially small positive length derivatives on retained cores, also with domain parameters [12, Assumption 3.12, Theorems 3.13, 6.4, and 8.16–8.17]. For a fixed datum our finite union of labels is an immersed Lagrangian with transverse double points; retaining its boundary-lift sectors provides the standard transverse-end Fredholm and determinant-line setup [3, Sections 4–5]. We next prove the extensions needed for (4.5) and for  $p_0$ .

At a component  $u$  use high Sobolev order on its compact part. At a jump take weighted sections tending to zero at the fixed intersection. At a smooth special point viewed as a cylindrical end, write a section as its movable limit  $a \in T_{u(z)}L$  plus a decaying section. Small positive weights are chosen below all nonzero indicial roots. This is the usual Fredholm formulation of a removable smooth boundary point: the zero mode is the tangent limit, and homogeneous or compact-core-forced solutions extend over the point. Stabilize any unstable nonconstant domain by interior marks constrained to transverse target slices at immersive points, taking finite automorphism orbits. One can do this also for an unbroken  $p_0$  map. The original interior mark is retained, but no boundary mark is artificially made an output.

**Lemma 4.10.** *For each component one can choose a finite-dimensional space  $E$  and an insertion  $I : E \rightarrow C_c^\infty(\Lambda^{0,1} \otimes u^*TT)$  supported in interior cores away from all markings, such that  $D_u \bar{\partial} - I$  is surjective at fixed domain. For every pure  $m$  component the space can be chosen so that*

$$(\xi, e) \mapsto (D_u \bar{\partial} \xi - Ie, \xi(z_0)) \quad (4.11)$$

*is surjective as well. Componentwise sums give a surjective nodal operator with all smooth matching equations imposed. The same holds with a prescribed pure root-output variation.*

*Proof.* The Fredholm cokernel is finite-dimensional. At first Sobolev order with target weighted  $L^p$ ,  $p > 2$ , smooth interior tests away from the finite special points are dense, and hence span its quotient. Choose finitely many such tests. To arrange (4.11), apply the same argument to the operator restricted to sections with  $\xi(z_0) = 0$ . Given any prescribed output value, first choose a section attaining it and then solve its remaining equation with zero output variation. Bootstrapping transfers these conclusions to the higher Sobolev orders. Replace  $E$  by the formal sum of its translates under the finite stabilizer and average the auxiliary metrics. This makes the choice equivariant without reducing its image; injectivity of the formal insertion is not needed.

At a smooth node the outgoing child is a pure  $m$  component. Starting at the root, choose its solution with any required external output; then solve each child with output value equal to the already chosen parent value, and continue outward. Formula (4.11) permits this at each step. Jump matching is at a fixed transverse intersection and adds no tangent matching variable. This proves nodal surjectivity. The pure constant disk operator is already surjective with arbitrary constant boundary value. A constant mixed component has only an  $X, L$  pair at the same intersection, together with removable smooth marks, and has the invertible transverse strip operator. Thus constants require no additional interior tests.  $\square$

## 4.8 A uniform gluing estimate with several necks

We give the uniform estimate explicitly, rather than invoking a fixed- $J$  gluing theorem for the domain-dependent equation. Fix a stabilized nodal map and the component augmentations of Lemma 4.10. All constants below are uniform on a small relatively compact neighborhood of its augmented nodal solutions. The finitely many component operators have a common indicial gap. Choose a small positive weight  $\delta$  below one tenth of that gap and of the decay rates in Lemmas 4.7 and 4.8.

On an end use weight  $e^{\delta s}$ . On a joined neck of length  $T_e$  use a smooth version of

$$w_e(s) = e^{\delta \min(s, T_e - s)}. \quad (4.12)$$

The smoothing takes place in a fixed-width middle interval and changes all norms by uniform factors. At a jump, sections have zero asymptotic constant. At a smooth node write

$$\xi_e = a_e + \xi_e^{\text{dec}}, \quad a_e \in T_{x_e} L,$$

using fixed sheet-straightening coordinates; include  $|a_e|$  separately in the norm and weight only the remainder. On a joined neck  $a_e$  may be the value at its middle boundary point. On the finite set of cores use the ordinary  $W^{m,2}$  norms, and on remainders use  $\sum_{j \leq m} \|w_e \nabla^j \xi_e^{\text{dec}}\|_{L^2}$ . Add the finite-dimensional domain and coefficient norms. Denote the resulting norm by  $\|\cdot\|_{m,\delta,T}$ , and use the corresponding weighted  $W^{m-1,2}$  target norm. Here  $m$  can be taken arbitrarily large, but finite.

**Lemma 4.11** (Uniform augmented inverse). *Fix any subset of nodes to remain broken. For all other lengths  $T_e$  sufficiently large, the linearized thickened equation at the preglued map is surjective, uniformly in every length. After adding finite-dimensional kernel coordinates measured on the cores and in the coefficient variables, it has an inverse  $Q_T$  satisfying*

$$\|Q_T(\eta, b)\|_{m,\delta,T} \leq C_m(\|\eta\|_{m-1,\delta,T} + |b|). \quad (4.13)$$

*The same holds at sufficiently small thickened solutions, at fixed domain, and with any prescribed pure root-output variation.*

*Proof.* Choose core observables giving an isomorphism from the augmented nodal kernel to a Euclidean space. Together with the nodal operator and its matching equations, these observables form a bounded isomorphism. The same isomorphism works in the required finite range of Sobolev orders and small weights: the indicial gaps do not change the index, and elliptic regularity identifies its kernels. Domain coordinates can either be included or held fixed. The output assertion uses the augmented operator (4.11).

Here is a bounded parametrix for the smoothed operator. Split a target section at each middle neck by a partition whose transition has fixed width. Transfer the two pieces to the corresponding semi-infinite ends and extend them by zero. Their one-sided weighted norms are bounded by a fixed multiple of the joined norm, since on the support of the left piece  $s \leq T_e/2 + O(1)$  and hence

$$e^{\delta s}/w_e(s) \leq e^{O(\delta)};$$

the right side has the identical estimate. Solve this forcing with the nodal isomorphism, retaining all smooth matching equations and the prescribed core coordinates. The resulting two sections at a smooth node have the same constant  $a_e$ .

On a component end cut off only the decaying part. Take the left cutoff equal to one through  $T_e/2 + 2$ , zero by  $3T_e/4$ , and with  $j$ th derivative  $O(T_e^{-j})$ ; take the reflected right cutoff. The support of each split forcing lies where its own cutoff is one. On the overlap set

$$\xi_e = a_e + \beta_l \xi_l^{\text{dec}} + \beta_r \xi_r^{\text{dec}} \quad (4.14)$$

at a smooth node; at a jump omit  $a_e$ . Thus the matching constant is included once, not twice. The joined weight is bounded by either one-sided weight where that end is used, so the decaying terms have uniform norm. If the joined norm uses the actual middle value instead of  $a_e$ , weighted Sobolev embedding gives

$$|\xi_e(T_e/2, 0) - a_e| \leq Ce^{-\delta T_e/2}(\|\xi_l^{\text{dec}}\|_{m,\delta} + \|\xi_r^{\text{dec}}\|_{m,\delta}).$$

Since  $\|w_e\|_{L^2} \leq Ce^{\delta T_e/2}$ , recentering by this constant is still uniformly bounded. This verifies the parametrix bound including the zero modes.

Apply the smoothed operator to (4.14). Commutators with the cutoffs have norm  $O(T_e^{-1})$ . The difference of the equation coefficients from the corresponding component coefficients on a cut end is exponentially small: its cutoff ends at  $3T_e/4$ , so the opposite-end errors of Lemma 4.7 are at most  $Ce^{-\pi T_e/4}$  there, and pregluing errors have the analogous map-decay bound. At a smooth match, the limiting operator annihilates a constant tangent vector. Its zero-order coefficient on such a vector is  $O(|du|)$ , so all errors involving  $a_e$  have the same exponential estimate after weighting; the choice of  $\delta$  makes this estimate summable. Core-supported obstruction maps, constraints, and kernel observables converge by their smooth core transport. It follows that the parametrix  $P_T$  satisfies

$$\|\mathcal{L}_T P_T - \text{id}\| \leq C_m \sum_{e \text{ smoothed}} (T_e^{-1} + e^{-cT_e}), \quad (4.15)$$

where  $\mathcal{L}_T$  includes the kernel observables. The number of nodes is fixed. Neumann inversion proves a uniformly bounded right inverse. The boundary-node Fredholm index formula subtracts  $\dim L$  for a movable smooth match and adds no matching dimension at a nondegenerate transverse end. It is exactly the nodal index with the chosen smoothing lengths held fixed. Hence adding the same kernel observables gives index zero, so this right inverse is an inverse. All these estimates hold with any other nodes kept broken. Small changes of the map and coefficients preserve the bound by a final Neumann inversion.  $\square$

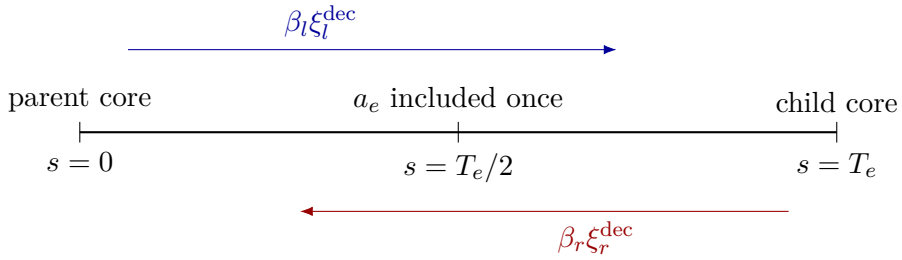


Figure 2: A smooth matching neck. The forcing is split in a fixed-width middle interval. The two decaying solutions are cut off over long intervals, while their common tangent value is retained once.

The preceding proof also handles the controlled inhomogeneous collars used later for ordinary continuation. More precisely, suppose that, after Hamiltonian transport of a limiting chord to an intersection, the equation has an additional term  $R_T(s, \tau, u)$  with bounded finite strip derivatives, of size  $O(r_l + r_r)$ , and with differences from the two partially broken equations  $O(r_r)$  and  $O(r_l)$  on the respective halves. Assume the differentiated versions in Lemma 4.7 as well. Each limiting end then has a decaying forcing  $O(e^{-\pi s})$ . The spectral inverse gives decay at any smaller rate than the indicial gap and  $\pi$ ; pregluing has an exponentially small residual in a smaller weight, and the same parametrix proof applies. No nondecaying additive forcing is permitted in this assertion.

**Proposition 4.12** (Local analytic chart contract). *For any prescribed finite integer  $r$ , choose sufficiently high Sobolev order and sufficiently small positive weights. Around each stable map in the finite system of Proposition 4.6, the component augmentations of Lemma 4.10 give the following local construction, also for  $p_0$ .*

*Allow finite-dimensional coefficient variables whose insertions are supported on the normalized component cores, depend smoothly on finitely many core jets and incidence markings, ignore the auxiliary stabilizing marks, and have the core-transport and exponential length-derivative properties of Lemma 4.7. At zero coefficients assume the fixed-domain augmentations above are surjective. Then the small pairs solving*

$$\bar{\partial}_J u = I(u, \mathbf{e}) \quad (4.16)$$

*form local  $C^r$  manifolds with corners modulo the finite map stabilizers. Corner coordinates are  $q_e = 1/T_e \geq 0$ , with  $q_e = 0$  meaning that node  $e$  is retained. Their normalized faces are exactly the corresponding component solution spaces, matched over  $L$  at smooth nodes and over the fixed intersection at jumps. Pure boundary outputs are submersive on every stratum, also with the domain fixed. Interior outputs need only be smooth.*

*These charts contain all sufficiently close solutions of (4.16), including all partially broken ones. Core observables, evaluation maps, and retrieval of transverse stabilizing slices are  $C^r$ . Changes between stabilizations retaining the components and nodes at the comparison point are  $C^r$  in these coordinates. Finally, outside sufficiently large fixed compact interior cores, derivatives on hyperbolic unit vectors can be made uniformly small on all these small-coefficient solutions.*

*Proof.* We spell out both the nonlinear estimate and the length-derivative estimate needed here. Stabilizing slice equations are imposed with marking positions free before division by disk automorphisms. Their transversality follows from the immersive incidence points, since neither the equation nor the insertions uses those auxiliary marks. Choose a finite collection  $b$  of core kernel observables, including coefficient and domain coordinates as necessary, which makes the nodal linearization an isomorphism. Near the nodal map they parametrize its augmented solution stratum. Preglue this nearby nodal family with the prescribed  $b$ , cutting the decaying map parts and retaining the matching values on  $L$ .

The residual of (4.16), including the prescribed core observables, has norm at most  $C \sum_e e^{-cT_e}$  in a weight smaller than the component decay rate. On each core this follows from the smooth insertion transport and Lemma 4.7; in each middle neck it follows from cutting exponentially decaying tails and the two one-sided coefficient comparisons. The same statement includes the controlled additive errors described above, when present.

The nonlinear remainder satisfies, on a ball of fixed small radius,

$$\|\mathcal{N}_T(\xi) - \mathcal{N}_T(\eta)\|_{m-1, \delta, T} \leq C(\|\xi\|_{m, \delta, T} + \|\eta\|_{m, \delta, T})\|\xi - \eta\|_{m, \delta, T}, \quad (4.17)$$

with  $C$  independent of the lengths. Indeed ordinary Sobolev multiplication on fixed-width cylinder charts has uniform constants;  $w_e \geq 1$  puts the weight on the differentiated factor, while Sobolev embedding bounds the undifferentiated one. At a smooth node use actual motions of the common point in  $L$ , not an affine variation that violates the boundary condition. In sheet-straightening coordinates a constant such motion still solves the constant-map equation. Thus every nonlinear term on the neck involving the constant has a derivative of a nonconstant part as a factor. Its weighted estimate is exactly the preceding product estimate, with  $|a_e|$  handled as a finite-dimensional variable. The insertions are on fixed cores, where ordinary high-order estimates apply. Starting at a larger  $m$  accommodates their finite jet dependence.

Apply the contraction theorem with the inverse in Lemma 4.11 and the prescribed observables. It gives a unique small correction, with radius and estimates uniform in all large lengths. It works unchanged with an arbitrary subset of nodes still broken, retaining their matching equations. Conversely, a nearby solution has the indicated stabilized domain by marked-domain compactness. Place its slice markings uniquely by incidence on the retained cores. Core closeness and Lemma 4.8 put it in the weighted implicit-function neighborhood: outside successively larger cores the decaying tails are small in any strictly smaller weight. Hence the implicit construction captures every nearby solution and has no extra local branch. Lemma 4.10 and the version of the inverse with output prescribed prove stratumwise output submersivity. Kernel coordinates for this statement may be chosen at fixed domain as in that lemma.

It remains to verify differentiability at simultaneous corners. Identify domains under a change of  $T_e$  by fixing vertex coordinates on both sides and stretching only a middle part of neck  $e$ . Trivialize compact component deformations while keeping their disjoint nodal coordinate germs fixed. Fix a finite range

$$m_0 > m_1 > \cdots > m_r > 3, \quad \delta_0 > \delta_1 > \cdots > \delta_r > 0,$$

with sufficiently large initial order. The nodal and glued inverses are bounded in this entire range.

We record the weight estimate used for a middle stretch. Let  $\chi_e$  be supported where  $\min(s, T_e - s) \geq \theta T_e$ , for a fixed  $0 < \theta < 1/2$ , with uniformly bounded strip derivatives up to the order in use. For two successive weights  $\delta' > \delta''$ , using the weighted target norm on the left,

$$\|\chi_e D_{\text{strip}} \xi\|_{m-1, \delta'', T} \leq C e^{-\theta(\delta' - \delta'')T_e} \|\xi\|_{m, \delta', T}.$$

Indeed, the ratio of the two weights on this support is bounded by a uniform multiple of  $e^{-\theta(\delta' - \delta'')T_e}$ . Here  $D_{\text{strip}}$  differentiates components in the sheet-straightening coordinates: at a smooth match,  $D_{\text{strip}}(a_e + \xi^{\text{dec}}) = D_{\text{strip}} \xi^{\text{dec}}$ . The estimate therefore applies to lower parameter variations as well as to the map remainder. A polynomial in  $T_e$  multiplying the left side is absorbed by reducing the exponential rate. Differentiating the domain identification produces these middle strip-derivative terms. Its remaining zero-order terms involving a movable constant contain a strip derivative of the map or a lower variation, and obey the same weighted product estimate. For a stretch on another neck  $f$ , the loss of weight there absorbs its polynomial in  $T_f$ , uniformly in all the other lengths.

Differentiating the equation with  $b$  fixed uses the same inverse. Explicit length derivatives of its core coefficients and insertions are exponentially small. On neck  $e$ , stretching derivatives multiply a positive strip derivative of the map, exponentially small near the middle in the next smaller weight. On any other neck, the differentiated harmonic coefficient has the factor  $e^{-\pi T_e}$  established in Lemma 4.7; it multiplies a strip derivative of the map. Constants occur only at smooth matches, where the coefficient is  $J_*$  and every relevant zero-order term has a factor  $du$ . These observations give, in the next norm of the finite range,

$$\|\partial_{T_e} u_T\|_{m_1, \delta_1, T} \leq C e^{-cT_e}.$$

Here derivatives mean the covariant derivative under the just specified core and middle-stretch identifications, including the movable constants.

The same argument iterates. At order  $j$  the unknown highest variation is acted on by the same augmented linearization; its right-hand side is a sum of products of lower variations, differentiated coefficients, and positive strip derivatives. The uniform product estimate, with a finite loss of Sobolev order, bounds every such term. A middle-stretch derivative may contribute a polynomial in its length; the difference between successive weights absorbs

this polynomial against the exponential tail. If a mixed derivative contains  $\partial_{T_e}$ , apply this induction with that derivative first. Every subsequent tangential or length differentiation preserves an exponential factor in  $T_e$ , in a weaker norm. Thus for every multi-index  $\alpha$  with  $1 \leq |\alpha| \leq r$  and every  $e$  in its support, and for the allowed remaining parameter derivatives  $D_b^\gamma$ ,

$$\|D_b^\gamma \partial_T^\alpha u_T\|_{m_r, \delta_r, T} \leq C_{\alpha, \gamma} e^{-c_r T_e}, \quad |\alpha| + |\gamma| \leq r. \quad (4.18)$$

Constants are uniform in the other large lengths and in which of the other nodes are held broken. This also bounds the restrictions to every retained core and their movable limiting values. Using an index  $e$  for which  $T_e$  is largest among the differentiated lengths absorbs every polynomial introduced by  $\partial_{q_e} = -T_e^2 \partial_{T_e}$ .

The limit as  $T_e \rightarrow \infty$  is the corresponding partially broken solution, by matching and implicit uniqueness. Integrating the exponential derivative bounds to that face, also after the allowed tangential differentiations, proves  $C^r$  extension in all  $q_e$ . At a smooth node held broken, the estimate initially controls the value and parameter variations uniformly on the punctured end. In a smaller unpunctured neighborhood the equation has fixed  $J_*$  and no obstruction support. Removable-boundary-point regularity and elliptic estimates, also for the differentiated equations, upgrade these bounds to all needed evaluations at the node. This completes the differentiability proof for evaluations and core observables.

Two systems of auxiliary slices are recovered smoothly on retained cores by their transverse incidence equations. Compare them, if necessary, through a third disjoint stabilization. At the comparison point all components are retained in each description; no chain of nodes is contracted. The real plumbing parameters for the same node consequently satisfy

$$\rho'_e = v_e \rho_e, \quad v_e > 0,$$

where  $v_e$  is a smooth unit in the marked-domain coordinates. After retrieving the slices it is  $C^r$  in the chart variables. Therefore

$$T'_e = T_e - \pi^{-1} \log v_e, \quad q'_e = \frac{q_e}{1 - \pi^{-1} q_e \log v_e},$$

which is  $C^r$  also at  $q_e = 0$ . This argument concerns retained real boundary nodes only. The angular interior-node problem addressed in the corrigendum [13, Section 1] never arises here. No compatibility with forgetting actual input markings is asserted or needed.

Finally all insertions stay on compact interior cores. Near ordinary boundary arcs, the hyperbolic metric makes a bounded smooth derivative small on its unit vectors as the boundary is approached. In jump ends and necks, hyperbolic unit vectors are bounded in strip coordinates and (4.9) applies. Choose the cores large enough and then shrink the coefficient and implicit neighborhoods. These observations give the stated uniform smallness off the cores, including for the nonzero-coefficient solutions used in the next section. Finite stabilizers act on this entire construction; passing to their quotients gives the announced local charts.  $\square$

## 5 Finite integration with exact boundary products

The local charts of the preceding section must be assembled so that Stokes' theorem gives the *same* integrations on every boundary factor. We do this at a fixed finite cut. The coefficient variables of the obstruction equations serve as continuous perturbation parameters; integrating their normalized densities removes them. Intrinsic incidence points on the disks identify those variables on overlaps. The point requiring care is that a chart may carry more variables than its neighbor. We arrange that every such additional variable has identically zero insertion on the overlap, so adding it is an ordinary product operation.

Throughout this section fix one of the finite boundary-closed systems of Proposition 4.6, including its relative classes, and fix an arbitrarily large finite differentiability order. A *map* includes its domain, boundary lifts, and original markings, but excludes auxiliary stabilizing markings and obstruction coefficients. All map isomorphisms preserve these original data. In the  $p$ -sector we retain the actual cyclically ordered input slots: cyclic reindexing is a symmetry between these labeled spaces, not an additional quotient.

**Theorem 5.1** (Finite integration package). *For this finite system there are oriented integration correspondences for the stable pure  $m$ , directed, mixed, and interior-output  $p$  sectors, including  $p_0$ , with the following properties.*

- (i) *Pullbacks of evaluation forms can be integrated in the virtual dimension. The pure boundary-output correspondences give differential forms by submersive pushforward. The interior output of a  $p$ -operation may be tested against any smooth ambient form; no submersivity of that evaluation is required.*
- (ii) *Stokes' theorem holds exactly with the normalized boundary. On a face with distinguished nodes its integration is the product of the component prescriptions, using an evaluation fiber product over  $L$  at a smooth node and a product over the fixed intersection generator at a jump node. These identifications agree on iterated corners.*
- (iii) *The prescriptions are equivariant under cyclic reindexing of  $p$  inputs, with the determinant and form-permutation signs. Only actual finite map isotropy contributes an orbifold integration factor.*
- (iv) *A regular zero-dimensional type with empty boundary and trivial isotropy has its ordinary signed count. In particular this applies to the rigid directed spaces and to the low-area rigid comparison triangles of the preceding section. The constant  $p_{1,0}$  correspondence is ordinary integration on  $L$ , with a fixed orientation sign.*

*All assertions hold coefficientwise after multiplying each relative-class term by its energy monomial and, when defined, by  $\exp(tB_{\text{rel}})$ . The choices need only be made at the specified finite cut. Operation signs and the unstable de Rham differential are specified in the next section.*

We prove the theorem by constructing and comparing the integrations. The framework for disk obstruction charts in [10, Theorem 7.1 and Condition 7.11] and the componentwise framework of [11, Theorems 2.5, 5.3–5.4 and Remark 6.1] motivate the construction. The intrinsic probes, the  $p_0$  treatment, and the precise overlap integrations below are part of our argument.

## 5.1 Intrinsic incidence probes

Write  $\mathbb{D}$  for the open unit disk, equipped with its hyperbolic metric. A *probe* consists of the following finite data:

- (a) two ordered codimension-two target slices  $S_1, S_2$ , with compact subpatches  $K_i \subset \text{int}(S_i)$ ;
- (b) a positive lower threshold for the least singular value of the normal incidence derivative at each point, computed using the hyperbolic metric on the disk, and bounds  $0 < d_- \leq d_+ < \infty$  for the distance between the two points;
- (c) an oriented, even-dimensional real vector space  $E$  and a linear map from  $E$  to smooth  $TT$ -valued real one-forms compactly supported in  $\mathbb{D}$ , using the fixed parallelization of  $TT$ . The map has at least one dummy coordinate with zero image.

Choose a compact set  $K \Subset \mathbb{D}$  containing all these supports, the origin, and every possible normalized second point at distance in  $[d_-, d_+]$  from the origin. Choose  $0 < \epsilon < \text{dist}(K, \partial\mathbb{D})$ .

An occurrence on an actual map  $\mathbf{u}$  is an ordered pair of distinct interior points  $(w_1, w_2)$  of one normalized disk component such that  $u(w_i) \in K_i$  and all the closed thresholds hold. There is a unique disk biholomorphism

$$\varphi : \mathbb{D} \longrightarrow D_v, \quad \varphi(0) = w_1, \quad \varphi(r) = w_2 \quad \text{for some } r \in (0, 1).$$

If the original interior output mark belongs to this component, require its  $\varphi$ -coordinate to be at distance at least  $\epsilon$  from  $K$ . There is no restriction involving the order or position of the boundary special points. We use every occurrence on the representative map, without quotienting the occurrence set by automorphisms.

The probe also carries a smooth scalar cutoff  $0 \leq \chi \leq 1$  along transverse incidence branches. Its support lies within strictly stronger versions of the closed conditions just imposed. For instance, choose bumps on the full slices supported inside  $K_i$ , a distance bump supported inside  $(d_-, d_+)$ , and smooth functions of the normal derivative matrices supported where the least singular values exceed the selected thresholds. Such derivative cutoffs exist on the finite-dimensional space of matrices; they can be made invariant under orthogonal changes of frames. For the interior mark use a smooth function on the normalized disk, zero on a neighborhood of  $K$  and equal to one near the circle, with support inside the avoidance condition. This factor is defined to be one if the component has no original interior mark. Given a desired transverse pair with strict margins, these choices can be made with  $\chi = 1$  near that pair.

**Lemma 5.2** (Finiteness and continuation of probes). *For a finite list of probes, every actual stable map has finitely many occurrences. If actual maps converge stably, each sequence of occurrences has a subsequence converging, in a whole-disk normalization, to an occurrence on one disk of the limiting map. Each limiting occurrence continues uniquely to nearby maps, including sufficiently close ambient maps in stabilized plumbing neighborhoods. The continuations are equivariant under map isomorphisms.*

*Proof.* There is a positive lower bound for the area in a hyperbolic radius-one ball about the first incidence point. Otherwise recenter a sequence of such balls by disk isometries. The harmonic-measure coefficients have uniform bounds of every required order on compact interior balls; only finitely many equation data occur at this cut. Interior small-energy regularity would force the derivative at the center to tend to zero, contrary to the positive lower incidence threshold. This argument is applied here to actual, unforced maps.

In a stably convergent sequence, the energy identity allows compact interior cores of the nonconstant limiting disks to carry all but less than this lower bound of the sequence's energy. A radius-one ball at an occurrence must therefore meet one of these cores. The component reparametrizations are restrictions of whole-disk biholomorphisms, as in the boundary plumbing construction. In the normalization of the core that it meets, the first point belongs to a fixed compact interior set; the upper bound  $d_+$  puts the second in another such set. After passing to a subsequence both converge there. Smooth core convergence preserves the closed slice, derivative, and distance conditions. If the limiting component carries the original interior mark, its coordinate converges too, preserving mark avoidance. We obtain an occurrence on a single limiting disk.

The derivative of the incidence equation with respect to the position  $w_i$  is an isomorphism onto the two-dimensional normal space of  $S_i$ . The implicit function theorem, using the full slice rather than its compact subpatch, therefore continues the point uniquely under sufficiently small  $C^2$  core perturbations and domain deformations. The same argument applies simultaneously to the ordered pair. For a fixed map, an infinite occurrence set would

have an accumulation point by the preceding compactness argument, contradicting this local uniqueness. Thus the set is finite. A sequence of occurrences on nearby actual maps that was not among the continued branches would likewise contradict compactness and uniqueness. Distinct continued branches remain distinct after shrinking.

Finally, all conditions and the normalization are invariant under disk biholomorphisms. Changing the core identifications continues the same incidence branches; central finite automorphisms merely permute them. This proves the equivariance assertion.  $\square$

For a chart centered at  $\mathbf{u}$ , let  $O_{\mathbf{u}}$  be the occurrences at its center, and continue this fixed finite set over the chart neighborhood. *Do not retest the closed conditions* at a nearby map  $\mathbf{x}$ . Set

$$E_{\mathbf{u}} = \bigoplus_{o \in O_{\mathbf{u}}} E_o.$$

Transport the one-forms for  $E_o$  by its actual normalization  $\varphi_o(\mathbf{x})$  and project to  $(0, 1)$ -type for the equation on  $\mathbf{x}$ . Denote the resulting insertion by  $I_o(\mathbf{x})$ . The coefficient space is a formal direct sum: the insertion maps need not be injective or have independent images.

These insertions satisfy the smooth core-transport hypotheses of Proposition 4.12. Incidence positions are recovered smoothly on the cores. Whole-disk normalizations commute with the projective insertions in real boundary plumbing, and their expressions on retained cores have no singular scaling. Projection to the varying  $(0, 1)$ -type has the length-derivative estimates established there. The derivative and mark cutoffs are smooth finite-order core observables. If the original interior mark enters the disk from a different component under smoothing, its coordinate tends to the unit circle, so its avoidance factor is identically one near the broken configuration. Thus on every normalized face both the insertions and their cutoff factors restrict *exactly* to the component rules.

## 5.2 A finite family of regular thickenings

At each actual map choose two distinct immersive interior points on each nonconstant disk, transverse slices through their images, and a probe with strict margins at this pair. Its compactly supported test forms can be chosen to span the finite-dimensional cokernel of the fixed-domain operator. On a pure  $m$  component choose them to span the cokernel of the operator augmented by its boundary output evaluation. Compact interior tests away from the original interior mark suffice by Proposition 4.12. Enlarge the coefficient space to even dimension and include a dummy zero direction. Include a probe also on a regular nonconstant component, since it will be useful for isotropy.

Constant components require no such tests. A pure constant component has the regular real-disk operator with freely varying value in  $L$ . A constant mixed component has only an  $X, L$  pair, two opposite jumps at one intersection, and possibly removable smooth  $L$ -marks. Its map operator is the invertible transverse strip operator. A stable constant directed component cannot occur: three distinct boundary labels have no common point, whereas a two-ended constant strip is unstable.

The chosen active branches persist on nearby actual maps. In particular, when several components join, the inverse construction of Proposition 4.12 uses their continuing probe blocks. All smooth matching equations are solved using pure child output submersivity; one may still prescribe the pure parent output. This proves the required fixed-domain augmentations for each joined component. A nearby nonconstant joined component contains a continuing probe: a cluster of constant components cannot become a nonconstant actual component with energy tending to zero, by the area gap. Compactness now gives a finite cover of all actual moduli at the cut by such neighborhoods. Take the union of their probe lists and allow every probe on every disk. Additional coefficient directions preserve the augmentations.

For small coefficients we use the space of pairs satisfying

$$\bar{\partial}_J \mathbf{x} = \sum_{o \in O_{\mathbf{u}}} \chi_o(\mathbf{x}) I_o(\mathbf{x}) e_o, \quad e_o \in E_o. \quad (5.1)$$

After auxiliary stabilization and quotient by actual map isomorphisms, Proposition 4.12 gives a regular orbifold with corners near  $(\mathbf{u}, 0)$ . Its dimension is the virtual dimension plus  $\dim E_{\mathbf{u}}$ . Pure boundary output is submersive on every stratum. Auxiliary incidence marks can be chosen in automorphism orbits away from the compact obstruction supports: nonconstant maps have immersive points in any remaining interior collar, by unique continuation. The auxiliary mark variables and slice constraints cancel in the tangent determinant. Orient the thickening by the virtual Fredholm and marked-domain determinant, followed by the oriented coefficient directions.

Here is the isotropy check, including the sector without boundary inputs. For  $p$ , root the disk tree at its interior mark. If a nonconstant root has a nonidentity finite stabilizer, an ordered pair of distinct interior anchors cannot be fixed: a disk automorphism fixing both is the identity. The stabilizer consequently permutes probe occurrences nontrivially. Their dummy coefficients, which are tangent kernel directions at zero, detect this permutation even after quotient by infinitesimal reparametrizations. A constant root has at least one boundary special point by stability; with only one, its stabilizer is trivial. A nontrivial rotation about the interior mark must permute at least two boundary special points nontrivially and acts effectively on their angular configurations modulo simultaneous rotation. Indeed the tangent space of these configurations is  $\mathbb{R}^r / \mathbb{R}(1, \dots, 1)$  for  $r \geq 2$ ; no nonidentity permutation induced by a finite rotation acts trivially on it. Fixed-domain augmented surjectivity allows all these domain variations in the thickening. An automorphism identical on the root is identical down the tree, because a finite disk automorphism fixing a boundary parent point is the identity. The same last argument makes boundary-rooted  $m$  stabilizers trivial. Thus the charts can be taken effective. For an unbroken nonconstant  $p_0$  map the two interior anchors provide exactly the stabilization and effective action just described.

### 5.3 Comparison of charts without changing the equation

**Lemma 5.3** (Joint charts and inactive coefficients). *The thickening neighborhoods can be shrunk with the following properties. Every incidence branch with potentially nonzero cutoff is tracked there. On an overlap, branches common to two charts have identical coefficient coordinates and insertions. Every branch belonging to only one list has identically zero cutoff near the overlap. After adding these inactive blocks to the other list, the two equations define the same local space of pairs, with the same evaluations, isotropy, and determinant orientation. The comparison is smooth to the prescribed finite order, including on faces with any specified nodes retained.*

*Proof.* First exclude untracked active branches for ambient maps, not merely actual ones. Choose fixed compact cores and sufficiently small gluing neighborhoods. Outside the cores the derivative on hyperbolic unit vectors can be required to be smaller than every probe's lower threshold. At ordinary boundary arcs this follows from smooth boundary control and the blowup of the hyperbolic metric; at long necks and jump ends it follows from the decay estimates in Lemma 4.8, where hyperbolic unit vectors have bounded strip-coordinate norm. These restrictions are valid also for the small-coefficient solutions, whose obstruction terms remain on the specified cores. Consequently any pair in the inner cutoff supports must lie on the continued compact cores. Core closeness and the implicit uniqueness in Lemma 5.2 identify it with a tracked branch. The strict margin between the cutoff support and the

closed occurrence conditions is essential: a branch failing the latter at the center stays outside the former on a neighborhood.

We next make precise that the underlying-map comparisons used here have no extra multiplicity. The core convergence in these neighborhoods, together with arbitrarily small oscillation in the omitted necks and ends, has unique stable-map limits up to isomorphism. To prove this, suppose a sequence admits two such limits. Stabilize using the markings for both limits and pass to a common stable marked-domain limit. Extra comparison marks may be displaced slightly to avoid coincidences. Forgetting one or the other marking system recovers the two given domain limits. A component contracted in one description has constant limiting map, by core convergence or the small oscillation on its omitted region. The maps on every retained component agree in the common refinement. After forgetting all auxiliary marks and contracting unstable constant components, both limits therefore give the same stable map. This uses no holomorphic equation on the retained cores.

It follows in particular that, after shrinking about a center, lifts of the same map to its slice gauge differ only by its central finite automorphisms. Otherwise compare a sequence of conflicting gauge lifts by the preceding marked convergence. Their limiting identifications are central automorphisms, which preserve the chosen slice systems with their induced relabelings. Transverse incidence then makes the continued marks exactly identical up to that relabeling, a contradiction. This also shows that underlying isomorphisms lift to the gauges. Stability is preserved nearby, either by the original domain specials or by the persistent immersive cores of nonconstant components.

On an overlap take the union of the two continued branch lists, identifying coinciding branches by transverse uniqueness. A branch missing from one list is outside its inner cutoff support throughout a smaller comparison neighborhood by the first paragraph; hence its contribution to Equation (5.1) is identically zero there. Adding its coefficient block thus takes a product with  $E_o$ , rather than changing the equation. Redundancy among the active insertions does not affect this assertion: coefficients remain part of the pair and are never quotiented by the kernel of their insertion.

For smoothness recover both sets of stabilizing marks by their transverse incidences, or compare through a third marking system disjoint from them at the point in question. Each stabilization retains every component of that point. Adding and forgetting comparison marks therefore contracts no component there. Every retained node represents the same single domain divisor on the two sides, so its real plumbing parameters differ by a smooth positive unit. The inverse-length changes are smooth by Proposition 4.12; there is no multiplication of plumbing parameters along a contracted chain. This argument also applies on a normalized face, keeping its distinguished nodes. Core observables, slice positions, and kernel coordinates compare smoothly, and the weighted neighborhood conditions are preserved after shrinking by the same decay estimates. It applies with small nonzero coefficients, since the padded equations are literally identical.

These comparisons retain the map isomorphisms, acting on the branch variables as well. The auxiliary mark and constraint determinants cancel, and adjoining a block appends its oriented determinant. Permuting entire blocks preserves orientation because their ranks are even. This proves all the claims.  $\square$

## 5.4 Buffered partitions depending only on maps

Choose for each probe a smooth compactly supported top-form density  $\Xi^h$  on its oriented coefficient space, of integral one and supported in  $|e| < \hbar$ . The same density is used at every

occurrence of that probe. In a chart let

$$\Xi_{\mathbf{u}}^h = \bigwedge_{o \in O_{\mathbf{u}}} \Xi_o^h(e_o). \quad (5.2)$$

It is closed, and its definition is unaffected by permutation of the even-dimensional blocks. Inactive blocks still carry their full densities.

We must localize on maps without selecting only some coefficient representatives of a map. The following elementary consequence of the chart construction supplies this point.

**Lemma 5.4** (Buffered partitions). *There are finitely many trimmed thickening charts  $V_\alpha$ , nonnegative map functions  $f_\alpha$ , and a common sufficiently small  $\hbar > 0$ , such that near all density supports:*

- (i) *the zero extensions of the  $f_\alpha$  pull back smoothly to every chart, and  $\sum_\alpha f_\alpha > 0$ ;*
- (ii) *each  $f_\alpha$  has support, with a buffer, strictly inside its chart, relative to the actual nodal corners;*
- (iii) *if a map is in that support, the chart represents every assignment at the common small support scale that solves its equation for that map. This includes every free choice of coefficients in any blocks made inactive during an overlap comparison.*

Thus  $\lambda_\alpha = f_\alpha / \sum_\beta f_\beta$  is a partition on the map projections of the relevant pairs. It is independent of which small coefficient assignment represents the map.

*Proof.* Around each center choose nested coordinate neighborhoods with compact closures inside a larger thickening chart, and trim inside the larger one. Here compactness refers to the finite-dimensional chart, with its actual corner faces retained. Its local coordinates can be chosen from the coefficient coordinates and finitely many smooth functionals of the map on retained cores, together with stabilized domain variables and inverse neck lengths. Indeed these functionals separate the augmented kernel: after fixing the coefficients, a homogeneous kernel section vanishing on open cores vanishes by elliptic uniqueness. Finitely many tests suffice on this finite-dimensional kernel, and the inverse function theorem supplies local coordinates.

Consequently bounds on finitely many map and domain observables keep a solution in a smaller coordinate neighborhood uniformly for all sufficiently small coefficients. Take a smooth bump of these observables, positive at the center and supported with a strict interior buffer. Average over the central finite group. The observables use continued slice gauges, whose only ambiguity is that group by Lemma 5.3; the averaged function is therefore a function of the map. These bumps do not depend on its coefficients. Select finitely many so that their positive neighborhoods cover the actual compact locus, and include their cyclic translates in the  $p$ -sector.

We justify the required uniformity before fixing the support radius. Consider a sequence of pairs in the trimmed closures whose coefficients tend to zero. There is a subsequence converging in a larger chart to  $(\mathbf{v}, 0)$ , where  $\mathbf{v}$  is actual. Every applicable open gauge neighborhood of  $\mathbf{v}$  then contains the maps in that sequence. Transfer their common branch coefficients to such a gauge and choose arbitrary coefficients tending to zero in its inactive blocks. The local chart parametrizes all small solutions of its equation, so every resulting pair tends to  $(\mathbf{v}, 0)$  in that chart. Uniqueness of the map limit and slice positions excludes any other nearby lift. Some  $f_\alpha$  is positive at  $\mathbf{v}$ ; thus failure of positivity at every sufficiently small coefficient scale would contradict this subsequence argument.

The same argument proves the all-assignments assertion. If it failed at arbitrarily small scales, take an offending map in a projected cutoff support and an alternative assignment at

that scale solving the same equation. A lift of the map in the originating cutoff support lies in its compact buffered region. Pass to its actual limit. The alternative assignment tends to zero; after branch transfer and arbitrary padding it must lie near  $(\mathbf{v}, 0)$  in the originating chart, by its local completeness. The strict buffer contradicts its asserted exclusion. This reasoning includes the inactive variables and any kernel among the active insertions. Finiteness of the chart collection permits one uniform scale. One may first obtain these conclusions for coefficients bounded by  $2\hbar$ , and then put the densities inside the  $\hbar$  balls.

Finally consider smoothness of the zero extensions at this fixed small scale. Near a projected cutoff support, the compact buffer and uniqueness of underlying-map limits put every comparison in the same open slice gauge. The map observables there compare smoothly by Lemma 5.3, and the all-assignments property permits every small coefficient transfer needed for that comparison. Away from the projected support the cutoff is identically zero on a neighborhood of the density support. Its extension is therefore smooth in either case. The same buffer applies to its derivatives. Normalizing the sum now gives the asserted partition.  $\square$

This construction uses no global parameter space of constant rank. Only the local pairs and the map functions in Lemma 5.4 enter the integration.

## 5.5 Patching, Stokes, and boundary composition

For a product  $\alpha$  of pulled-back evaluation forms, in its specified order, define

$$\mathcal{I}_{\mathcal{M}}(\alpha) = \sum_{\gamma} \int_{V_{\gamma}} \lambda_{\gamma} \alpha \wedge \Xi_{\gamma}^{\hbar}. \quad (5.3)$$

Integrals are ordinary orbifold integrals: in a local finite quotient they are divided by its actual group order. The density is put last. The integral vanishes unless the remaining form degree is the virtual dimension. Compact support in the coefficient directions and the buffered map support make every integral compactly supported away from the artificial trimming boundary. At a pure boundary output the same formula, using submersive fiber integration, gives a differential form. We characterize its order convention by placing the output test form first. Submersivity on every stratum ensures that its boundary pushforwards are defined by the same rule.

**Lemma 5.5** (Exact patching and Stokes). *Equation (5.3) is independent of the buffered partition and of the slice gauges used to compute it, for the chosen probes and densities. It is unchanged by adjoining inactive blocks. It satisfies*

$$\mathcal{I}_{\mathcal{M}}(d\alpha) = \mathcal{I}_{\partial^{\text{norm}} \mathcal{M}}(\alpha),$$

where the right side has the induced boundary orientation. Its prescription on a normalized face is exactly the product prescription from its component moduli, with the corresponding matching conditions.

*Proof.* On a buffered overlap identify common branch coordinates and take the union  $O$  of the branch lists. By Lemma 5.3 its pair space is locally the first chart times the missing inactive blocks, and also the second chart times its missing inactive blocks. Over every map in the cutoff support these products contain the full coefficient supports by Lemma 5.4. All map forms and cutoffs pull back from the same map. Fubini therefore gives, for each missing block,

$$\int_{E_o} \Xi_o^{\hbar} = 1,$$

and identifies the two localized integrals with the single integral over the joint list  $O$ . This comparison retains finite automorphism quotients: locally both descriptions are quotients of the same slice gauge with the same branch action. It therefore introduces no extra incidence multiplicity. Branch continuation may permute entire blocks; their even dimensions preserve the density and determinant orientation. One can localize this argument on finitely many map neighborhoods with buffers of the preceding kind, so it proves equality on the whole overlap. No quotient is taken by redundancy among active coefficients.

Multiplying by a second map partition and using these overlap equalities proves partition independence. It also proves the cancellation needed for Stokes. Each  $\Xi_\gamma^h$  is closed. Apply ordinary Stokes to  $\lambda_\gamma \alpha \wedge \Xi_\gamma^h$  and sum. The terms involving  $d\lambda_\gamma$  can be compared on joint charts as just above, and there they sum to zero because  $\sum_\gamma d\lambda_\gamma = 0$ . There are no trimming-boundary terms. The only terms left are the actual normalized faces, each with its induced orientation. This proves the displayed identity. The same argument with an output test form proves its pushforward version, including the usual signs from differentiating that form.

Now fix a face with a distinguished breaking. The actual maps on it are precisely the component maps with the indicated matching. Probe occurrences on its normalized disks, their continued insertions, and their cutoff factors are the component prescriptions already proved above. Hence its pair equation is the product of the component pair equations, with a value match at a smooth node or the fixed generator at a jump. A face chart and product charts may retain different inactive branches; adding their joint lists gives the identical equations and identical product densities.

For completeness the neighborhoods on the two sides cover the same portion of these equations at one sufficiently small common support scale. An escaping sequence of component products with scale tending to zero has, by component compactness, an actual product limit. This is a covered actual nodal map. Gauge transfer and the local completeness of the face chart then put all those assignments in its neighborhood. Conversely a sequence in a covered face tending to zero coefficients breaks into covered actual component limits and enters their product neighborhoods. The buffered all-assignments argument proves the assertion simultaneously for all the finitely many faces and corners.

Partition independence now allows refinement to the product of the component partitions. Ordinary fiber integration gives exactly their composition. At each smooth node it is the pure child boundary-output submersion that makes the fiber product regular and supplies this Fubini identity; at jump nodes the matching base is a point. Rooting a  $p$  tree at the interior mark gives the same description with  $m$  branches attached in their linear input orders. Repeating with several distinguished nodes gives the same common component equation, proving compatibility on all iterated corners.  $\square$

*Completion of Theorem 5.1.* The construction and Lemma 5.5 give the integrations, submersive outputs, and exact boundary products. The spin and capping determinants orient the virtual problems; appending the even coefficient spaces preserves the determinant gluing comparisons. The next section computes their signs. Our probes ignore boundary slot positions and are therefore intrinsic under cyclic reindexing of  $p$  inputs. Pulling back a partition by that symmetry changes no integral by partition independence. This proves cyclic equivariance on the labeled spaces, with no division by the order of the reindexing group. The effective isotropy discussion above includes  $p_0$ .

For a regular rigid actual map without isotropy, the ordinary linearized problem is invertible after its domain slice. For every sufficiently small choice of the probe coefficients there is a unique nearby solution, with the same orientation sign. Its parameter integral is that sign times the product of the density integrals, hence that sign. Compactness excludes new solutions away from these neighborhoods as the support scale shrinks. An empty actual

moduli space contributes zero for the same reason. This proves agreement with ordinary signed counts for a regular zero-dimensional type with empty boundary. Directed rigid spaces have empty boundary because all their component problems are regular and match over points: every broken configuration has negative expected dimension. The low-area no-extra-mark comparison triangles have only the same regular strip and triangle sectors, and the energy bound excludes pure disk bubbling. They have the same property. The ordered jumps exclude disk automorphisms; a nonconstant strip has no stabilizing translation.

The constant disk with one interior and one boundary mark has no domain moduli or automorphisms. Its moduli space is  $L$ , its two evaluations are  $i_L$  and the identity, and it has no probe occurrences. Thus the raw  $p_{1,0}$  pairing is the ordinary integral of the pulled-back ambient test form followed by the input form on  $L$ , up to the fixed global operation sign selected in the next section. In particular its scalar factor is nonzero and its weight is one.

Finally, energy and the relative  $B$ -integral are constant on the retained relative-class refinements, also for the ambient maps in the thickenings: they are homotopy invariants with the specified boundary conditions and fixed jumps. They add under every component splitting. Their monomial and exponential factors therefore multiply in the boundary products and have zero exterior derivative on each chart. Stokes and all overlap identities consequently hold coefficientwise in  $t$ . The finite cut bounds the number of compositions needed later, independently of Taylor order; the initial differentiability can be chosen high enough for all the required pushforwards and primitives.  $\square$

## 6 Signs and removal of the disk obstruction

The finite integrations of Theorem 5.1 provide geometric correspondences. We now specify their signs and use the correspondence with one interior mark to remove curvature. The topological input from Proposition 3.1 is precisely that

$$i_{L*} : H_2(L; \mathbb{Q}) \longrightarrow H_2(T; \mathbb{Q}), \quad i_{L*} : H_6(L; \mathbb{Q}) \longrightarrow H_6(T; \mathbb{Q}) \quad \text{are injective.} \quad (6.1)$$

Together with the grading modulo four, these two injections suffice; no injectivity in the other homological degrees is required.

Throughout this section all sign exponents are read modulo two. For an operation whose boundary inputs are written counterclockwise after its output, the suspended degree of an input  $x$  is  $|x| - 1$ . We write  $A^{\bar{r}}(L)$  for forms whose ordinary degrees are congruent to  $r$  modulo four. Formal Taylor variables have degree zero. The notation  $\equiv_N$  means equality after discarding terms of energy greater than  $N$ , coefficient by coefficient in the Taylor variable.

### 6.1 Cappings, degrees, and local gluing orientations

At a transverse intersection choose capping operators joining the two graded tangent planes, with matching lifts of their gradings. Use the reverse path for the other end of the same generator. Choose spin frames on the two branes at the intersection, extend them across the cap, and use the reversed frames for the reciprocal cap. On a brane boundary arc use a spin trivialization matching its endpoint frames. A basis of each cap determinant line is fixed up to positive scaling. These are the ordinary spin cappings with background class zero; we use their determinant gluing and excision constructions [3, Section 5]. The particular comparisons needed below are spelled out, including their order of factors.

**Lemma 6.1** (Degrees). *The operation  $m_k$  has degree  $2 - k$  modulo four, and has that degree integrally on a strictly directed graph word. A pure disk of Maslov index  $\mu$  with  $k$  boundary inputs has virtual dimension*

$$8 + \mu + k - 2 \quad \text{with a boundary output,} \quad 8 + \mu + k - 1 \quad \text{with an interior output.}$$

In particular the curvature of a cochain in  $A^{\bar{1}}(L)$  has only ordinary form degrees 2 and 6.

*Proof.* The cap attached to an input has a negative end and has index equal to the generator degree. To check this convention, split a transverse plane pair into positive rotations by angles  $\alpha_j \in (0, \pi)$ . In one coordinate use a half-disk with negative puncture at zero and the two boundary conditions on its real radii. Writing  $a = \alpha_j/\pi$ , the section  $z^a$  decays at the puncture. Choose the plane path on the rounded outer boundary with angle  $a \arg z$ . Dividing by  $z^a$  removes the rotating boundary condition and leaves the real disk operator with a removable boundary point and a free real constant mode. Its index is one. Equivalently, moving the end weight to the removable-point problem crosses no other indicial root. Thus the minimal positive rotations contribute eight; further winding adds its Maslov index. With the grading lifts  $\vartheta_1, \vartheta_2$  this gives

$$8 + \vartheta_2 - \vartheta_1 - \sum_j \alpha_j/\pi.$$

The reciprocal output cap has complementary index, so the two cap indices sum to eight and have the same parity.

After capping all transverse ends, the disk Maslov index is a multiple of four; it is zero for the integral matching on a graph word. Fredholm indices add under transverse capping. Adding the  $k - 2$  domain parameters for  $k + 1$  boundary marks and then taking the evaluation pushforward gives degree  $2 - k$  with the indicated grading. For instance, with jumps only, if the input degrees are  $d_j$  and the output degree is  $d_0$ , the caps have total index  $\sum_j d_j + 8 - d_0$ . The uncapped operator therefore has index  $\mu + d_0 - \sum_j d_j$ ; virtual dimension zero gives  $d_0 = \sum_j d_j + 2 - k - \mu$ . Inserting a smooth form and taking an evaluation pushforward gives the same formula with its form degree in place of the jump degree. For a pure disk the uncapped real disk index is  $8 + \mu$ . There are  $k + 1$  real marked-point parameters in the boundary-output case and  $k + 2$  in the interior-output case, and the disk automorphism group has dimension three. This proves both dimension formulas. In the pure case the ordinary form degree of the output is

$$\sum_{j=1}^k |x_j| + 2 - k - \mu.$$

Putting every input in degree 1 modulo four gives degree 2 modulo four. On the eight-manifold  $L$  these are exactly degrees 2 and 6.  $\square$

Here is the determinant order used for the geometric correspondences. At a smooth slot put  $p_j = 0$ , and at a jump put  $p_j$  equal to the generator degree modulo two, represented by 0 or 1. Let  $X_j$  be the degree of the form at that slot, zero at jumps;  $X_0$  denotes the degree of an output test form when there is one. Put

$$KW_0W_1 \cdots W_k, \quad K = o(D_{\text{cap}}) o(\text{PSL}(2, \mathbb{R}))^*, \quad W_j = U_j C_j^*. \quad (6.2)$$

Here  $U_j$  is the oriented tangent of the mark, positive counterclockwise at a boundary mark, and  $C_j$  is the cap determinant factor, trivial at a smooth slot. Thus  $K$  has odd parity and a boundary factor  $W_j$  has parity  $1 + p_j$ . For an interior-output operation  $W_0$  is just the even tangent factor of the interior mark. All rearrangements of determinant lines use the Koszul rule.

Consider a normalized boundary face that splits off  $s$  consecutive inputs, beginning at slot  $i$ . Let  $D'$  be the child and  $A'$  the parent. Their respective arities are  $s$  and  $b = k - s + 1$ . Let  $z$  denote the new node slot, and put

$$l = i - 1, \quad r = k - i - s + 1.$$

Subscripts  $<$ ,  $D'$ , and  $>$  on  $P$  or  $X$  denote sums over the preceding, child, and trailing input blocks;  $P$  is always a sum of the  $p_j$ . The case  $s = 0$  is included, with an empty child input block.

**Lemma 6.2** (Boundary orientation). *Give a product face the order parent, child, followed at a smooth node by the dual tangent of the matching copy of  $L$ . The boundary orientation relative to this product has exponent*

$$\sigma = i + p_0 + P_{<} + (p_z + s + P_{D'})(r + P_{>}) + \varepsilon + \rho_z. \quad (6.3)$$

Here  $\varepsilon$  is one fixed smooth disk gluing sign, and  $\rho_z$  is a fixed additional contraction sign for the chosen transverse cap pair at  $z$ ; it is zero at a smooth node. For an interior-output parent and a nonwrapping placement of a pure child, the exponent is

$$i - 1 + sr + \varepsilon. \quad (6.4)$$

The same signs hold for the thickened charts and their normalized faces.

*Proof.* We first justify that the residual contraction is local. Cap the exterior jumps before gluing the distinguished node, using associativity of determinant gluing. At a smooth node the spin boundary framings concatenate and one contracts against the oriented tangent diagonal. At a transverse node one compares strip gluing with gluing the two reciprocal caps at that node. The extra determinant pairing therefore uses only this fixed cap pair.

More explicitly, choose the spin framing on a boundary collar and fill that collar by a trivial real disk. The remaining clutching is transferred to a complex sphere away from the distinguished end. This is the excision construction of the spin disk orientation. The sphere determinant and its interior matching factor are complex-oriented even lines, so moving them past the node contraction creates no sign. The boundary framing after gluing is the concatenation in its spin homotopy class, with the node cap reversed when appropriate. Thus changes in exterior boundary paths or in their Maslov contributions do not alter this local comparison. Its signs are the fixed  $\varepsilon$  and  $\rho_z$  in the statement.

The domain factor is local for the same reason. Put the child node at infinity in an upper half-plane coordinate and insert the child at a parent mark  $a$  by  $\zeta = a + \rho z$ ,  $\rho > 0$ . Use the outward normal, directed toward the broken configuration, first. The child's residual affine automorphisms are translation and dilation. They cancel with the parent-node position and the radial smoothing parameter. In a cross-section to these two directions all retained real mark tangents are transported by positive scaling, while interior-mark pairs are transported with their complex orientations. One can make this calculation before quotienting the parent automorphisms and then quotient by the same parent directions on both sides. The sign is consequently independent of the other marks, including the presence of an interior mark on the parent. Changing upper half-plane charts preserves these orientations. Auxiliary interior stabilization marks and slice constraints add matching even-dimensional factors and do not change the comparison.

It remains to perform the permutation in Equation (6.2). Write

$$A = 1 + p_0 + l + P_{<}, \quad R = r + P_{>}, \quad C = s + P_{D'}.$$

Starting with outward normal, parent factors, child factors, and the smooth matching factor when present, group the local factors as

$$N' K_{A'} W_{A'z} K_{D'} W_{D'z}.$$

The parent node factor crosses the preceding parent slots, the child  $K$  crosses the remaining parent slots, and the child node factor crosses those same slots. Finally the child input block crosses the trailing parent slots. Their total permutation sign is

$$\begin{aligned} (1 + p_z)A + (A + R) + (1 + p_z)(A + R) + CR \\ = A + (p_z + C)R = i + p_0 + P_{<} + (p_z + s + P_{D'}) (r + P_{>}). \end{aligned}$$

The matching tangent has dimension eight and contributes no further permutation parity. This proves Equation (6.3). For an interior-output parent all boundary slots are smooth and its output factor is even, so the same computation has  $A = l$ ,  $R = r$ ,  $C = s$ , and  $p_z = 0$ . It gives Equation (6.4).

For the thickening, append the oriented obstruction coefficient directions to the Fredholm determinant. They have even dimension and split by components on the face, so the Fredholm and tangent exact sequences give the same permutation. Compact lower-order terms in the linearized equations, including derivatives of the map-dependent core insertions, can be homotoped to zero with fixed asymptotic operators and compatibly with the split face. Determinant orientations persist under this homotopy. True chart isomorphisms preserve the cappings, spin data, and external slots; stabilization pairs and permutations of probe blocks are even. Their orientation compatibility on the smooth locus therefore extends across the corners by this same gluing comparison, also when components and their smoothing parameters are permuted.  $\square$

## 6.2 Normalization of the operations

For a boundary word  $W$  of arity  $k$ , put

$$\begin{aligned} e_W &= k + P - 1 - p_0, \\ G_W &= \binom{e_W}{2} + p_0 e_W + P + \varepsilon + \sum_{\text{input jumps } j} \rho_j, \\ S_W &= \sum_{j=1}^k (j + P_{<j}) X_j + G_W. \end{aligned} \tag{6.5}$$

The expression  $\binom{e}{2} = e(e-1)/2$  is used for every integer  $e$ , including negative integers. This matters for zero-input terms. Define a stable  $m$  term by multiplying the geometric integral by  $(-1)^{S_W}$ , with the pulled-back forms ordered as output test first, then inputs. Thus at a smooth output its definition is

$$\int_L y_0 \wedge m_{k,\beta}(x_1, \dots, x_k) = (-1)^{S_W} \int_{\mathcal{M}_{k,\beta}}^{\text{vir}} \text{ev}_0^* y_0 \wedge \text{ev}_1^* x_1 \wedge \dots \wedge \text{ev}_k^* x_k,$$

and at a jump output it defines the scalar coefficient of that generator, omitting the output form. We sum these terms with their energy monomial and, on the permitted words, the relative weight  $\exp(tB_{\text{rel}})$ . Add the unstable pure unary operation  $m_{1,0} = d$ .

For a pure interior-output operation tested by  $\eta \in A^*(T)$ , write  $p_k^\eta(x_1, \dots, x_k)$  for the resulting scalar integral. Its operation sign is

$$\sum_{j=1}^k j X_j + \binom{k-1}{2} + \varepsilon, \tag{6.6}$$

again with the output test form first. There is no unstable constant  $p_0$  term. The stable constant  $p_{1,0}$  term is retained.

**Proposition 6.3** (Finite signed identities). *Within the finite cost system the normalized operations satisfy the curved suspended identities*

$$\sum_{s=0}^k \sum_{i=1}^{k-s+1} (-1)^{\sum_{j<i} (|x_j|-1)} m_{k-s+1}(x_1, \dots, x_{i-1}, m_s(x_i, \dots, x_{i+s-1}), x_{i+s}, \dots, x_k) = 0. \quad (6.7)$$

For a closed interior test form, the corresponding Stokes identity for  $p^\eta$  has the same insertion signs and no output differential. These assertions hold coefficientwise in  $t$  with the relative weight, as well as without that weight.

*Proof.* The virtual Stokes and product formulas are exact by Theorem 5.1. We verify their normalization. The parity condition for an  $m$  Stokes integral is

$$X_0 + \sum_j X_j = p_0 + P + k + 1.$$

Multiply its Stokes equation by  $(-1)^{S_W + X_0 + 1}$ . Integration by parts turns the  $dy_0$  term into the output differential with coefficient one, because

$$\int_L dy_0 \wedge u = (-1)^{X_0 + 1} \int_L y_0 \wedge du.$$

For an input differential  $dx_i$ , the change of  $S_W$  is  $i + P_{<i}$ , while the Leibniz sign before that input is  $X_0 + X_{<i}$ . The normalized exponent is therefore

$$1 + X_{<i} + i + P_{<i} = \sum_{j<i} (p_j + X_j - 1) = \sum_{j<i} (|x_j| - 1).$$

Consider next the split in Lemma 6.2. Let  $X_z$  be the degree of the *output* form after the child pushforward, not the degree of a test form on that output. The degree formula gives

$$X_{D'} + X_z = s + P_{D'} + p_z =: u.$$

The Fubini comparison contributes  $uX_{>}$ : move the child input forms past the trailing forms and then move its output form back into the node slot. Integration uses the test form first on the even-dimensional matching base. For the form-dependent parts  $\Phi_W = S_W - G_W$ , the preceding input coefficients cancel, the child input coefficients differ by  $l + P_{<}$ , and the trailing coefficients differ by  $u + 1$ . Hence

$$\Phi_W + \Phi_{A'} + \Phi_{D'} = (l + P_{<})X_{D'} + (l + 1 + P_{<})X_z + (u + 1)X_{>}.$$

After adding  $uX_{>}$  this becomes  $(l + P_{<})u + X_z + X_{>}$ . Use  $X_z = X_{D'} + u$  and the Stokes parity equation above. Subtracting the boundary term from Stokes and including  $S_W, S_{A'}, S_{D'}$  and Equation (6.3), the difference from the desired insertion exponent is

$$k + P + u(b + P_{<} + P_{>}) + \varepsilon + \rho_z + G_W + G_{A'} + G_{D'}. \quad (6.8)$$

We check this cancellation without specifying the bare signs. Set

$$a_0 = p_0, \quad q = p_z, \quad v = l + P_{<}, \quad w = r + P_{>}, \quad c = s + P_{D'}.$$

Then

$$e_{A'} = v + w + q - a_0, \quad e_{D'} = c - 1 - q, \quad e_W = e_{A'} + e_{D'}.$$

The binomial terms in the sum of the three  $G$ 's give  $e_{A'}e_{D'}$ . The terms involving the output parity give  $(a_0 + q)e_{D'}$ , the three input-parity sums give  $q$ , and the remaining constants give

$\varepsilon + \rho_z$ : every exterior input contraction sign occurs twice, while the node sign occurs once. Consequently Equation (6.8) reduces to

$$v + w + c + (c + q)(v + w + 1) + (v + w)(c - 1 - q) + q = 0 \pmod{2}.$$

This verifies the split term, including  $s = 0$  and negative values of  $e_{D'}$ .

For pure  $p$  the parity condition for the Stokes integral changes by one, to  $X_0 + \sum X_j = k$ , and the boundary exponent changes by one, from  $i + sr + \varepsilon$  to  $i - 1 + sr + \varepsilon$ . These changes cancel in the preceding normalization. The input-differential computation is unchanged. There is no  $d\eta$  contribution for a closed test form. This proves the asserted interior-output identity on nonwrapping blocks; cyclic reindexing supplies the other blocks, with the sign computed below.

Finally, Lemma 4.1 gives constancy of  $B_{\text{rel}}$  on each retained relative class and its additivity on a split. Thus  $\exp(tB_{\text{rel}})$  is constant for Stokes and is the product of the two component weights for Fubini. It introduces no extra differential or boundary term. The same proof therefore works separately at every Taylor coefficient.  $\square$

**Lemma 6.4** (Cyclic sign). *Moving the first boundary input of  $p_k^\eta$  to the last slot multiplies the operation by*

$$(-1)^{(X_1-1)\sum_{j>1}(X_j-1)}. \quad (6.9)$$

*This is a reindexing identity on the labeled moduli spaces and does not impose a further quotient by cyclic rotations.*

*Proof.* The boundary tangent factors give  $k - 1$ , permuting the input forms gives  $X_1 \sum_{j>1} X_j$ , and changing Equation (6.6) gives  $(k - 1)X_1 + \sum_{j>1} X_j$ . Their sum is  $(X_1 - 1) \sum_{j>1} (X_j - 1)$  modulo two. The interior mark is even and stays first. Cyclic equivariance of the raw integration is part of Theorem 5.1; hence this calculation also applies to its normalized faces and finite isotropy factors.  $\square$

### 6.3 The interior-output identity

Let  $b$  be a formal cochain with coefficients in  $A^1(L)[[t]]$ , only finitely many energy exponents at most  $N$ , and all energies at least  $\delta_L$ , the pure disk gap from Lemma 4.2. Set

$$\Delta(b) = \sum_{k \geq 0} m_k(b, \dots, b), \quad (6.10)$$

including  $db$  in the unary term. Through energy  $N$  the sum is finite. Every  $b$  input is even after suspension.

**Proposition 6.5** (Obstruction pairing identity). *For every closed form  $\eta$  on  $T$ , the finite signed correspondences give*

$$\sum_{r \geq 0} p_{r+1}^\eta(\Delta(b), b, \dots, b) \equiv_N 0. \quad (6.11)$$

*The statement uses the weighted operations on the pure  $L$  sector and holds coefficientwise in  $t$ .*

*Proof.* For  $k > 0$  apply the interior-output Stokes identity to  $k$  copies of  $b$  and divide by  $k$ . A child carrying a block of  $s$  of the  $k = r + s$  inputs occurs in  $k$  cyclic placements. For  $0 < s < k$  these are the choices of the start of the block. For  $s = 0$  the zero-input bubble attaches in any of the  $k$  cyclic gaps. For  $s = k$  all inputs lie on the child, but its output node can still occupy any of those  $k$  gaps; these give the  $k$  possible linear input orders on the child.

Thus the factor  $k$  occurs also at both extremes. An input differential likewise occurs in  $k$  slots. After rotating its resulting curvature slot to the first position, all signs are positive: the other inputs have suspended parity zero, so Equation (6.9) contributes no sign, even though the curvature itself has suspended parity one.

For  $k = 0$  use coefficient one. There is one normalized boundary type  $p_1^\eta(m_0)$ , with the interior mark distinguishing the parent. Its constant parent is stable, having one interior mark and one boundary node. This argument does not add the nonexistent unstable constant  $p_0$ . The usual finite-isotropy integration already counts a face with its distinguished node; there is no extra rotation factor. In particular a finite-order disk automorphism fixing that boundary node is the identity on the incident disk, and the distinction propagates along attached disks. If an automorphism of a  $p_0$  tree permutes indistinguishable edges before the edge is distinguished, its ordinary orbifold factor and the normalized-boundary multiplicity give exactly the preceding count. Summing these equations over  $k$  gives Equation (6.11).

All equations used lie within the finite cost cut. At total energy at most  $N$  there are at most  $N/\delta_L$  inserted  $b$ 's. Their arity cost is less than  $N$ , since  $a < \delta_L$  in Proposition 4.6. The interior-output equation therefore has cost less than  $2N$ , and splitting preserves cost. This is contained in the chosen cut of  $4N$ .  $\square$

**Theorem 6.6** (Finite weighted bounding cochains). *For every integer  $N \geq 1$  and every finite list of graph labels, the finite operations can be chosen with a cochain*

$$b_{L,N}(t) = \sum_{\lambda \in F_N} \mathbf{q}^\lambda b_\lambda(t), \quad F_N \subset [\delta_L, N] \text{ finite}, \quad b_\lambda(t) \in A^{\bar{1}}(L)[[t]],$$

such that  $\Delta(b_{L,N}) \equiv_N 0$ . This assertion holds for all Taylor coefficients simultaneously. Only a finite order of differentiability, chosen sufficiently high depending on  $N$  and the finite list, is needed. No compatibility as  $N$  varies is asserted or required.

*Proof.* Begin with  $b = 0$  and proceed in increasing positive energy. Suppose all curvature coefficients below  $\lambda$  have been canceled. Summing Equation (6.7) on strings of  $b$ 's gives the Bianchi identity

$$\sum_{u,v \geq 0} m_{u+v+1}(b^{\otimes u}, \Delta(b), b^{\otimes v}) \equiv_N 0.$$

At energy  $\lambda$ , every nonconstant operation or additional  $b$  input raises the energy. The only term involving the first uncanceled curvature is therefore the constant unary term. It follows that

$$d\Delta_\lambda(t) = 0. \tag{6.12}$$

The identical energy argument applied to Equation (6.11) leaves only  $p_{1,0}^\eta(\Delta_\lambda(t))$ . Its moduli space is the ordinary copy of  $L$ , with its boundary evaluation the identity and its interior evaluation  $i_L$ . The constant disk has  $B_{\text{rel}} = 0$ . By Theorem 5.1 its raw integral is ordinary integration. For the even-degree form  $\Delta_\lambda$  the sign in Equation (6.6) is one fixed sign, independent of  $\lambda$ , of its Taylor coefficient, and of the choice of complementary even-degree test form. Hence

$$\int_L i_L^* \eta \wedge \Delta_\lambda(t) = 0. \tag{6.13}$$

Only its nonzero sign, not a numerical convention for it, is needed for this implication.

By Lemma 6.1, write the curvature coefficient as the sum of its degree-two and degree-six parts. For a closed degree-two coefficient  $u$ , Equation (6.13) for all closed ambient six-forms says that  $i_{L*}(u \cap [L]) \in H_6(T; \mathbb{C})$  pairs to zero with all of  $H^6(T; \mathbb{C})$ , and is therefore zero. The injection on  $H_6$  in Equation (6.1) makes  $u \cap [L] = 0$ . Poincaré duality on  $L$  gives  $[u] = 0$ . For a degree-six coefficient the same argument uses closed ambient two-forms and injectivity

on  $H_2$ . Rational injectivity remains injectivity after scalar extension to  $\mathbb{C}$ . Applying this argument to each Taylor coefficient proves that  $\Delta_\lambda(t)$  is exact coefficientwise.

Fix a Riemannian metric on  $L$  and a linear primitive operator on exact forms, for example  $h = d^*G$  with  $G$  the Hodge Green operator. For a closed exact form  $u$  it satisfies  $dhu = u$ . Replace

$$b \quad \text{by} \quad b - \mathbf{q}^\lambda h \Delta_\lambda(t).$$

The correction has degrees one and five. Its unary differential cancels precisely the energy- $\lambda$  curvature. Every other new contribution has greater energy: a nonconstant vertex has positive area, and a constant stable  $m$  vertex has at least two inputs, so any such occurrence of the correction is accompanied by another positive-energy input. Thus no earlier coefficient is changed. The lower energy bound  $\delta_L$  is preserved.

For completeness, this recursion is finite without assuming that the symplectic period group is discrete. There are only finitely many positive energies of nonconstant pure terms in the relevant cut by Proposition 4.6; call their set  $\mathcal{E}_N$ . Each is at least  $\delta_L$ . Every energy introduced by a curvature operation or by taking its primitive is a sum of energies in  $\mathcal{E}_N$ . Below  $N$  such a sum has at most  $\lfloor N/\delta_L \rfloor$  summands, so only finitely many sums occur. Zero-area stable vertices add no new exponents. In the expression trees, after ignoring the linear  $d$  and  $h$  edges, every such vertex has at least two inputs; their number is bounded by the number of positive-energy vertices and the external arity. Thus only finitely many primitive and composition steps are needed.

This bound is independent of Taylor order. The formal weight changes the scalar at a vertex; coefficientwise products in a fixed finite arity use finite Cauchy sums. The same finite expression trees and the same primitive operator handle every Taylor coefficient. Choose the finite chart regularity in Proposition 4.12 high enough for the bounded number of pushforwards, compositions, differentials, and primitives. Their finite losses of regularity then leave the regularity needed for the identities throughout the recursion. This proves the simultaneous formal assertion and completes the induction through energy  $N$ .  $\square$

## 6.4 Twisted mixed operations

Write  $b = b_{L,N}(t)$ . For a strictly directed graph word  $X_0 \prec \cdots \prec X_d$ , graph inputs  $x_1, \dots, x_d$ , and  $y \in \mathbb{C}\langle X_d \cap L \rangle$ , define

$$\mu_{d|1}^b(x_1, \dots, x_d; y) = \sum_{\ell \geq 0} m_{d+1+\ell}(x_1, \dots, x_d, y, b^{\otimes \ell}), \quad (6.14)$$

truncated to total energy at most  $N$ . Its degree is  $1 - d$  modulo four. The next result shows that  $\mu_{0|1}^b$  is a differential through the cutoff on the finite free intersection-generator space at  $X_0$ .

**Corollary 6.7** (Finite module identities). *For external graph arity at most  $N$ , the operations in Equation (6.14) satisfy the right module identities through energy  $N$  over the directed graph operations. All coefficient entries have nonnegative energy, and the errors in these finite matrix identities have energy greater than  $N$ , coefficientwise in  $t$ . On words without a  $C$ -slope label the operations use the relative weight; their specialization at  $t = 0$  is exactly the restriction of the unweighted operations on the full finite list, including its  $C$ -slope labels, twisted by  $b_{L,N}(0)$ .*

*Proof.* Apply Equation (6.7) to the mixed words with the appended  $b$ 's and sum over their number. The  $b$ 's have suspended parity zero, so these summations introduce no new insertion signs. A split entirely in the graph inputs gives a directed category operation. A mixed child gives a composition of two operations  $\mu^b$ . The remaining pure splits, together with

the differentials of appended  $b$ 's, sum to an insertion of  $\Delta(b)$  and vanish through energy  $N$ . Explicitly, putting  $\gamma_j = |x_j| - 1$ , the remaining identity is

$$\begin{aligned} 0 \equiv_N & \sum_{s=1}^d \sum_{i=1}^{d-s+1} (-1)^{\sum_{j<i} \gamma_j} \mu_{d-s+1|1}^b(x_1, \dots, m_s(x_i, \dots, x_{i+s-1}), \dots, x_d; y) \\ & + \sum_{i=0}^d (-1)^{\sum_{j \leq i} \gamma_j} \mu_{i|1}^b(x_1, \dots, x_i; \mu_{d-i|1}^b(x_{i+1}, \dots, x_d; y)). \end{aligned}$$

The case  $d = 0$  states  $(\mu_{0|1}^b)^2 \equiv_N 0$ . There is no directed curvature term, because the graphs are weakly exact and all their nonidentity words are strictly directed.

At energy at most  $N$  the twist uses at most  $N/\delta_L$  appended copies of  $b$ . If  $d \leq N$ , its stable  $m$  cost is bounded by

$$N + a(d + \ell) \leq N + a(N + N/\delta_L) < 3N.$$

The same bound applies to the total cost of a split by additivity. Every term of the asserted identity is therefore present in the  $4N$  system. Discarding terms above  $N$  cannot change any coefficient at or below  $N$ , as operations and  $b$  have nonnegative energies. On the jump sector the generator sets are finite, so these are finite matrix identities in each fixed external arity.

The weighted and unweighted constructions use the same finite geometric integrations and signs. Pure  $L$  terms admit the relative weight, and setting  $t = 0$  in the constructed bounding cochain cancels their unweighted curvature. The unweighted mixed identities then apply also to words containing  $C$ -slope labels. On the common words the weight specializes to one, proving the claimed compatibility.  $\square$

## 7 Exact coefficients and comparison with ordinary graph Floer theory

The finite constructions of the preceding sections need not be compatible as the energy cutoff increases. We first pass their matrix coefficients to a field in which all errors of unbounded valuation vanish. We then compare the resulting directed graph category with ordinary weakly exact Floer theory. This comparison concerns the graph objects only. Its purpose is to make the module constructed from  $L$  accessible to the ordinary mirror theorem, without putting  $L$  into that theorem's source category.

### 7.1 The coefficient fields

Write

$$\Lambda = \left\{ \sum_{\lambda \in \mathbb{R}} a_\lambda \mathbf{q}^\lambda : a_\lambda \in \mathbb{C}, \quad \#\{\lambda < C : a_\lambda \neq 0\} < \infty \text{ for every } C \in \mathbb{R} \right\}.$$

The valuation of a nonzero series is the least exponent with nonzero coefficient; set  $v(0) = +\infty$ . A nonempty support has a least element, and the support condition makes addition and multiplication well defined. After extracting its leading term, a nonzero series is invertible by a geometric series. Thus  $\Lambda$  is a field.

**Lemma 7.1.** *The field  $\Lambda$  is complete and algebraically closed. Its value group is  $\mathbb{R}$ , its residue field is  $\mathbb{C}$ , and its cardinality is at most the continuum.*

*Proof.* In a Cauchy sequence, the coefficients below any fixed real bound eventually stabilize. Their stabilized values define a series satisfying the same support condition, proving completeness. The assertions about the value group and residue field follow from the monomials and the constant coefficient.

Here is a short algebraic-closure argument that also records why the choice of complex coefficients matters. Completeness implies that the valuation is henselian. Let  $E/\Lambda$  be a finite Galois extension of degree  $d_E$ . Its uniquely extended valuation is Galois invariant. For  $x \in E^\times$ ,

$$d_E v(x) = v(N_{E/\Lambda}(x)) \in \mathbb{R}.$$

Divisibility of  $\mathbb{R}$ , and torsion-freeness of an ordered group, imply that  $v(x)$  already belongs to  $\mathbb{R}$ . The residue extension has degree at most  $d_E$ : linearly independent residue classes lift to linearly independent elements after rescaling a putative relation to have a coefficient of valuation zero. It is therefore just  $\mathbb{C}$ . Consequently every Galois automorphism acts trivially on the residue field.

For any  $0 \neq x \in E$ , choose  $a \in \Lambda^\times$  with  $v(a) = v(x)$ . All conjugates of  $x/a$  have the same nonzero residue. Their sum has nonzero residue, because the residue characteristic is zero. Hence  $\text{Tr}_{E/\Lambda}(x) \neq 0$ . The  $\Lambda$ -linear trace therefore has zero kernel, which forces  $d_E = 1$ . In characteristic zero every finite extension lies in a finite Galois extension, so  $\Lambda$  is algebraically closed.

Finally every support is countable, being a countable union of its finite parts below integral bounds. Both a countable list of real exponents and a countable list of complex coefficients have at most continuum many possibilities.  $\square$

Fix a nonprincipal ultrafilter  $\mathfrak{u}$  on  $\mathbb{N}$ . Let  ${}^*\Lambda = \Lambda^{\mathbb{N}}/\mathfrak{u}$ , with its valuation in  ${}^*\mathbb{R} = \mathbb{R}^{\mathbb{N}}/\mathfrak{u}$ . Inequalities below mean inequalities on an ultrafilter-large set. Define

$$\begin{aligned} O_b &= \{x \in {}^*\Lambda : v(x) \geq -c \text{ for some } c \in \mathbb{R}\}, \\ \mathfrak{m}_\infty &= \{x \in {}^*\Lambda : v(x) > c \text{ for every } c \in \mathbb{R}\}, \\ \kappa &= O_b/\mathfrak{m}_\infty, \quad R = \kappa[[t]]. \end{aligned} \tag{7.1}$$

Zero is included in both displayed subsets.

**Lemma 7.2.** *The ring  $O_b$  is a valuation ring with maximal ideal  $\mathfrak{m}_\infty$ . Its residue field  $\kappa$  is algebraically closed, and constant sequences induce an embedding  $\Lambda \hookrightarrow \kappa$ . The fields and rings in Equation (7.1) have cardinality at most the continuum.*

*Suppose a fixed finite collection of polynomial identities in matrix entries holds, for the  $N$ -th collection of entries, up to errors of valuation greater than  $N$ . If all those entry sequences are bounded below in valuation, their images in  $\kappa$  satisfy the identities exactly. The same assertion holds coefficientwise for formal power series in  $t$ .*

*Proof.* The elements of  ${}^*\mathbb{R}$  bounded above and below by standard reals form a convex subgroup. Coarsening the valuation by this subgroup gives exactly  $O_b$ ; positive coarsened valuation gives  $\mathfrak{m}_\infty$ . Alternatively, an element of  $O_b$  outside  $\mathfrak{m}_\infty$  has valuation bounded on both sides, so its inverse also belongs to  $O_b$ . Every nonzero constant sequence has this property, proving the asserted embedding.

The ultrapower is algebraically closed: a polynomial has finitely many coefficients, and choosing a root coordinatewise proves this assertion directly. A monic polynomial over  $\kappa$  lifts to a monic polynomial over  $O_b$ . Each root of the lift in  ${}^*\Lambda$  lies in  $O_b$ ; otherwise its highest power would have strictly smaller coarsened valuation than every other term. Reducing the factorization proves algebraic closure of  $\kappa$ . The cardinality bound follows from  $(2^{\aleph_0})^{\aleph_0} = 2^{\aleph_0}$ .

Reduction  $O_b \rightarrow \kappa$  respects finite sums and products, and a sequence with valuation greater than  $N$  belongs to  $\mathfrak{m}_\infty$ . For a fixed coefficient of  $t$ , a polynomial expression in power series uses only finitely many coefficients. This proves the last assertion.  $\square$

## 7.2 The exact directed category and its module

Choose finite lists exhausting the graph labels, and let the energy and external-arity cutoffs tend to infinity through  $N$ . The almost complex data, cap lines, and generator conventions are fixed once and for all; only the finite perturbations and the solutions  $b_{L,N}$  of the curvature equation are chosen anew. Apply Corollary 6.7, and truncate each post-twist matrix entry through energy  $N$ . For any fixed word, all sufficiently large lists contain that word. Entries for the finitely many missing lists may be set equal to zero.

**Proposition 7.3.** *These choices produce the following uncurved, nonunital  $A_\infty$  data:*

- (i) *an integer-graded directed category  $\mathcal{D}_\Lambda$ , with objects all graph labels and*

$$\mathrm{Hom}_{\mathcal{D}_\Lambda}(X, Y) = \begin{cases} \bigoplus_{x \in X \cap Y} \Lambda x, & X \prec Y, \\ 0, & \text{otherwise;} \end{cases}$$

- (ii) *its scalar extension  $\mathcal{D}_\kappa$ , and a right  $\mathbb{Z}/4$ -graded module  $\mathcal{M}_0$  over the reduction modulo four of  $\mathcal{D}_\kappa$ , with value complex*

$$\mathcal{M}_0(X) = \bigoplus_{x \in X \cap L} \kappa x;$$

- (iii) *on the labels with scalar slope, a weighted integer-graded directed category  $\mathcal{D}_R$ , and a right  $\mathbb{Z}/4$ -graded module  $\mathcal{M}_R$ , with finite free value complexes*

$$\mathcal{M}_R(X) = \bigoplus_{x \in X \cap L} Rx.$$

*Reduction modulo  $t$  identifies  $\mathcal{D}_R$  and  $\mathcal{M}_R$  with the corresponding restrictions of  $\mathcal{D}_\kappa$  and  $\mathcal{M}_0$ , including all operations.*

*The graph category is given by actual Novikov-convergent ordinary signed counts. In particular it does not depend on any asserted convergence of the sequence  $b_{L,N}$ .*

*Proof.* All Taylor coefficients of the finite post-twist entries have nonnegative valuation. Indeed unforced components have nonnegative area, and each inserted  $b_{L,N}$  has valuation at least  $\delta_L > 0$ . For each fixed arity, the category and module equations are finite matrix identities. Corollary 6.7 makes their errors have valuation greater than  $N$ , coefficientwise in  $t$ , for all sufficiently large  $N$ . Lemma 7.2 therefore makes them exact after reduction. Taking the constant Taylor coefficient commutes with these polynomial identities. The common choice of the finite weighted and unweighted systems gives the exact specialization assertion. No comparison between  $b_{L,N}$  and  $b_{L,N+1}$  is used.

For a directed word, all rigid moduli are regular, and Theorem 5.1 identifies the finite virtual integral with their ordinary signed count. Compactness gives only finitely many contributing classes below a fixed energy bound. Consequently the count defines an element of  $\Lambda$ , or coefficientwise of  $\Lambda[[t]]$  when the relative weight is present. Its difference from the finite-cut entry has valuation tending to infinity. Thus its image in  $\kappa$ , respectively  $R$ , is precisely the coefficient just constructed. Integer degrees on the graph category and modulo-four degrees on its module follow from Lemma 6.1.  $\square$

We write the module action in boundary order. An arrow  $x : X \rightarrow Y$  acts contravariantly on values:

$$\mathcal{M}_0(Y) \longrightarrow \mathcal{M}_0(X).$$

Our suspended convention has insertion sign  $\sum_{j < i} (|x_j| - 1)$ , as in Proposition 6.3. On cohomology the associative, unsuspended composition of successive arrows  $x, y$  is  $(-1)^{|x|} m_2(x, y)$ . In particular no additional sign occurs when the first arrow has degree zero. The same convention applies to the binary module action.

**Lemma 7.4.** *For every graph label  $X$ ,*

$$\sum_{r \in \mathbb{Z}/4} (-1)^r \dim_{\kappa} H^r(\mathcal{M}_0(X)) = [X] \cdot [L] = \nu[X] \cdot \alpha. \quad (7.2)$$

*For ordinary integer-graded graph Floer complexes the analogous Euler characteristic is the oriented intersection number.*

*Proof.* A finite-dimensional complex with an even grading period has the same alternating dimension as its cohomology: each boundary space appears twice with opposite signs. It remains to check the sign of an intersection generator.

At a transverse intersection choose unitary coordinates with the first tangent plane real. Let  $\alpha_1, \dots, \alpha_8 \in (0, \pi)$  be the positive plane-rotation angles, and let  $\vartheta_1, \vartheta_2$  be the grading lifts, taken modulo four when necessary. The generator degree is

$$|x| = 8 + \vartheta_2 - \vartheta_1 - \sum_{\ell=1}^8 \alpha_{\ell}/\pi. \quad (7.3)$$

The positive-rotation basis of the second plane differs from its marked orientation by  $(-1)^{\vartheta_2 - \vartheta_1 - \sum \alpha_{\ell}/\pi}$ . Concatenating the first-plane basis with that rotation basis has ambient symplectic orientation sign  $(-1)^{8 \cdot 7/2} = 1$ . Since eight is even, the local intersection sign is exactly  $(-1)^{|x|}$ . Summing proves the result, using  $[L] = \nu\alpha$ .  $\square$

### 7.3 Ordinary graph data extending the prescribed sector

An ordinary category used below has all ordered words of graph objects, including repeated labels; its morphisms for repeated or nontransverse positions use Hamiltonian chords. The relevant established ingredients are the transverse Floer compactness, determinant orientations, continuation, and gluing constructions; see, for example, [24]. We must verify that our prescribed domain-dependent data can be included without changing their strict-word counts. The verification also explains the closed weakly exact setting used here.

**Proposition 7.5.** *There is an ordinary integer-graded, zero-background spin, weakly exact Floer category  $\mathcal{C}$  of the graph labels and a nonunital  $A_{\infty}$  functor*

$$\mathcal{D}_{\Lambda} \longrightarrow \mathcal{C}$$

*which is the identity on objects and induces an isomorphism on morphism cohomology whenever  $X \prec Y$ . Every finite collection in  $\mathcal{C}$  is comparable, by ordinary quasi-equivalences, with the corresponding ordinary graph category constructed from any other regular compatible Floer data. These comparisons require no polygons with boundary on  $L$ .*

*Proof.* We first extend the data, then establish compactness and gluing, and finally compare conventions and choices.

**Data on the compactified domain families.** For each additional ordered pair choose regular strip data  $(H_\tau d\tau, J_\tau)$  with nondegenerate chords. On strict pairs retain exactly the unforced pair data already chosen. For a word of length at least three, use a Hamiltonian-valued one-form  $K$  and a compatible  $J$  on its punctured disks. Require the tangential restriction of  $K$  on each boundary arc to vanish. On every strict word use the original  $K = 0$  harmonic data, unchanged.

The requisite extension space can be described explicitly. Replace each puncture by its strip-height interval. Near a boundary node use the real plumbing coordinates

$$r_l = e^{-\pi s}, \quad r_r = e^{-\pi(T-s)}, \quad r_l r_r = \rho = e^{-\pi T}, \quad 0 \leq \tau \leq 1.$$

We require smooth coefficients in  $(r_l, r_r, \tau)$ , with the other component parameters as spectators. For one-forms, smoothness refers to their coefficients in the vertical strip coframe, dual to

$$-r_l \partial_{r_l} + r_r \partial_{r_r}, \quad \partial_\tau.$$

At the added interval prescribe the pair data  $(H_\tau d\tau, J_\tau)$ ; on the two nodal faces prescribe the component data. At external ends use the one-sided version.

These conditions are independent of the chosen plumbing chart. Indeed a change of holomorphic boundary-node coordinate has positive real linear term. Its radial variables therefore change by smooth positive factors. A displacement of a node origin in a smoothed fiber is  $O(\rho)$ ; dividing by  $r_l$  produces  $O(r_r)$ , and conversely. Thus the vertical strip transitions extend smoothly and have the same limiting height and strip directions. This is also a direct description of the real radial blowup of the boundary plumbing.

The harmonic data obey these conditions by Lemma 4.7. For any new word, the prescribed face restrictions agree at every intersection: both restrictions are the data attached to the same component tree. They can consequently be extended over corner charts by extension of face functions and inclusion–exclusion, and then patched by a partition of unity. The zero tangential coefficient on a boundary arc is preserved by extending that coefficient with its boundary defining factor. Compatible structures may be interpolated in a local chart, or by interpolating metrics followed by compatible retraction. This inductive extension never changes a strict word, since every component of a strict word is again strict.

On new stable words, Hamiltonian one-form variations supported at an interior domain point yield the usual universal transversality. Their Hamiltonian vector fields span the target tangent space, and domain dependence localizes the variation regardless of whether the map is multiply covered. Elliptic unique continuation then gives surjectivity of the universal linearization. Relative Sard–Smale choices give regularity while preserving prescribed faces. There are only countably many words and relative classes, so the choices can be made simultaneously. Strict words already have the required regularity.

**Energy and compactness.** On a neck, the difference from the pair model is  $O(r_l + r_r)$ , with all derivatives needed here. The curvature of  $H_\tau d\tau$  is zero. Thus the Hamiltonian curvature of  $K$  has an integral supremum bound uniform in the conformal parameters of a fixed word:

$$\int_R^{T-R} (e^{-\pi s} + e^{-\pi(T-s)}) ds \leq \frac{2}{\pi} e^{-\pi R}.$$

The finitely many core charts give the remaining bound. The energy identity for  $du - X_K$  therefore differs from  $\int u^* \omega$  by a uniformly bounded curvature term and the fixed external chord terms. In particular the topological area is bounded below, and an upper area bound gives a uniform analytic energy bound.

There are no nonconstant unforced spheres, since  $\pi_2(T) = 0$ . There are no unforced disks on an individual graph, since  $\pi_1(X) \rightarrow \pi_1(T)$  is injective and  $\pi_2(T, X) = 0$ . Small-scale

bubbles of an inhomogeneous equation would be exactly such unforced disks or spheres, so none occur. Translation on a growing neck limits to the prescribed pair strip equation. Nonconstant strip breakings have a positive energy gap for each fixed pair. On the remaining small-energy neck portions, solutions approach isolated chords; Hamiltonian transport of the pair equation reduces the estimate to a transverse intersection with height-dependent almost complex structure. Its asymptotic operator has a spectral gap. The exponential estimates then apply, with the exponentially decaying  $O(r_l + r_r)$  coefficient errors included. Thus necks lose neither energy nor matching data.

The spectral-gap and neck estimates are those of Lemma 4.8, after Hamiltonian transport. This is the ordinary compactness argument, with precisely the same domain degenerations as the compactified word family. The areas are homotopy invariant for the fixed chord ends and add on broken trees: the matched chord intervals occur with opposite orientations. Core closeness and the small neck neighborhoods also give eventual homotopy to the limiting tree, so only finitely many relative classes occur below a fixed area bound. Consequently the rigid counts define Novikov series and the one-dimensional compactifications have exactly the expected broken boundaries.

For later use one may allow a nonzero tangential component on boundary arcs, provided it has the same prescribed pair limits and face restrictions. Its coefficient in the strip-length direction is then  $O(r_l + r_r)$  near the added intervals; its integral, and hence the additional boundary term in the energy identity, remains uniformly bounded. The preceding conclusions still apply.

**Gluing.** The transverse gluing construction applies to these data as well. On the left half of a length- $T$  neck the difference from the partially broken left datum is  $O(r_r)$ , and on the right half it is  $O(r_l)$ . In coordinates transporting the pair Hamiltonian flow, the limiting equation is the transverse unforced strip equation; the additional forcing at its constant solution is  $O(r_l + r_r)$ . Choose an exponential weight smaller than both the asymptotic spectral gap and the coefficient decay rate. Cutting off the decaying component solutions gives an exponentially small pregluing error in this weight.

To construct a right inverse, split a forcing term in a fixed-width region around the middle of each neck, solve on the two component ends, and splice the resulting sections with cutoffs whose derivatives are  $O(1/T)$ . The component weights dominate the joined weight. The error consists of these cutoff derivatives and the coefficient differences just estimated; its operator norm tends to zero. A Neumann series therefore corrects the parametrix. Fixing core kernel projections makes the augmented linearization invertible, with a uniform bound. The ordinary uniform nonlinear estimates then give the unique nearby glued solution. Exponential decay gives the converse: every nearby curve lies in this gluing neighborhood. The same construction can leave any subset of nodes broken.

Length derivatives in coordinates fixed on the corresponding component halves have exponential bounds: explicit coefficient derivatives contain the opposite-end radial factor, while reparametrization derivatives in the middle multiply exponentially decaying map derivatives. Differentiating the inverse construction in slightly weaker weights gives the boundary smoothness needed for zero- and one-dimensional moduli. Thus the compactification and orientation gluing give the ordinary  $A_\infty$  relations. This is the transverse case of the splice used earlier, with no movable smooth-node mode. It requires no virtual perturbation for the graph category.

**Comparison of choices.** Two systems can be put on two differently labeled copies of their object lists. Prescribe each system on its own words, and extend on mixed words by the preceding construction. There is no conflict on faces. Ordinary data with exact strip collars

are admissible here too. Interpolating strip data with fixed external ends define continuation maps. Their homotopies have bounded curvature on compact interpolation regions and the same no-bubbling property. Gluing interpolations and deforming their concatenation proves composition and independence. Attaching continuation strips at the three triangle ends, and varying the remaining compact data, proves compatibility with binary products.

It follows that the inclusions of either copy are cohomologically full and faithful, and the two copies of each object are isomorphic on cohomology. The resulting quasi-equivalences are the ordinary comparison of data. Ordinary self-Floer units give cohomological units in the combined category. The same argument includes Hamiltonian-transported copies, as made explicit in Lemma 7.6 below.

For the strict-word inclusion one finally identifies determinant conventions. Lemma 7.7 proves that this is a strict change of generator signs. The equations on the strict sector are then literally those of  $\mathcal{D}_\Lambda$ . This completes the construction and comparison.  $\square$

The proof applies as well after adjoining finitely many of the auxiliary vertical tori used in Lemma 4.3. They are graded spin tori with injective fundamental-group maps, so the same disk and sphere exclusions, regular perturbations, energy estimates, and continuation argument apply. We use this extension only to compute the comparison triangle products below.

## 7.4 Hamiltonian gauge and conventions

**Lemma 7.6.** *Replacing any finite set of graph labels by specified Hamiltonian images, with their transported gradings and spin structures, preserves the ordinary category up to the above comparisons. Under the corresponding identification of polygons, topological area weights differ only by fixed input and output factors  $\mathbf{q}^{a_x}$ , one for each chord.*

*Proof.* On a word disk let  $G_z(v, x)$ ,  $0 \leq v \leq 1$ , be normalized Hamiltonians, equal on each open boundary arc to the Hamiltonian specified for that label. The functions can be interpolated using harmonic arc weights indexed by arc positions; the face and radial-collar rules are the same as above. Let  $\phi_z^v$  be their time- $v$  flows and put

$$F_v(z, x) = (z, \phi_z^v(x)), \quad \Gamma = \int_0^1 F_v^*(d_z G_z(v, \cdot)) dv.$$

Here  $d_z$  differentiates with the target variable held fixed. With the convention  $\iota_{X_G} \omega = -d_x G$ , Cartan's formula gives

$$F_1^* \omega = \omega + d\Gamma. \tag{7.4}$$

Indeed

$$\frac{d}{dv} F_v^* \omega = dF_v^*(-d_x G_z) = dF_v^*(d_z G_z),$$

since the omitted full differential has zero exterior differential.

Conjugating maps by  $\phi_z^1$  gives Floer data of the admitted kind. Every variation of a Hamiltonian flow is Hamiltonian. The moduli spaces therefore correspond, with their determinant orientations transported using the isotopy of the generator caps. On an open boundary arc the generating Hamiltonian is independent of its arc coordinate, so  $\Gamma$  vanishes on the boundary tangent. Stokes in Equation (7.4) leaves only the integrals over external chord intervals. Those depend just on the fixed chord, with the input and output signs dictated by boundary orientation. They are absorbed by rescaling the chord basis.

One obtains the same conclusion when ordinary perturbations are presented by moving boundaries. Extend their generating functions over the disk with the pair limits and the face conditions, and conjugate back to fixed boundaries. The resulting one-form may have a

tangential component, which is allowed by the energy estimate above. On a boundary tangent  $V$ ,  $\Gamma(V)$  is the normalized Hamiltonian generating  $\partial_V \phi_z^1$ , pulled back by  $\phi_z^1$ . This follows either from the mixed tangent component of Equation (7.4) or directly by differentiating the flow. The moving-boundary Hamiltonian correction therefore gives the same fixed-boundary area up to the stated end factors.  $\square$

There is a harmless variance choice in comparing Floer conventions. Our strip has the previous label at height zero and the next at height one, with inputs at positive ends. Its input degree is the index of the closing *negative*-end cap, giving Equation (7.3). Changing tensor notation does not change that index. If a reference orders the geometric boundary labels oppositely, use its  $A_\infty$  opposite, reversing tensors with the suspension Koszul signs. These are the same geometric disks; no reflection of a holomorphic map is involved. For example, with matched grading lifts, the degree assigned to the opposite corner is

$$\vartheta_A - \vartheta_B + \sum_{\ell} \alpha_{A,B,\ell}/\pi = 8 + \vartheta_A - \vartheta_B - \sum_{\ell} \alpha_{B,A,\ell}/\pi.$$

Thus it agrees with our cap index for the arrow  $B \rightarrow A$ . On the mirror side a variance reversal is matched by perfect duality. The ordinary graph theorem may consequently be used in the covariant form needed below.

**Lemma 7.7.** *After matching cap determinant lines, the strict directed operations of Section 7 and ordinary spin Floer operations with the same boundary and suspension order are strictly isomorphic by sign changes of generators. These changes preserve degrees and Novikov valuations.*

*Proof.* Use the same negative- and positive-end cappings, reciprocal contractions, and spin trivializations for the two constructions. A change of basis at one end is tracked also in the cap contraction; the remaining freedom is precisely the ordinary sign of the generator line. Having matched those lines, write

$$(-1)^{r_k(a_1, \dots, a_k)}$$

for the ratio of the two signs on a rigid  $k$ -input operation, where  $a_j = |x_j| - 1$  is the suspended integer degree. The output suspended degree is  $1 + \sum_j a_j$ .

We justify that this ratio depends only on these indices. Both constructions orient the capped disk by its spin boundary framing and orient sphere factors complexly. There is no background class. Integral graph gradings make the capped Maslov contribution zero. The two rules use the same spin homotopy class along each boundary segment, so their ratio acquires no relative-class twist. At fixed degree, a transverse graded pair can be varied through connected data: after fixing the first unitary frame, deform the second transverse plane to the imaginary plane in its graph chart, transporting its lift and orientation. Positive changes of frame are connected. Along a segment, lifted determinant winding determines the homotopy class of a unitary frame path, since  $SU(8)$  is simply connected. Capping and lower-order changes therefore leave only permutations of determinant, marking, and suspension factors; those depend solely on the indices and the order of the inputs. The same argument applies to the one-dimensional determinant comparisons.

Both conventions have the same suspended insertion sign. Determinant gluing at a two-vertex boundary consequently implies that

$$r_s(a_i, \dots, a_{i+s-1}) + r_{k-s+1} \left( a_1, \dots, a_{i-1}, 1 + \sum_{j=i}^{i+s-1} a_j, a_{i+s}, \dots, a_k \right) \quad (7.5)$$

is independent of the chosen block and its length  $s \geq 1$ . This is a determinant-line identity, not an inference from a potentially empty moduli space. It holds for arbitrary integer tuples: choose the ordered graded end planes with the prescribed indices, choose the node and output degrees for two rigid formal vertices, and join the boundary paths in the graded-plane cover. The one-dimensional gluing comparison then gives Equation (7.5); strip vertices include their translation direction in this comparison.

Choose  $f : \mathbb{Z} \rightarrow \mathbb{Z}/2$  recursively so that

$$f(a+1) + f(a) = r_1(a).$$

Rescaling a generator of suspended degree  $a$  by  $(-1)^{f(a)}$  replaces  $r_k$  by

$$r'_k = r_k + f(1 + \sum_j a_j) + \sum_j f(a_j).$$

It makes  $r'_1 = 0$  and preserves the block identity. Compare a unary child at input  $i$  with a unary parent. The identity then gives

$$r'_k(a_1, \dots, a_i + 1, \dots, a_k) = r'_k(a_1, \dots, a_i, \dots, a_k).$$

Hence  $r'_k$  is a constant  $c_k$ . Comparing a binary child with a unary parent gives  $c_k = c_2 + c_{k-1}$ , with  $c_1 = 0$ ; thus  $c_k = (k-1)c_2$ . A common sign change of every arrow by  $(-1)^{c_2}$  changes a  $k$ -input operation by  $(-1)^{(k+1)c_2} = (-1)^{(k-1)c_2}$ , removing all remaining signs. The resulting identification is strict.  $\square$

## 7.5 Later copies act invertibly

For a graph label  $X$ , let  $X^+$  be its next-layer copy. The close perfect Morse perturbation has a unique degree-zero generator, the maximum, denoted

$$w_X \in \text{Hom}_{\mathcal{D}_\Lambda}^0(X, X^+).$$

This is the arrow used later to pass from the directed category to the full line-bundle model.

**Proposition 7.8.** *The arrow  $w_X$  is closed. In the ordinary graph category its class is an isomorphism, and its action is a chain isomorphism*

$$\mathcal{M}_0(X^+) \xrightarrow{w_X} \mathcal{M}_0(X).$$

*In particular it induces an isomorphism in each modulo-four cohomology degree.*

*Proof.* Self-Floer cohomology of a weakly exact graph with the transported brane structure is its ordinary cohomology over  $\Lambda$ . For a sufficiently small exact perturbation this is the usual Morse comparison; ordinary continuation identifies all the choices, and no disk or sphere terms occur. The chosen perfect Morse pair has exactly the total Betti number of generators. Its differential must therefore be zero: the total dimension of cohomology equals that of the underlying finite complex. The degree-zero group is one-dimensional. Its nonzero generator  $w_X$  is consequently a nonzero scalar multiple of a Hamiltonian continuation isomorphism, and hence itself an isomorphism. The sign identification above preserves these assertions.

We now prove the stronger statement about the newly constructed module. Put  $Y = X^+$ , and use Lemma 4.3:  $X$  is horizontal, and near every point of  $X \cap L$  the other sheet is a vertical patch  $q = q_j$ . The graph  $Y$  is  $p = \nabla f(q)$ , with oscillation less than  $\epsilon_X$ . Here  $\epsilon_X$  is smaller than both the disk gap  $\delta_L$  and the fixed crossing energy needed to leave the chosen tube. The intersections of  $X$  and  $Y$  with  $L$  are paired by these vertical patches. The generator  $w_X$

is at the unique maximum of  $f$ , with the degree and brane structures transported through the coordinate change.

Let  $F_j^{\text{vert}}$  be the full comparison vertical torus at  $q_j$ . Every triangle for  $(X, Y, F_j^{\text{vert}})$  with first input  $w_X$  has area

$$\Delta_j = f(\max) - f(q_j) \leq \text{osc}(f) < \epsilon_X, \quad (7.6)$$

by lifting to the cover and applying Stokes. The two Floer complexes with the vertical torus each have one generator and no differential. Product with the isomorphism  $w_X$  is therefore nonzero. Its coefficient is

$$c_j \mathbf{q}^{\Delta_j}, \quad c_j \in \mathbb{C}^\times.$$

All triangles contributing to it have the single area in Equation (7.6), so this assertion concerns the ordinary signed count itself.

The data of Proposition 4.4 for these comparison triangles were chosen to agree with those for  $(X, Y, L)$  inside the tube. Every such low-area triangle stays in one vertical patch; there are none between unmatched patches. There are also no nonconstant low-area strips for  $(X, L)$  or  $(Y, L)$ : inside the tube their ends belong to the same patch and Stokes gives area zero. Thus replacing the comparison torus by  $L$  identifies the low-area rigid triangle spaces literally. Their orientations agree up to a fixed nonzero sign at each patch. Indeed the boundary arc lies in a contractible common vertical patch, so its spin trivialization and caps can be identified, allowing only a fixed grading shift and basis sign, independent of the triangle. Theorem 5.1 makes the finite virtual count in this regular sector equal to the ordinary signed count.

No term with a bounding-cochain insertion occurs at these energies, because each insertion has valuation at least  $\delta_L > \epsilon_X$ . Consequently, for every sufficiently large cutoff, the matrix  $A_N$  of action by  $w_X$  has the form

$$A_N = \text{diag}(c'_j \mathbf{q}^{\Delta_j}) + E_N, \quad c'_j = \pm c_j \neq 0, \quad v((E_N)_{ij}) > \epsilon_X.$$

The determinant's uniquely lowest-valuation term is the product of the displayed diagonal entries. In fact any changed row of a determinant term replaces a factor of valuation  $\Delta_j < \epsilon_X$  by one of valuation greater than  $\epsilon_X$ . Diagonal corrections have the same strict inequality. This comparison is row by row; it does not require  $\sum_j \Delta_j < \epsilon_X$ . Thus

$$v(\det A_N) = \sum_j \Delta_j$$

for every sufficiently large  $N$ , with the fixed nonzero leading coefficient  $\prod_j c'_j$ . In particular  $\det A_N$  is not in  $\mathfrak{m}_\infty$ , so the limiting matrix over  $\kappa$  is invertible. The exact module identity makes it a degree-zero chain map, hence a chain isomorphism.

If  $X \cap L$  is empty,  $Y$  was chosen disjoint from  $L$  as well. Both value complexes are then zero, and the assertion holds.  $\square$

## 8 The theta algebra and its projective family

The scalar graphs make a computable part of the directed category. Their products will provide the ring over which the new brane's cohomology becomes a graded module. We first compute these products over the formal coefficient ring. We then realize the ring, by an abstract field embedding, as a classical theta section ring. This second step uses convergence only of the explicit theta series.

Keep the symmetric complex matrix  $M$  commuting with  $D$ , the integer  $h_0 \geq 3$ , and  $R = \kappa[[t]]$  from Lemma 7.2. Thus  $\Lambda \hookrightarrow \kappa$ , the field  $\kappa$  is algebraically closed, and  $|\kappa| \leq |\mathbb{C}|$ . Throughout this section  $G_i$  means the layer-zero graph

$$p = h_0 i q + \nabla f_i(q), \quad i \geq 1.$$

The periodic potentials have sufficiently small Hessians. We use the integer grading on arrows between these graphs, before folding modulo four.

### 8.1 Intersections and normalized triangle weights

A lifted sheet of  $G_i$  has equation  $p = h_0 i q + \nabla f_i(q) + a_i$ , with  $a_i \in \mathbb{Z}^8$ . If  $i > j$ , put  $m = h_0(i - j)$  and  $u = (a_j - a_i)/m$ . Its intersection with a lifted sheet of  $G_j$  is the unique solution of

$$m(q - u) + \nabla(f_i - f_j)(q) = 0. \quad (8.1)$$

Indeed the left side is the gradient of the proper strictly convex function  $m|q - u|^2/2 + f_i(q) - f_j(q)$ . The intersection generators are consequently indexed by

$$\bar{u} \in (m^{-1}\mathbb{Z}^8)/\mathbb{Z}^8,$$

and there are  $m^8$  of them. All have integer degree zero. A holomorphic strip between these two graphs lifts to fixed sheets: its ends are their same unique intersection. Each sheet is exact on the cover, so Stokes' theorem makes the strip's area zero. Thus every such strip is constant. In particular the arrow differential vanishes; the grading also makes every operation on decreasing-slope words except the binary product zero. The binary product is therefore associative on the actual arrow spaces.

For  $i > j > k$ , set

$$m = h_0(i - j), \quad m' = h_0(j - k), \quad u = \frac{a_j - a_i}{m}, \quad v = \frac{a_k - a_j}{m'}, \quad w = \frac{mu + m'v}{m + m'}.$$

For a symmetric matrix  $U$  define

$$\begin{aligned} T_{m,m'}^U(u, v) &:= \frac{1}{2}(mu^t U u + m'v^t U v - (m + m')w^t U w) \\ &= \frac{mm'}{2(m + m')}(u - v)^t U (u - v). \end{aligned} \quad (8.2)$$

These quadratic expressions are precisely the triangle weights after a change of basis. Here is the change, including the boundary primitive terms.

For  $B = dq^t M \wedge dp$ , a primitive on the cover is  $-p^t M dq$ . On the graph use the prescribed relative primitive  $\xi_i = -(\nabla f_i)^t M dq$ . Hence the potential of  $-p^t M dq - \xi_i$  on sheet  $i$  is

$$-\frac{h_0 i}{2} q^t M q - a_i^t M q.$$

For  $\omega$  the analogous potential is  $-h_0 i |q|^2/2 - a_i^t q - f_i(q)$ . At the intersection  $q_{ij}$  the difference of the sheet- $i$  and sheet- $j$  potentials is respectively  $mu^t M u/2 + \delta_{ij}^B$  and  $m|u|^2/2 + \delta_{ij}^\omega$ , where

$$\begin{aligned} \delta_{ij}^B &= -\frac{m}{2}(q_{ij} - u)^t M (q_{ij} - u), \\ \delta_{ij}^\omega &= -\frac{m}{2}|q_{ij} - u|^2 - f_i(q_{ij}) + f_j(q_{ij}). \end{aligned} \quad (8.3)$$

These quantities are unchanged by simultaneous integral translation of  $q_{ij}$  and  $u$ , and thus depend only on the projected corner. In the counterclockwise boundary integral the two input

differences occur with positive sign and the output difference with negative sign. Replacing each intersection generator by

$$\mathbf{q}^{-\delta_{ij}^\omega} \exp(-t\delta_{ij}^B) [q_{ij}] \quad (8.4)$$

cancels all three corrections. A triangle with these lifted boundaries therefore contributes its sign times

$$\mathbf{q}^{T_{m,m'}^{18}(u,v)} \exp(tT_{m,m'}^M(u,v)).$$

The factors in Equation (8.4) are units of  $R$ . Their Novikov valuations are fixed finite numbers; the coefficient-limit construction is unchanged by this normalization.

**Lemma 8.1** (The count of a lifted triangle). *For every fixed triple of lifted sheets with  $i > j > k$ , the unweighted oriented triangle count has absolute value one. After choosing the intersection orientation bases equivariantly, its sign depends only on  $(i, j, k)$ .*

*Proof.* Shrink the potentials  $f_i, f_j, f_k$  to zero. Their pair differences remain strictly convex, so all lifted corners continue uniquely. Deform the triangle data to the standard complex structure  $J_*$  at the affine end. The primitive computation above bounds area in this fixed lifted class uniformly along the deformation. There are no spheres or disks on a single graph, and no nonconstant strips can break off, by the uniqueness and exactness just proved. A three-marked disk has no domain boundary without such bubbling or breaking. Thus the parameterized count is a compact oriented cobordism. If necessary, achieve transversality by Hamiltonian one-form variations supported in the triangle interior, fixed at the regular endpoints and at the strip ends. The usual continuation energy estimate adds only a uniformly bounded curvature term; these variations introduce none of the excluded bubbles. This also handles parameter values with triple constant maps. Anchoring one corner fixes compatible lifts throughout the compactness argument.

At the affine endpoint the three vertices have real parts  $w, u, v$  in boundary order. Their differences are complex scalar multiples of the real vector  $u - v$ . If  $u \neq v$ , they bound a positively oriented straight triangle in a complex affine line, with area  $T_{m,m'}^{18}(u, v) > 0$ . Uniformization gives a holomorphic triangle with the three prescribed boundary marks. If  $u = v$ , the three vertices coincide and the solution is constant.

Any two solutions differ by a holomorphic map with boundary on the three linear tangent planes and zero corner limits. The planes have their quadratic exact primitives, and Stokes' theorem gives this difference zero area. It is therefore zero. The same argument gives zero kernel for the linearized operator with the three domain marks fixed. Its index is zero by the degree calculation, so it is surjective. This proves regularity, including for the constant solution, and the asserted absolute count. At the affine endpoint the tangent planes, framed spin data, and linearized operators depend only on the slopes. Choose the cap bases using these common frames, and transport them along the deformation. Deck translation preserves these choices. The resulting sign is thus independent of the offsets.  $\square$

Write this sign as  $\epsilon(i, j, k) \in \{1, -1\}$ . The unsigned formulas are associative: for three inputs, the two sums range over the same three lifted points modulo common translation, and their exponents agree by telescoping the quadratic expression in Equation (8.2). The sums can be rearranged because the quadratic area gives only finitely many terms below each energy bound. Their iterated products are nonzero at  $t = 0$ , where every coefficient is a sum of positive Novikov monomials. Comparing this associativity with the signed Floer associativity gives

$$\epsilon(i, j, k)\epsilon(i, k, l) = \epsilon(j, k, l)\epsilon(i, j, l) \quad (i > j > k > l).$$

Rescale every arrow  $G_i \rightarrow G_j$  with  $j > 1$  by  $\epsilon(i, j, 1)$ , and leave arrows to  $G_1$  unchanged. The displayed identity makes every product sign  $+1$ . Both the monomial normalization and this sign normalization commute with reduction modulo  $t$ .

## 8.2 The ring and its cohomology modules

Define the graded  $R$ -module  $\mathcal{S}$  by  $\mathcal{S}_0 = R$  and

$$\mathcal{S}_\ell = \bigoplus_{\bar{u} \in ((h_0\ell)^{-1}\mathbb{Z}^8)/\mathbb{Z}^8} Re_{\ell, \bar{u}} \quad (\ell \geq 1).$$

Let  $1 \in R$  be its multiplicative identity. For  $m = h_0\ell$  and  $m' = h_0\ell'$ , define

$$e_{\ell, \bar{u}} e_{\ell', \bar{v}} = \sum_{\substack{u \in \bar{u} + \mathbb{Z}^8, v \in \bar{v} + \mathbb{Z}^8 \\ \text{modulo simultaneous } \mathbb{Z}^8}} \mathbf{q}^{T_{m, m'}^{18}(u, v)} \exp(t T_{m, m'}^M(u, v)) e_{\ell + \ell', \overline{(mu + m'v)/(m + m')}}. \quad (8.5)$$

In this formula an input residue denotes any representative of its coset; the indicated quotient makes the sum independent of those representatives. The output belongs to  $((m + m')^{-1}\mathbb{Z}^8)/\mathbb{Z}^8$ . Fixing one representative of  $u$  leaves a discrete lattice of possible  $v - u$ . The expression  $T_{m, m'}^{18}$  is a positive definite quadratic form on that lattice. Consequently each coefficient of  $t^a$  in each output coefficient is a locally finite Novikov sum: inserting  $(T^M)^a/a!$  changes its coefficients, not its support. Thus Equation (8.5) is defined in  $\Lambda[[t]]$  before applying  $\Lambda[[t]] \hookrightarrow R$ .

**Proposition 8.2** (The directed theta ring). *Equation (8.5) makes  $\mathcal{S}$  a commutative associative graded  $R$ -algebra. The normalized arrows give isomorphisms*

$$\mathcal{S}_{i-j} \simeq \text{Hom}_{\mathcal{D}_R}(G_i, G_j), \quad i > j,$$

*compatible with composition, and the arrow spaces have zero differential. Each  $\mathcal{S}_\ell$  is free of rank  $(h_0\ell)^8$  for  $\ell > 0$ . All these identifications reduce to the unweighted ones at  $t = 0$ .*

*Proof.* Symmetry under exchanging  $(m, u)$  and  $(m', v)$  proves commutativity. Associativity follows from the quadratic telescoping and convergence argument above. A compatible lift triple of sheets is determined, modulo deck translation and common integral  $p$ -translation, by the two input lifts in Equation (8.5); its third corner is their weighted average  $w$ . Lemma 8.1, the action computation, and the removal of  $\epsilon$  therefore identify this multiplication with the binary Floer operation. The remaining assertions were established above.  $\square$

Set  $\mathcal{S}_s = \mathcal{S}/t\mathcal{S}$ .

**Proposition 8.3** (The graded cohomology modules). *For  $r \in \mathbb{Z}/4$ , the groups*

$$V^r = \bigoplus_{j \geq 1} H^r(\mathcal{M}_R(G_j)), \quad V_s^r = \bigoplus_{j \geq 1} H^r(\mathcal{M}_0(G_j)), \quad (8.6)$$

*with the summands placed in ordinary degree  $j$ , are graded modules over  $\mathcal{S}$  and  $\mathcal{S}_s$ , respectively. Each ordinary graded piece of  $V^r$  is finite over  $R$ , and each such piece of  $V_s^r$  is finite-dimensional over  $\kappa$ . The actions are induced by chain maps compatible with reduction modulo  $t$ .*

*Proof.* The arrow  $G_{j+\ell} \rightarrow G_j$  gives the right-module action from Proposition 7.3,  $\mathcal{M}_R(G_j) \rightarrow \mathcal{M}_R(G_{j+\ell})$ . It has degree zero, hence is a chain map without an additional cohomological sign. The module identity makes the two binary actions for a pair of arrows chain homotopic; passing to cohomology and using Proposition 8.2 gives the associative graded action. Let  $1 \in \mathcal{S}_0$  act identically. Reduction gives the identical construction for  $\mathcal{M}_0$ . Finiteness follows from the finite free value complexes over the Noetherian ring  $R$  and their finite-dimensional reductions.  $\square$

The notation  $V_s^r$  refers to cohomology *after* reducing the complex. It does not identify this group with  $V^r/tV^r$ ; the possible torsion contribution will be retained in the lifting argument.

### 8.3 Analytic theta functions and prescribed field embeddings

Consider the nonempty convex domain

$$\mathcal{P}_M = \left\{ (\tau, s) \in \mathbb{C}^2 : \Pi = \tau 1_8 + \frac{sM}{2\pi i}, \quad \Im \Pi > 0 \right\}. \quad (8.7)$$

Its periods are symmetric and commute with  $D$ . Nonemptiness follows by taking  $s = 0$  and  $\Im \tau > 0$ ; convexity is the convexity of positive definite real symmetric matrices. The following theta description applies to every symmetric  $\Pi$  with  $\Im \Pi > 0$ , in particular to the whole marked domain  $\mathcal{U}$ . Denote by  $\Theta_\Pi$  the theta line on  $A_\Pi = \mathbb{C}^8/(\mathbb{Z}^8 + \Pi\mathbb{Z}^8)$  with multipliers

$$F(z + b) = F(z), \quad F(z + \Pi b) = e^{-\pi i b^t \Pi b - 2\pi i b^t z} F(z), \quad b \in \mathbb{Z}^8.$$

For  $m > 0$  a basis of its  $m$ th power is

$$\vartheta_{m, \bar{u}}(z, \Pi) = \sum_{u \in \bar{u} + \mathbb{Z}^8} \exp(\pi i m u^t \Pi u + 2\pi i m u^t z), \quad \bar{u} \in (m^{-1}\mathbb{Z}^8)/\mathbb{Z}^8. \quad (8.8)$$

Positive definiteness gives normal Gaussian convergence on compact sets. The multiplier equations follow by reindexing. Conversely a periodic holomorphic section has a Fourier expansion in  $e^{2\pi i n^t z}$ ; its second multiplier equation relates the coefficients whose indices differ by  $m\mathbb{Z}^8$ . One freely chosen coefficient for each coset gives exactly Equation (8.8), proving that this is the full basis. The convergence is normal jointly in  $z$  and  $\Pi$ , so these sections also vary holomorphically over  $\mathcal{U}$ . Restrict now to  $\mathcal{P}_M$ . Completing the square in the product gives Equation (8.5) with substitutions

$$\mathbf{q}^\lambda \rightsquigarrow e^{2\pi i \tau \lambda}, \quad t \rightsquigarrow s. \quad (8.9)$$

Indeed the extra exponent is  $2\pi i \tau T_{m, m'}^{1s} + s T_{m, m'}^M$ .

These substitutions currently describe analytic functions, not a map on the entire formal coefficient field. The following statement supplies exactly the field map that is needed.

**Proposition 8.4** (Realizing the theta constants). *There is a countable union  $E_M$  of proper closed analytic subsets of  $\mathcal{P}_M$  such that every  $(\tau, s) \in \mathcal{P}_M \setminus E_M$  admits a field embedding*

$$\iota_{\tau, s} : \text{Frac}(R) \hookrightarrow \mathbb{C}$$

under which

$$\mathcal{S} \otimes_{R, \iota_{\tau, s}} \mathbb{C} \simeq \bigoplus_{\ell \geq 0} H^0(A_\Pi, \Theta_\Pi^{h_0 \ell}). \quad (8.10)$$

The isomorphism sends  $e_{\ell, \bar{u}}$  to  $\vartheta_{h_0 \ell, \bar{u}}$ . Likewise, for some  $\tau$  in the upper half-plane an embedding  $\kappa \hookrightarrow \mathbb{C}$  identifies  $\mathcal{S}_s \otimes_\kappa \mathbb{C}$  with the same section ring for  $\Pi = \tau 1_8$ .

*Proof.* List the countably many scalar multiplication coefficients of Equation (8.5) as  $c_1, c_2, \dots \in \Lambda[[t]]$ , and their analytic functions as  $c_1^{\text{an}}, c_2^{\text{an}}, \dots$ . We first prove that, for every polynomial  $P \in \mathbb{Q}[X_1, \dots, X_N]$ ,

$$P(c_1, \dots, c_N) = 0 \iff P(c_1^{\text{an}}, \dots, c_N^{\text{an}}) \equiv 0 \text{ on } \mathcal{P}_M. \quad (8.11)$$

Only finitely many integers  $m, m'$  occur in this expression. Their area exponents are non-negative rational numbers with a common denominator  $D_0$ . This follows directly from

Equation (8.2), since the input lattice denominators are  $m$  and  $m'$ . Multiplying finitely many such series preserves the common denominator. Put  $Q = e^{2\pi i\tau/D_0}$ . There is a constant  $C_M$  such that

$$|T_{m,m'}^M(u, v)| \leq C_M T_{m,m'}^{1s}(u, v)$$

for all these summands; the operator norm bound for  $M$  even gives a constant independent of  $m, m'$ . Thus the series and all their finite products converge normally on a sufficiently small bidisk in  $(Q, s)$  centered at  $(0, 0)$ . For example their absolute terms are bounded by  $|Q|^{D_0 T^{1s}} e^{|s| C_M T^{1s}}$ , with only polynomially many lattice points of bounded area. They extend holomorphically across  $Q = 0$ .

The resulting joint power series in  $Q, s$  is exactly the series obtained formally by expanding  $\exp(tT^M)$  and replacing  $\mathbf{q}^{n/D_0}$  by  $Q^n$ . For any coefficient of  $t^a$ , distinct  $\mathbf{q}^{n/D_0}$  have uniquely determined coefficients in  $\Lambda$ . The injection  $\Lambda \hookrightarrow \kappa$  preserves that equality. Consequently vanishing in  $\kappa[[t]]$  is equivalent to vanishing of this joint holomorphic germ. On the punctured bidisk it is the analytic theta expression, with a branch of  $\tau$  of sufficiently large imaginary part. The identity theorem on connected  $\mathcal{P}_M$  proves Equation (8.11).

Let  $F = \mathbb{Q}(c_1, c_2, \dots) \subset \text{Frac}(R)$ . It is a countable field. For each rational polynomial whose formal value is nonzero, exclude the zero set of its analytic value. By Equation (8.11) these are countably many proper closed analytic subsets. At every remaining point, evaluation defines an injective homomorphism on  $\mathbb{Q}[c_1, c_2, \dots]$  and hence an embedding  $F \hookrightarrow \mathbb{C}$ ; all nonzero denominators remain nonzero.

To extend this embedding, choose a transcendence basis of  $\text{Frac}(R)$  over  $F$ . Its cardinality is at most  $|\mathbb{C}|$ , since  $|\kappa((t))| \leq |\mathbb{C}|^{\aleph_0} = |\mathbb{C}|$ . Over the countable image of  $F$ , the field  $\mathbb{C}$  has transcendence degree  $|\mathbb{C}|$ : the algebraic closure of a field generated by an infinite set of cardinality  $\lambda$  over a countable field has cardinality  $\max(\lambda, \aleph_0)$ . Choose that many algebraically independent target elements, map the basis to them, and extend across the remaining algebraic extension using algebraic closedness of  $\mathbb{C}$ . This gives  $\iota_{\tau,s}$ . Its prescribed values on all multiplication constants, together with the theta bases, give Equation (8.10) degree by degree.

For the special fiber, perform the same argument with the constants  $c_i|_{t=0}$  and the one-variable cusp expansion in  $Q$ . Their analytic functions are the structure constants at  $s = 0$ . Avoid their countably many nonidentical relations in the upper half-plane and extend the resulting countable-field embedding to  $\kappa$  by the same cardinality argument.  $\square$

*Remark 8.5.* The embedding in Proposition 8.4 prescribes the values of whole theta structure constants. It need not fix the original complex coefficient field, send the formal element  $t$  to the numerical parameter  $s$ , or preserve the Novikov valuation. None of those additional requirements is used. In particular the proposition makes no analytic evaluation of the new brane's Floer series. It remains valid after any enlargement of  $\kappa$  that retains the injection of  $\Lambda$  and the cardinality bound. The same constants and the same excluded analytic sets can be used.

## 8.4 Normal generation and the family over the trait

We include the needed normal-generation argument to specify the algebraic properties of the special fiber without making a finite-generation assumption on the formal ring.

**Lemma 8.6** (Elliptic normal generation). *Let  $E$  be a smooth complex elliptic curve and  $L$  a line bundle of degree  $h \geq 3$ . Then  $L$  is very ample and every multiplication map*

$$H^0(E, L) \otimes H^0(E, L^a) \longrightarrow H^0(E, L^{a+1}), \quad a \geq 1,$$

*is surjective.*

*Proof.* Riemann–Roch and Serre duality give  $H^1(E, A) = 0$  and  $h^0(E, A) = \deg A$  for every positive-degree line bundle  $A$ . For every subscheme  $Z \subset E$  of length two,  $\deg L(-Z) > 0$ , so evaluation to  $L|_Z$  is surjective. This separates both points and tangent vectors and proves very ampleness.

First consider  $a = 1$ . If  $h = 3$ , the complete linear system embeds  $E$  as a plane cubic. No nonzero quadratic polynomial vanishes on that integral cubic, so  $\mathrm{Sym}^2 H^0(E, L) \rightarrow H^0(E, L^2)$  is injective. Both sides have dimension six, and the map is therefore surjective. For  $h > 3$ , apply induction to  $L(-p)$  for any point  $p$ . Products of its sections span  $H^0(E, L^2(-2p))$ . There are sections  $s_0, s_1$  of  $L$  with vanishing orders zero and one at  $p$ , respectively, by the same positive-degree cohomology calculation. Modulo sections vanishing to order two, the products  $s_0^2, s_0 s_1$  span the two-dimensional space of first jets of  $L^2$  at  $p$ . They complete the image to  $H^0(E, L^2)$ .

Choose a basepoint-free pencil  $W \subset H^0(E, L)$ : take a nonzero section and then a second section nonvanishing at each of its finitely many zeros. The pencil sequence is

$$0 \longrightarrow L^{-1} \longrightarrow W \otimes \mathcal{O}_E \longrightarrow L \longrightarrow 0.$$

After tensoring by  $L^a$ , with  $a \geq 2$ , its first term is  $L^{a-1}$  and has zero  $H^1$ . Hence even multiplication by  $W$  surjects onto  $H^0(E, L^{a+1})$ .  $\square$

At  $\Pi = \tau 1_8$ , the variety  $A_\Pi$  is the product of eight elliptic curves and  $\Theta_\Pi^{h_0}$  is the external tensor product of degree  $h_0$  lines. Künneth’s formula for sections and Lemma 8.6 give generation of its section ring in degree one. The product embedding followed by the Segre embedding is the embedding of its complete linear system. Its section-ring Proj is therefore that smooth integral eightfold, with grading line  $\Theta_\Pi^{h_0}$ .

**Proposition 8.7** (The smooth theta family). *The ring  $\mathcal{S}$  is a Noetherian graded  $R$ -algebra generated by the finite free module  $\mathcal{S}_1$ . Its relative Proj*

$$\mathcal{X} = \mathrm{Proj} \mathcal{S} \longrightarrow \mathrm{Spec} R$$

*is smooth, projective, and flat, and its total space is regular. The special fiber  $\mathcal{X}_s = \mathrm{Proj} \mathcal{S}_s$  is a smooth geometrically integral eightfold. Its grading line  $\mathcal{O}(1)$  is very ample, and the canonical map*

$$\mathcal{S}_s \longrightarrow \bigoplus_{\ell \geq 0} H^0(\mathcal{X}_s, \mathcal{O}(\ell))$$

*is an isomorphism. Under any realization in Proposition 8.4, the geometric generic fiber and its grading line become  $(A_\Pi, \Theta_\Pi^{h_0})$ .*

*Proof.* Apply the special-fiber embedding of Proposition 8.4. After the faithfully flat field extension  $\kappa \hookrightarrow \mathbb{C}$ , the preceding elliptic argument identifies the ring, Proj, and grading line with the claimed product and its full section ring. Surjectivity of multiplication descends degree by degree. Formation of Proj commutes with field extension; smoothness, geometric integrality, and the complete-linear-system embedding descend as well. Proper coherent cohomology commutes with field extension, so the canonical section-ring isomorphism descends.

For each  $\ell$ , the cokernel of  $\mathrm{Sym}_R^\ell \mathcal{S}_1 \rightarrow \mathcal{S}_\ell$  is a finite  $R$ -module. Its reduction modulo  $t$  is zero by special-fiber normal generation. Nakayama’s lemma makes the cokernel zero. Thus  $\mathcal{S}$  is a quotient of a polynomial ring on the finite free module  $\mathcal{S}_1$  over the Noetherian DVR  $R$ . It is Noetherian and its Proj is projective, with  $\mathcal{O}(1)$  relatively very ample.

The direct sum of the free graded pieces makes  $\mathcal{S}$  flat over  $R$ . For a homogeneous element  $f$ , the affine chart ring  $(\mathcal{S}[f^{-1}])_0$  is the degree-zero direct summand of an  $R$ -flat localization. It is flat; these charts prove flatness of  $\mathcal{X}$ . A flat morphism of finite presentation with

geometrically smooth special fiber is smooth along that fiber. The complement of the smooth locus is closed, and its image is closed because  $\mathcal{X}$  is proper over  $\text{Spec } R$ . If it were nonempty, this image would contain the closed point of the trait, contradicting the preceding smoothness. Hence the morphism is smooth everywhere. The base is regular, so its smooth total space is regular. Finally the generic-fiber assertion follows by taking  $\text{Proj}$  in Equation (8.10).  $\square$

## 9 Ordinary graph branes and a simultaneous algebraic mirror

We now work at the ordinary point  $t = 0$ , retaining the integer grading on the graph category  $\mathcal{D}_\Lambda$ . The purpose of this section is to compare this directed category with actual line bundles. The comparison is first made on finite lists over the complete Novikov field. Finite algebraic models then assemble those comparisons after one further field extension. Only the ordinary graph branes enter family Floer theory; the module constructed from  $L$  will enter through the algebraic localization in the next section.

Put

$$Y_\Lambda = E(\mathbf{q})^8, \quad H_Y = \bigotimes_{a=1}^8 H_{Y,a},$$

where  $E(\mathbf{q})$  is the Tate elliptic curve with multiplicative period  $\mathbf{q} = \mathbf{q}^1$ , and  $H_{Y,a}$  is a principal theta line from the  $a$ th factor. Write

$$\Sigma = \{h_0 k 1_8 : k \geq 1\} \cup \{S_-\}, \quad N_\sigma = \bigotimes_{a=1}^8 H_{Y,a}^{-\sigma_{aa}}.$$

Thus  $N_{h_0 k 1_8} = H_Y^{-h_0 k}$  and  $N_{S_-} = \bigotimes_{a \leq 4} H_{Y,a}^{-1} \otimes \bigotimes_{a > 4} H_{Y,a}$ . A *slope type* means an element of  $\Sigma$ ; the many labels of one slope type are its Hamiltonian perturbations in the successive layers.

### 9.1 The finite ordinary comparison

We use the following precise scope of established family Floer theory. For a finite transverse collection of embedded, integer-graded, tautologically unobstructed branes in a closed non-singular Lagrangian torus fibration with  $\pi_2$  of the base zero, compatible relative Pin data give a faithful  $A_\infty$  functor to twisted analytic perfect complexes, after choosing a sufficiently fine family cover [1, Theorem 2.10, §2.4.1, and Appendix A]. The corresponding cohomology maps are also surjective [2, Theorem 1.1 and Corollary 1.2]. The finite  $A_\infty$  construction, rather than only its map on morphism cohomology, is the input used here.

All its hypotheses hold for our graphs. The fibration

$$T \longrightarrow \mathbb{R}_q^8 / \mathbb{Z}^8$$

has closed total space and no singular fibers. Both its base and total space are aspherical. For  $X = \{p = A_X q + \nabla f_X(q)\}$ , the map on fundamental groups is  $b \mapsto (b, A_X b)$ , hence is injective. The relative homotopy exact sequence gives  $\pi_2(T, X) = 0$ . Consequently every disk on  $X$  has area zero in its relative class, while every nonconstant tame holomorphic disk has positive area. The same argument excludes spheres. These branes are therefore tautologically unobstructed for every tame almost complex structure, with zero bounding cochain. Their spin structures come from the base frame, their local systems are trivial and valuation-unitary, and their integer graph gradings are obtained by continuation from constant symmetric graph planes. The fixed squared complex volume form represents the fibration grading: its phase on the constant vertical frame is constant.

The two possible twists are trivial with these choices. The global zero section may be used in every family chart, so all differences of local sections and their primitive cocycle are zero. The global base frame trivializes the ordinary-spin comparison cocycle as well; the relative background is zero. These are the section and base-spin terms distinguished in [2, Remark 2.16]. The rigid dual has uniformizing coordinates  $z = (z_1, \dots, z_8)$  and periods  $z \mapsto \mathbf{q}^b z$ ,  $b \in \mathbb{Z}^8$ , so it is  $Y_\Lambda^{\text{an}}$ . The preceding verification applies also to  $C$ : no extra  $B$ -weight is being imposed in this section.

Choose the covariant convention for the mirror functor. If a family model uses the contravariant convention of [2, Equation (1.1)], compose once with perfect duality. This changes all numerical signs together. Restricting an ordinary finite functor to strictly ordered words gives a nonunital functor on the directed sector. Every word in its boundary identities remains strictly ordered. Proposition 7.5 lets us use the prescribed operations of  $\mathcal{D}_\Lambda$ : compare the two finite sets of ordinary data on separately labeled copies, take quasi-inverses, and then restrict to strict words. This comparison involves only ordinary weakly exact Floer theory.

**Lemma 9.1** (Numerical type of a graph image). *For any such finite construction, the image of a graph  $X$  is a line bundle, up to an integer shift. Before fixing the uniform convention sign, its numerical type is  $\varepsilon A_X$ , with one  $\varepsilon \in \{1, -1\}$  for the whole construction. More precisely, it is a shift of*

$$\bigotimes_{a=1}^8 H_{Y,a}^{\varepsilon(A_X)_{aa}} \otimes P_X, \quad P_X \in \text{Pic}^0(Y_\Lambda).$$

*Proof.* A graph meets every torus fiber once and transversely. Its local family complex is consequently free of rank one in one integer degree, with zero differential. We determine its transition characters; this is the additional information needed to identify the numerical line type.

On the real universal cover of the base, use

$$\ell(q) = A_X q + \nabla f_X(q) = dg(q), \quad g(q) = \tfrac{1}{2} q^t A_X q + f_X(q),$$

and the fiber paths  $s\ell(q)$  from the zero section. In family continuation, a coefficient carries the character of the fiber boundary closed by these paths [1, §§2.2 and 8.2]. For compatible real lifts this character is zero. Indeed, lift the projection of a comparison strip along its prescribed local base path. A further lift of its graph boundary lies in  $p = \ell(q) + a$  for one constant  $a \in \mathbb{Z}^8$ . The endpoint paths cancel this same offset at both ends, and the other boundary lies over the prescribed base path. Closing the strip therefore leaves no additional fiber winding. Equivalently, such a winding would be a nonzero fiber loop bounding in  $\mathbb{R}^8 \times (\mathbb{R}^8/\mathbb{Z}^8)$ , contrary to injectivity of its fiber fundamental group.

The family trivialization factor contributes no missing character. Its exponent is  $df_{ij} - dg_j + dg_i$  in [1, Equation (8.16)]. Our common zero section gives  $f_{ij} = 0$ , and the common real lift gives  $dg_i = dg_j$ . One can make the calculation for the undisplaced section, already transverse to every fiber, or use continuation of small perturbations. For the latter, local identifications from  $q$  to  $q + v$  change the angle by  $\ell(q + v) - \ell(q)$  and preserve this lifted-path calculation. In absolute multiplicative coordinates, every term of a local transition thus has character exponent zero. Its sum is a scalar in  $\Lambda$ , nonzero by continuation of the one-generator cohomology.

All local generator degrees agree on the real cover. A homotopy between the degree-zero transition maps has negative degree and is zero; the higher descent maps vanish for the same reason. The scalar transitions form an ordinary locally constant cocycle. Contractibility of the real cover permits scalar rescaling of the generators to trivialize this cocycle. The section and frame also trivialize the twisting data there. Hence no invertible analytic function is left undetermined in the following period calculation.

A base period  $b \in \mathbb{Z}^8$  changes the angle lift by  $A_X b$ . In the frame just chosen, the period multiplier is therefore

$$j_X(b, z) = c_b z^{-\varepsilon A_X b}, \quad c_b \in \Lambda^*. \quad (9.1)$$

The scalar is independent of the local polytope because the transition frames agree and the real base is connected. The multiplier cocycle identity says

$$c_{b+c} = c_b c_c \mathbf{q}^{-\varepsilon b^t A_X c}.$$

For a diagonal integral symmetric matrix  $A$ , the product theta line of type  $\varepsilon A$  has multiplier

$$j_{\varepsilon A}(b, z) = \mathbf{q}^{-\varepsilon b^t A b / 2} z^{-\varepsilon A b}.$$

Dividing Equation (9.1) by this multiplier leaves the scalar character  $d_b = c_b \mathbf{q}^{\varepsilon b^t A_X b / 2}$ , for which  $d_{b+c} = d_b d_c$ . Such a character defines a line in  $\text{Pic}^0$ : writing  $d_b = \prod_a d_{e_a}^{b_a}$ , it is the quotient of the principal product theta line translated by  $(d_{e_a}^{-1})_a$  and the original theta line. Thus the analytic image has exactly the asserted type. Projective rigid GAGA algebraizes this line over  $\Lambda$ ; its applicable form over a complete, possibly nondiscretely valued field is recalled in [5, p. 502].  $\square$

**Lemma 9.2** (Normalization of signs, shifts, and copies). *Suppose a finite list contains three distinct layer-zero scalar slopes and the layer-zero representative of every slope type on the list. After a common shift its images are unshifted lines*

$$L_X = N_{A_X} \otimes P_{A_X}, \quad P_{A_X} \in \text{Pic}^0(Y_\Lambda), \quad (9.2)$$

*with one degree-zero factor per slope type. The finite directed functor is an isomorphism on cohomology for every strict arrow.*

*Proof.* For distinct slope types the matrix difference is definite. Using the small layer-zero perturbations, each lifted sheet pair intersects once; there are  $|\det(A_X - A_{X'})|$  projected intersections, all of one degree. The differential is therefore zero. Hamiltonian continuation gives the same cohomology for any copies. The total Ext dimension of the corresponding lines is the same determinant: their quotient is an external product of nonzero-degree elliptic lines, with an arbitrary  $\text{Pic}^0$  factor, and elliptic Riemann–Roch and Künneth give exactly this dimension. Faithfulness is consequently fullness for these pairs.

Two copies of the same slope have the self-Floer cohomology of an eight-torus. Their images before shifts have degree-zero quotient. A nontrivial such quotient has zero cohomology: on the product of elliptic curves a  $\text{Pic}^0$  line is an external product, and at least one nontrivial degree-zero factor has both cohomology groups zero. Faithfulness therefore forces the quotient to be trivial. Its Ext groups have the torus Betti dimensions, so faithfulness is again fullness. We may identify all copies with the same line, leaving their shifts to determine.

Write a layer-zero image as  $L_i[s_i]$ . For  $i > j$  the source Hom from  $G_i$  to  $G_j$  is nonzero only in degree zero. If  $\varepsilon = 1$  in Lemma 9.1, its line quotient is anti-ample and has Ext only in degree eight. Thus  $s_j - s_i = 8$  for every such pair. Three slopes contradict the additivity of shift differences. Hence  $\varepsilon = -1$ . The quotients are now ample and have Ext only in degree zero, so all scalar shifts agree. The layer-zero arrows  $G_j \rightarrow C$  also have degree zero. Their line quotients have the positive diagonal type  $h_{0j} 1_8 - S_-$ , so the shift for  $C$  is the same. Finally, the source cohomology between copies of a slope has support in  $[0, 8]$ , with both endpoints nonzero. Since their underlying lines agree, fullness forces their shift difference to vanish. A common shift of the entire construction now gives Equation (9.2).  $\square$

## 9.2 Algebraic enhancements and uniform generation

We will need maps into chosen algebraic models, including their higher components. Here is a concrete enhancement of the GAGA comparison just used. Take a finite affine algebraic cover  $\mathfrak{U}$  of  $Y_\Lambda$ , an affinoid family cover on the analytic side, and a common finite admissible affinoid refinement  $\mathfrak{V}$  subordinate to both. Such a refinement exists by quasi-compactness. All algebraic and analytic intersections used here are affine and affinoid, respectively, by separatedness.

For honest bundles  $P, Q$ , first form Čech Hom complexes using all index tuples, with Alexander–Whitney cup composition. Assign each refinement open one containing open from each original cover. Analytification followed by restriction gives actual cochain maps

$$\check{C}(\mathfrak{U}, P^\vee \otimes Q) \longrightarrow \check{C}(\mathfrak{V}, (P^\vee \otimes Q)^{\text{an}}) \longleftarrow \check{C}(\mathfrak{U}_{\text{an}}, (P^\vee \otimes Q)^{\text{an}}).$$

The all-tuple convention allows arbitrary assignments of containing opens. These maps commute with differentials and cup composition. Acyclicity of affine and affinoid intersections and coherent cohomology GAGA make them quasi-isomorphisms. Projection to strictly increasing tuple complexes is also a cup dg map and a quasi-isomorphism, giving the usual bounded Čech models. Thus this is an enhanced comparison on lines and bounded bundle complexes. Passing to finite cones and retracts gives the comparison on their split-closed hulls. In the family-cover category, the rank-one images of Lemma 9.1 have ordinary strict descent, so analytic descent identifies their Homs and compositions with exactly this analytic Čech model. No algebraization of an arbitrary analytic gerbe is involved.

**Lemma 9.3** (Nine lines generate uniformly). *There are integers  $1 \leq k_1 < \cdots < k_9$  such that, for every choice  $P_j \in \text{Pic}^0(Y_\Lambda)$ , the nine lines*

$$H_Y^{-h_0 k_j} \otimes P_j, \quad 1 \leq j \leq 9,$$

*split-generate  $\text{Perf}(Y_\Lambda)$ . The same assertion holds after any algebraically closed field extension of  $\Lambda$ .*

*Proof.* Choose a vector bundle  $V$  which classically generates  $\text{Perf}(Y_\Lambda)$ . The classical-generator theorem [4] gives a perfect generator; a bounded vector bundle resolution and the direct sum of its terms give such a  $V$ . We construct nine evaluation steps uniformly in the degree-zero factors.

Put  $K_0 = V$  and  $S_j = (\text{Pic}^0 Y_\Lambda)^j$ . Suppose  $K_{j-1}$  is a vector bundle on  $Y_\Lambda \times S_{j-1}$ . Pull it to  $Y_\Lambda \times S_j$ , tensor by the inverse Poincaré line from the last factor and by  $H_Y^{h_0 k_j}$ . Relative Serre vanishing and generation for the projection to the proper parameter scheme  $S_j$  give, for one sufficiently large  $k_j > k_{j-1}$ , a locally free pushforward  $W_j$ , formation of which commutes with base change, and an evaluation surjection. Tensoring back gives

$$0 \longrightarrow K_j \longrightarrow W_j \otimes (H_Y^{-h_0 k_j} \otimes \mathcal{P}_j) \longrightarrow K_{j-1} \longrightarrow 0.$$

Here  $W_j$  is pulled back from  $S_j$ , and the kernel is locally free because the quotient is locally free. This completes the induction with exponents independent of the points  $P_j$ .

At any nine parameter points the resulting exact sequence is

$$0 \rightarrow K_9 \rightarrow W_9 \otimes L_9 \rightarrow \cdots \rightarrow W_1 \otimes L_1 \rightarrow V \rightarrow 0, \quad L_j = H_Y^{-h_0 k_j} \otimes P_j,$$

where now the  $W_j$  are finite-dimensional vector spaces. If  $C_V$  is the complex of the nine displayed line-bundle terms, there is a triangle

$$K_9[8] \longrightarrow C_V \longrightarrow V \longrightarrow K_9[9].$$

Since  $V$  is locally free and  $\dim Y_\Lambda = 8$ ,

$$\mathrm{Ext}^9(V, K_9) = H^9(Y_\Lambda, V^\vee \otimes K_9) = 0.$$

The triangle splits, so  $V$  is a summand of  $C_V$ . The nine lines therefore split-generate. All the evaluation sequences and their base-change properties were constructed over the entire parameter schemes. They and the splitting argument remain valid after the stated field extensions. The generator also remains a generator: by adjunction, a quasi-coherent complex right-orthogonal to its pullback has zero pushforward to  $Y_\Lambda$ ; the affine projection is conservative on quasi-coherent complexes, so that complex is zero.  $\square$

Fix these nine layer-zero labels as a core. Enlarge any finite list to contain the core and the layer-zero representatives of all its slope types. Lemma 9.2 and the enhanced GAGA comparison give a nonunital  $A_\infty$  functor from its ordered sector to an algebraic enhancement on the lines  $N_\sigma \otimes P_\sigma$ , with isomorphisms on strict-arrow cohomology. The images split-generate  $\mathrm{Perf}(Y_\Lambda)$  by Lemma 9.3. We may replace this enhancement by any chosen quasi-equivalent line model, using minimal-model transfer and quasi-inverses over the field [24]. Isomorphic copies of a line may similarly be replaced by a single chosen line with several labels. This statement concerns the target enhancement: it does not regard the finite directed source as a full ordinary subcategory.

### 9.3 Finite algebraic models over the Picard variety

The preceding finite constructions need not agree when the list grows. We next arrange algebraic formulas for their possible targets so that a simultaneous functor can be obtained without imposing analytic compatibility.

Let  $\hat{Y}_\Lambda = \mathrm{Pic}^0(Y_\Lambda)$ , with a fixed Poincaré line. Choose a finite affine open cover of  $\hat{Y}_\Lambda$ . For each slope type  $\sigma$ , the possible line is  $N_\sigma \otimes P_\sigma$ . In a word containing several labels of the same slope type, use *one* parameter for that type. Thus the parameter base for two distinct types is a product of two chosen affine charts, whereas for two copies of the same type it is just the one chart.

**Lemma 9.4** (Pairwise contractions with algebraic parameters). *For every pair of types and every choice of their affine parameter charts, there is a contraction of its bounded algebraic Čech Hom complex onto a finite projective graded cohomology module. The contractions can preserve the actual constant units. Choosing them pairwise and pulling them back to finite products of parameter charts gives strictly unital  $A_\infty$  line models whose operations at each arity are algebraic maps of finite projective modules and commute with arbitrary specialization.*

*Proof.* Fix an ordered finite affine cover of  $Y_\Lambda$ . Over an affine parameter base  $S = \mathrm{Spec} A$ , let  $C_{\sigma\tau}$  be the associated strictly increasing Čech complex of

$$(N_\sigma \otimes P_\sigma)^\vee \otimes (N_\tau \otimes P_\tau)$$

on  $Y_\Lambda \times S$ . These pair complexes carry the usual cup compositions after pullback to a common finite parameter product. Their terms are projective  $A$ -modules, although generally of infinite rank. Indeed, on an affine intersection  $U = \mathrm{Spec} B$  of the chosen cover, sections of the line are a finite projective module over  $B \otimes_\Lambda A$ . This ring is free as an  $A$ -module because  $B$  is a  $\Lambda$ -vector space. Such a section module is therefore an  $A$ -direct summand of a free module. Finite sums preserve this property. Affineness also shows that the terms, differential, and cup composition commute with base change.

The cohomology modules are finite projective and commute with specialization. If  $\sigma \neq \tau$ , the quotient line has either ample or anti-ample diagonal type: scalar types are totally

ordered, and  $h_0 k 1_8 - S_-$  is positive definite. Its fiber cohomology is concentrated in degree zero or eight, with constant dimension  $|\det(\sigma - \tau)|$ , by elliptic Riemann–Roch and Künneth. Proper coherent cohomology and base change then give the assertion. For a repeated type the quotient is canonically  $\mathcal{O}_Y$ , since its two labels share the same parameter. Its cohomology is simply  $H^*(Y_\Lambda, \mathcal{O}_Y) \otimes_\Lambda A$ .

We recall why a bounded complex of projective modules with projective cohomology admits the required contraction. Starting in its highest degree, split the surjection from the top term to its cohomology. Its boundary submodule is then projective. Split the preceding differential onto that submodule; its kernel is projective. Split this kernel onto its projective cohomology, and continue downward. This decomposes the complex into its cohomology and two-term identity complexes. Hence there are  $A$ -linear maps

$$i : H(C) \rightarrow C, \quad p : C \rightarrow H(C), \quad h : C \rightarrow C[-1]$$

with

$$pi = \text{id}, \quad ip = \text{id} - dh - hd, \quad h^2 = hi = ph = 0. \quad (9.3)$$

For a repeated type use the same structure-sheaf contraction for all its label pairs. In degree zero choose the constant function as the representative of  $1 \in H^0(Y, \mathcal{O}_Y)$ ; the decomposition can retain this summand. Thus  $h(1) = 0$  and the units are preserved.

Choose these contractions for each pair and chart choice, once and for all. For a finite word pull them back from the relevant pair parameter bases to the product indexed by its distinct types. The identities in Equation (9.3) survive every pullback, so no flatness qualification is needed at specialization. Homotopy transfer expresses the operation of arity  $d$  as the finite sum over rooted binary trees with  $d$  leaves: insert  $i$  at leaves, cup composition at vertices,  $h$  at internal edges, and  $p$  at the root, with the usual graded signs. These are multilinear maps between finite projective cohomology modules. They are algebraic, and every tree formula commutes with specialization. The side conditions and the preserved units give a strictly unital transfer. Using a fixed pair contraction, independent of other types in a word, makes these models compatible on all finite lists and on lists with repeated labels.  $\square$

## 9.4 A simultaneous ordered functor

**Proposition 9.5.** *There is an algebraically closed extension  $\kappa_1/\kappa$  of cardinality at most the continuum, a line  $L_\sigma = N_\sigma \otimes P_\sigma$  on  $Y_{\kappa_1} = Y_\Lambda \otimes_\Lambda \kappa_1$  for each slope type, and a nonunital integer-graded  $A_\infty$  functor*

$$J : \mathcal{D}_\Lambda \otimes_\Lambda \kappa_1 \longrightarrow \mathcal{B}. \quad (9.4)$$

Here  $\mathcal{B}$  is a strictly unital algebraic  $A_\infty$  model of the full Hom category of these lines, with a separate label for each graph copy. The functor is a quasi-isomorphism on every strict arrow. It sends each  $w_X : X \rightarrow X^+$  to a cohomological isomorphism. The split-closed hull of  $\mathcal{B}$  is  $\text{Perf}(Y_{\kappa_1})$ .

*Proof.* Take an increasing exhaustion by finite lists containing the core and the layer-zero representatives of their types. Choose one finite mirror functor on each list. By the preceding normalization it has one parameter point  $p_{\sigma,n} \in \hat{Y}_\Lambda$  per type; assign arbitrary points to absent types. Choose one affine chart containing each such point.

Fix a nonprincipal ultrafilter  $\mathcal{U}$  on the positive integers and set

$$\kappa_1 = \kappa^{\mathbb{N}}/\mathcal{U},$$

the ordinary field ultrapower, with the constant embedding of  $\kappa$ . It is algebraically closed: roots of a fixed-degree polynomial can be chosen in each coordinate wherever its leading

coefficient is nonzero. Its cardinality is at most  $(2^{\aleph_0})^{\aleph_0} = 2^{\aleph_0}$ . For each fixed type, one of the finitely many chart choices occurs on a  $\mathcal{U}$ -large set. Its parameter sequence defines a  $\kappa_1$ -point of that affine chart. This uses only finitely many coordinate functions and their algebraic relations. Denote its Poincaré line by  $P_\sigma$ .

At every finite stage transfer the target of the chosen functor to the specialized model of Lemma 9.4, using the selected charts at that stage. The enhanced comparison and quasi-inverse transfer give actual higher functor components with this target. For a fixed word, its labels belong to all sufficiently large lists, and all its chart choices agree with the selected limiting charts on a  $\mathcal{U}$ -large set: only a finite intersection of large sets is involved. On that set, every target operation is the specialization of one fixed algebraic module map.

Source Hom complexes for the word have fixed finite bases over  $\Lambda$ . Present each target finite projective as the image of a finite idempotent matrix over its parameter ring. Functor components can then be written in finite matrices into those images. Taking their coefficient sequences gives maps over  $\kappa_1$  into the evaluated idempotent images. For each arity the  $A_\infty$  functor equation has finitely many terms, and the target tree formulas commute with this evaluation. Its coordinatewise validity on a large set therefore proves the equation over  $\kappa_1$ . Entries outside the relevant large set can be assigned arbitrarily.

The first components remain quasi-isomorphisms. To see explicitly the finiteness of this assertion, fix a contraction of each source complex to its finite-dimensional cohomology over  $\Lambda$ . At every applicable stage, the first functor component induces an invertible map from this fixed cohomology to the finite projective target fiber. Choose an inverse. The two inverse equations are finite matrix equations, including the idempotent equations if the target is represented as a summand. They pass to the ultrapower, giving the claimed isomorphism on cohomology.

The resulting target is genuinely a full line model. Base-changing the pair contractions and performing their same finite tree transfers computes all derived Homs and compositions of the chosen lines on  $Y_{\kappa_1}$ , including arrows in both directions and endomorphisms. By Proposition 7.8, each  $w_X$  was a nonzero degree-zero cohomology class. Its image lies in the one-dimensional degree-zero Hom between two copies of the same line, and remains nonzero by the strict-arrow quasi-isomorphism. It is consequently invertible. Finally, the nine core lines split-generate by Lemma 9.3, proving the hull assertion.  $\square$

## 9.5 Extending the formal coefficients

We make the extension in Proposition 9.5 throughout the preceding algebraic constructions. On the ordinary fiber it is ordinary scalar extension by  $\kappa \rightarrow \kappa_1$ . For Taylor coefficients it is the coefficientwise map

$$R = \kappa[[t]] \longrightarrow R_1 = \kappa_1[[t]]. \quad (9.5)$$

This is a local faithfully flat map. Indeed  $R$  is a DVR,  $R_1$  is torsion-free over it and hence flat, and its residue extension is the nonzero field extension  $\kappa \rightarrow \kappa_1$ . In particular, for every finite free periodic value complex  $C$  constructed above,

$$H^r(C \otimes_R R_1) = H^r(C) \otimes_R R_1, \quad (C \otimes_R R_1)/t = C/tC \otimes_{\kappa} \kappa_1.$$

The same equalities hold for the chain actions and their induced cohomology actions. This uses the completed coefficient ring in Equation (9.5); no identification with  $\kappa_1 \otimes_{\kappa} \kappa[[t]]$  is asserted.

Each graded theta-ring piece is finite free, so its base extension has the same basis and multiplication formula with extended coefficients. Relative Proj and its grading line base-change, and smoothness and projectivity persist. The field-realization argument continues to apply: the theta structure constants are unchanged, and both  $\kappa_1$  and  $\text{Frac}(R_1)$  have cardinality

at most the continuum. All passage to the ultrapower and all subsequent extension here concern finite algebraic matrices and models. We have not extended an analytic family Floer construction to this field or evaluated a Floer series for  $L$  there.

Henceforth rename  $\kappa_1$  and  $R_1$  as  $\kappa$  and  $R$ , and write  $Y_\kappa = Y_\Lambda \otimes_\Lambda \kappa$ . With these coefficients, Proposition 9.5 supplies the simultaneous ordered map needed for localization.

## 10 Periodic modules and a nonpolarization class

We now use the ordinary mirror only at  $t = 0$ . The aim is to interpret the four graded modules  $V_s^r$  as section modules of coherent sheaves and then to detect a characteristic class by Euler pairings. The new Lagrangian need not be an object of the ordinary mirror category: its module will be extended from the directed category by the repeated graph copies.

Retain the enlarged coefficient field  $\kappa$  and the ordinary line model  $\mathcal{B}$  of Proposition 9.5. Its objects have the graph labels, and its split-closed pretriangulated hull is  $\text{Perf}(Y_\kappa)$ . The layer-zero labels  $G_k$  and  $C$  correspond to ordinary, unshifted lines  $L'_k$  and  $P_C$ , respectively. Here  $L'_k$  has numerical type  $H_Y^{-h_0k}$ . The common shift has already been removed in the construction of  $\mathcal{B}$ .

### 10.1 Periodic module conventions

For an integer-graded complex  $K$ , define its direct-sum fold by

$$\text{Fold}(K)^r = \bigoplus_{n \equiv r \pmod{4}} K^n, \quad r \in \mathbb{Z}/4. \quad (10.1)$$

Fold  $\mathcal{B}$  in this way and denote the result by  $\overline{\mathcal{B}}$ . Fold  $\mathcal{D}_\kappa$  and adjoin formal strict units, obtaining  $\overline{\mathcal{D}}$ . The nonunital ordered map of Proposition 9.5 extends strictly on these new units to a functor

$$J : \overline{\mathcal{D}} \longrightarrow \overline{\mathcal{B}}. \quad (10.2)$$

Extend  $\mathcal{M}_0$  by the same unit convention. For strict arrows,  $J$  induces cohomology isomorphisms. Each  $w_X : X \rightarrow X^+$  maps to a cohomological isomorphism, and its action on  $\mathcal{M}_0$  is a quasi-isomorphism by Proposition 7.8.

We use strictly unital right  $A_\infty$  modules and their dg category of premorphisms. All degrees in the rest of this section are modulo four unless explicitly stated otherwise. The parity signs are the usual ones, since the period is even. Write  $h_x$  for the right representable at an object  $x$ .

**Lemma 10.1** (Periodic module calculus). *For an uncurved strictly unital  $A_\infty$  category over a field, the dg Hom of strictly unital modules computes derived Hom after objectwise quasi-isomorphisms are inverted. Direct sums are coproducts, and*

$$\text{Hom}^*(h_x, N) = H^*(N(x)).$$

*The normalized bar tensor is exact on triangles and preserves objectwise quasi-isomorphisms; tensoring with the diagonal gives the identity. These assertions hold with direct sums in bar length and products in premorphism length in the period-four grading.*

*Proof.* Filter a premorphism by its number of arrow inputs. Its components form a product, complete for the filtration by components of length at least  $n$ . The length-preserving part of the differential is the internal Hom differential. If the target module is objectwise acyclic, that associated-graded Hom complex is acyclic: over a field each acyclic value complex splits, and its contraction contracts Hom from any complex. This also applies to the normalized

arrow inputs, with the strict units divided out. Starting at length zero, solve a boundary equation recursively in increasing length. At the next length the already solved equations make the residual component an internal cocycle, so it can be solved there. Completeness of the product gives the desired premorphism. Thus Hom into an objectwise acyclic target is acyclic. In particular an objectwise acyclic module is contractible, by applying this conclusion to its identity, and quasi-isomorphisms have contractible cones.

Direct sums are formed on values and operations. Their premorphism Hom into a fixed target is the product of the individual Hom complexes, giving the coproduct assertion. The ordinary strictly unital Yoneda homotopy evaluates a premorphism on the identity at  $x$  and contracts its remaining bar components by inserting that identity. Each component of this homotopy involves only a finite word, so it works unchanged with periodic degrees. It gives the displayed Yoneda isomorphism, naturally for the cohomology actions.

The normalized bar tensor uses a direct sum of finite tensor words; its differential does not increase word length. If one factor is objectwise acyclic, its length-preserving associated-graded differential is acyclic. A cycle involves only boundedly many lengths. Kill its highest-length component by an internal boundary and continue downwards, proving that the bar tensor is acyclic. Cones then prove invariance under quasi-isomorphisms and exactness on triangles. The diagonal contraction is the Yoneda bar contraction: insert an identity in the end representable slot. Strict unitality removes the other higher-unit terms. All these identities are finite at each word, so none requires convergence with respect to integer cohomological degree.  $\square$

## 10.2 Localization by later copies

Define a right module over the full folded line model by coinduction:

$$\widetilde{\mathcal{M}}(c) = \mathbf{R}\mathrm{Hom}_{\overline{\mathcal{D}}}(J^*h_c^{\overline{\mathcal{B}}}, \mathcal{M}_0). \quad (10.3)$$

The restricted Yoneda functor takes  $c$  covariantly to a  $\overline{\mathcal{D}}$ -module; dg Hom into  $\mathcal{M}_0$  makes (10.3) a right module. The unit of  $Jx$  gives the canonical Yoneda map

$$u_x : h_x^{\overline{\mathcal{D}}} \longrightarrow J^*h_{Jx}^{\overline{\mathcal{B}}}.$$

Precomposition defines an evaluation comparison

$$H^*(\widetilde{\mathcal{M}}(Jx)) \longrightarrow H^*(\mathcal{M}_0(x)). \quad (10.4)$$

**Proposition 10.2** (Evaluation after localization). *The comparison (10.4) is an isomorphism for every graph label  $x$ . It respects the cohomology actions of graph arrows under  $J$ . In particular  $\widetilde{\mathcal{M}}$  has finite-dimensional total cohomology at every object of  $\overline{\mathcal{B}}$ .*

*Proof.* Fix  $x_0 = x$  and put  $x_{j+1} = x_j^+$ . The layer ordering makes  $x_j$  strictly later than any fixed label once  $j$  is large. Form the sequential telescope of right representables, defined by the triangle

$$\bigoplus_{j \geq 0} h_{x_j} \xrightarrow{1-\text{shift}} \bigoplus_{j \geq 0} h_{x_j} \longrightarrow T_x \longrightarrow \left( \bigoplus_{j \geq 0} h_{x_j} \right) [1], \quad T_x = \mathrm{hocolim}_j h_{x_j}. \quad (10.5)$$

The shift maps are induced by  $w_{x_j}$ . In  $H^0(\overline{\mathcal{B}})$ , let  $b_j : Jx_j \rightarrow Jx$  be the inverse of the composite from  $Jx$  to  $Jx_j$ , with  $b_0 = \mathrm{id}$ . The Yoneda classes defined by  $b_j$  give compatible maps  $h_{x_j} \rightarrow J^*h_{Jx}^{\overline{\mathcal{B}}}$ : their composites with  $w_{x_j}$  agree on cohomology. Exactness of Hom applied to (10.5) therefore supplies a map

$$T_x \longrightarrow J^*h_{Jx}^{\overline{\mathcal{B}}} \quad (10.6)$$

whose restriction to  $h_x$  is  $u_x$ .

Evaluate this map at a label  $a$ . Cohomology of the telescope is the direct limit of the cohomology of its value complexes. For all large  $j$ , the arrows  $a \rightarrow x_j$  are strict, and hence  $J$  induces an isomorphism

$$H^*\overline{\mathcal{D}}(a, x_j) \xrightarrow{\sim} H^*\overline{\mathcal{B}}(Ja, Jx_j).$$

Composing with  $b_j$  identifies the right side with  $H^*\overline{\mathcal{B}}(Ja, Jx)$ . These identifications commute with the transition maps. Thus (10.6) is an objectwise quasi-isomorphism.

Apply derived Hom into  $\mathcal{M}_0$ . By Lemma 10.1, the two direct sums in (10.5) become products of the complexes  $\mathcal{M}_0(x_j)$ . Products of vector spaces are exact, so their cohomology is the product of the value cohomologies. The difference map on these products is surjective: all inverse-system transition maps

$$H^*\mathcal{M}_0(x_{j+1}) \longrightarrow H^*\mathcal{M}_0(x_j)$$

are isomorphisms, and one solves the difference equation recursively. Consequently there is no derived inverse-limit term. The cohomology of  $\mathbf{R}\mathrm{Hom}(T_x, \mathcal{M}_0)$  is the inverse limit of these groups, and projection to its zeroth factor is an isomorphism. In view of (10.6), it is exactly (10.4).

For naturality, an arrow  $x \rightarrow y$  gives two composites from  $h_x$  to  $J^*h_{Jy}$ . By Yoneda both represent its image under  $J$ , since the maps  $u_x, u_y$  represent the identity classes. Precomposition therefore makes (10.4) natural for the cohomology actions. Finally the value complexes of  $\mathcal{M}_0$  are finite-dimensional by Proposition 7.3.  $\square$

### 10.3 A perfect object in the folded hull

**Proposition 10.3.** *The module  $\widetilde{\mathcal{M}}$  is perfect over  $\overline{\mathcal{B}}$ . It is represented by an object  $\mathfrak{E}$  in the split-closed pretriangulated hull of the direct-sum fold of an algebraic dg model of  $\mathrm{Perf}(Y_\kappa)$ . For each graph label  $X$ , its value cohomology identifies with the periodic Hom from the corresponding ordinary line to  $\mathfrak{E}$ , and these identifications respect cohomology compositions.*

*Proof.* The integer-graded category  $\mathcal{B}$  is smooth: its diagonal bimodule is perfect, since its split-closed hull is  $\mathrm{Perf}(Y_\kappa)$  and  $Y_\kappa$  is smooth projective. This is the smooth projective Morita comparison: the perfect kernel of the geometric diagonal gives the diagonal bimodule; see [27]. One may express this by a finite presentation using cones, shifts and a retract of exterior Yoneda cells. Such a presentation remains a presentation of the correct diagonal after folding. Indeed for any two complexes  $K, L$ ,

$$\mathrm{Fold}(K \otimes L)^r = \bigoplus_{a+b \equiv r \pmod{4}} K^a \otimes L^b = (\mathrm{Fold}(K) \otimes \mathrm{Fold}(L))^r.$$

The differentials and actions agree, including signs, because four is even. Therefore an exterior Yoneda cell folds to the exterior cell of the folded representables, while the diagonal folds to the actual diagonal of  $\overline{\mathcal{B}}$ . Folding preserves quasi-isomorphisms, finite cones and retracts. Thus  $\overline{\mathcal{B}}$  is smooth in the periodic grading.

Tensor a split-perfect diagonal presentation with  $\widetilde{\mathcal{M}}$ . By Lemma 10.1, this recovers  $\widetilde{\mathcal{M}}$ , and every exterior cell becomes a representable tensored with one of its value complexes. By Proposition 10.2 that complex has finite-dimensional cohomology. Over a field it is therefore perfect as a periodic complex: split off its acyclic part and take its four finite cohomology spaces. The tensor presentation proves perfectness.

For the geometric model, take bounded vector bundle complexes on  $Y_\kappa$ , with dg Hom given by a fixed finite affine-cover Čech complex and cup composition. The ordinary comparison with the line model is cohomologically fully faithful and split-generating. Direct-sum folding

preserves the first property because cohomology commutes with direct sums. It preserves the second because each generation relation uses finitely many cones and a retract. Taking the split-closed pretriangulated hull gives the asserted object  $\mathfrak{E}$ . The representables  $h_{G_k}$  and  $h_C$  correspond to the folds of  $L'_k$  and  $P_C$ . The evaluation and action assertions are Proposition 10.2 and Yoneda.  $\square$

The object  $\mathfrak{E}$  is only asserted to belong to this periodic hull. We shall use its coherent periodic cohomology; no integer-graded lift is needed.

## 10.4 Identifying the polarization and section actions

Let  $\mathcal{S}_s$  and  $\mathcal{X}_s = \text{Proj } \mathcal{S}_s$  be as in Proposition 8.7.

**Lemma 10.4.** *There are an isomorphism  $\phi : Y_\kappa \xrightarrow{\sim} \mathcal{X}_s$  and a line  $B_0$  on  $Y_\kappa$  such that, with  $A_1 = \phi^* \mathcal{O}_{\mathcal{X}_s}(1)$ ,*

$$L'_k = B_0 \otimes A_1^{-k}, \quad k \geq 1. \quad (10.7)$$

*Under these identifications, the maps from  $\mathcal{S}_{s,i-j}$  to the cohomology arrows  $G_i \rightarrow G_j$  are ordinary section multiplication.*

*Proof.* For  $j \geq 1$ , the degree-one arrow space identifies with

$$\mathcal{S}_{s,1} \xrightarrow{\sim} H^0(Y_\kappa, (L'_{j+1})^\vee \otimes L'_j).$$

Here strict-arrow full faithfulness applies, and the quotient line has ample product type  $H_Y^{h_0}$ , so its ordinary Ext occurs only in degree zero. The complete linear system is very ample: each elliptic factor has degree  $h_0 \geq 3$ , and Picard-zero twists preserve this property. It defines a closed immersion

$$f_j : Y_\kappa \longrightarrow \mathbb{P}(\mathcal{S}_{s,1}^\vee).$$

These morphisms are independent of  $j$ . Indeed commutativity of the section algebra, transported through compositions, says that the evaluation functionals for two consecutive quotients satisfy

$$\lambda_j(a)\lambda_{j+1}(b) = \lambda_j(b)\lambda_{j+1}(a) \quad (a, b \in \mathcal{S}_{s,1}).$$

Both functionals are everywhere nonzero by global generation. Their cross-product equations give the same projective quotient, so  $f_j = f_{j+1}$ . Write  $f$  for this common closed immersion.

Every homogeneous relation in  $\mathcal{S}_s$  vanishes on the image of  $f$ : express it as a sum of products of degree-one sections, realize the products by successive line arrows, and evaluate. It follows that  $f$  factors through  $\mathcal{X}_s \subset \mathbb{P}(\mathcal{S}_{s,1}^\vee)$ . Both varieties have dimension eight, and  $\mathcal{X}_s$  is smooth and integral by Proposition 8.7. Thus the closed image is all of  $\mathcal{X}_s$ ; reducedness makes this an isomorphism, denoted by  $\phi$ .

Each successive quotient line, with its evaluation, is the pullback of the universal projective quotient  $\mathcal{O}(1)$ . This identifies all the quotients with  $A_1$ , including their degree-one section maps. Choose the line identifications successively and set  $B_0 = L'_1 \otimes A_1$ . This gives (10.7). Since  $\mathcal{S}_s$  is generated in degree one, compatibility with degree-one multiplication gives compatibility with every section arrow.  $\square$

## 10.5 Coherent periodic cohomology

Write  $D_4(\text{QCoh } Y_\kappa)$  for the derived category of period-four complexes of quasi-coherent sheaves, with quasi-isomorphisms of sheaf complexes inverted. We allow its idempotent completion.

**Lemma 10.5** (Finite Čech realization). *The folded perfect hull in Proposition 10.3 maps fully faithfully on cohomology into the idempotent completion of  $D_4(\mathrm{QCoh} Y_\kappa)$ . Its objects have coherent periodic cohomology sheaves. For a vector bundle  $P$  and an object  $E$  of this hull there is a natural, finite-column spectral sequence*

$$E_2^{a,b} = H^a(Y_\kappa, P^\vee \otimes \mathcal{H}^b(E)) \implies \mathrm{Hom}^{a+b}(P, E), \quad b \in \mathbb{Z}/4. \quad (10.8)$$

*It is natural for ordinary bundle maps in  $P$  and for idempotents in  $E$ .*

*Proof.* Fix a finite affine open cover of  $Y_\kappa$  and use ordered Čech cochains. All its intersections are affine, since  $Y_\kappa$  is separated. Send a bounded vector bundle complex  $P$  to its sheaf Čech resolution, folded into period four. A Čech Hom cochain acts on these resolutions by the cup map, with the usual total-complex signs. Cup composition makes this a dg construction. It extends to finite twisted complexes and hence, on cohomology, to their retracts in the idempotent-completed target.

We check full faithfulness first on folded bounded vector bundle complexes  $P, Q$ . Precompose a cup map by the augmentation  $P \rightarrow \check{C}(P)$ . The resulting maps from  $P$  to  $\check{C}(Q)$  are precisely the global total Čech Hom cochains of the original dg model. They already compute derived Hom. To verify this last assertion, filter the sheaf Čech target by its finitely many columns. Each column is a finite sum of terms  $j_*K$ , where  $j : U \hookrightarrow Y_\kappa$  is an affine intersection of the cover. Such  $j$  is an affine morphism: its intersection with any affine open of the separated variety is affine. Therefore  $j_*$  is exact on quasi-coherent sheaves; restriction is exact as well. Their ordinary adjunction consequently passes to the periodic derived categories. On  $U$ , the fold of the bounded complex  $P|_U$  consists of finite projectives and computes outgoing Hom without further resolution. Indeed its finite ordinary-degree filtration before folding reduces this assertion to the exactness of Hom from one projective module. The comparison follows for each column by adjunction and for their total by the finite column triangles. The sheaf Čech augmentation is a quasi-isomorphism, also periodically: locally its augmented cover complex contracts by inserting a cover open containing the testing neighborhood. These facts prove full faithfulness on  $P, Q$ . An exact functor fully faithful on those objects is fully faithful on their finite cone hull, by the triangle Hom sequences, and then on its retracts.

An ordinary folded perfect complex has coherent periodic cohomology. The cohomology exact sequence of a cone preserves coherence, so this holds for the finite twisted-complex hull. For a retract, take the images of the resulting idempotents on cohomology sheaves; these images are again coherent.

For a pre-retract object choose its actual periodic sheaf-complex representative  $K$  obtained from the preceding construction. Resolve it by the same sheaf Čech complex. Exactness of restriction and of the affine pushforwards shows that taking internal cohomology first gives the Čech cochains of  $P^\vee \otimes \mathcal{H}^b(K)$ . Taking their cohomology gives the left side of (10.8). The affine derived-Hom argument just given identifies the total cohomology with the right side. There are only finitely many cover columns, so the filtration and its abutment are finite in every residue degree, although the internal grading is periodic.

The construction is functorial for chain maps and for roofs in the derived category: a quasi-isomorphism induces an isomorphism already on the internal-cohomology page. Thus an idempotent acts on the spectral sequence and its finite abutment filtration. Taking its images proves (10.8) for a retract. Cup maps from ordinary bundle morphisms make the asserted source naturality explicit.  $\square$

Apply this lemma to  $\mathfrak{E}$  and abbreviate  $\mathcal{H}^b = \mathcal{H}^b(\mathfrak{E})$ .

**Proposition 10.6** (Coherent section-module tails). *For every  $r \in \mathbb{Z}/4$ , the module  $V_s^r$  is finitely generated over  $\mathcal{S}_s$  and has associated sheaf*

$$\widetilde{V}_s^r \cong \phi_*(B_0^\vee \otimes \mathcal{H}^r). \quad (10.9)$$

For all sufficiently large  $j$ , simultaneously for the four values of  $r$ , its degree- $j$  piece is

$$(V_s^r)_j \cong H^0(Y_\kappa, A_1^j \otimes B_0^\vee \otimes \mathcal{H}^r), \quad (10.10)$$

compatibly with multiplication by  $\mathcal{S}_s$ .

*Proof.* By Propositions 10.2 and 10.3,  $(V_s^r)_j$  is  $\text{Hom}^r(L'_j, \mathfrak{E})$ , with the given section actions. Use (10.7) in (10.8). Serre vanishing kills all terms with  $a > 0$  for sufficiently large  $j$ , uniformly over the four coherent sheaves  $B_0^\vee \otimes \mathcal{H}^b$ . This gives (10.10). Naturality in the line source, together with Lemma 10.4, identifies the multiplication actions.

The ring  $\mathcal{S}_s$  is the section ring of  $\mathcal{O}_{\mathcal{X}_s}(1)$ , as established in the theta construction. Serre's section-module theorem therefore makes each displayed tail finitely generated and identifies its Proj sheaf. The finitely many earlier pieces of  $V_s^r$  are finite-dimensional. Adjoining bases for them to tail generators proves finite generation of the whole module and gives (10.9); see [25].  $\square$

## 10.6 Euler pairings detect a nonpolarization class

Fix a prime  $\ell$  and use rational  $\ell$ -adic cycle cohomology, with the Tate twists implicit in Chern characters. Put

$$\begin{aligned} l &= c_1(\mathcal{O}_{\mathcal{X}_s}(1)), \\ \zeta_s &= \sum_{r \in \mathbb{Z}/4} (-1)^r \text{ch}(\widetilde{V}_s^r), \\ \eta_{\mathfrak{E}} &= \sum_{b \in \mathbb{Z}/4} (-1)^b \text{ch}(\mathcal{H}^b). \end{aligned} \quad (10.11)$$

These are rational algebraic Chern-character classes. By (10.9),

$$\phi^* \zeta_s = e^{-c_1(B_0)} \eta_{\mathfrak{E}}. \quad (10.12)$$

For a vector bundle  $P$ , define the periodic Euler pairing by

$$\chi(P, \mathfrak{E}) = \sum_{r \in \mathbb{Z}/4} (-1)^r \dim_\kappa \text{Hom}^r(P, \mathfrak{E}).$$

The groups are finite by (10.8) and properness. Since the period is even and the column filtration finite, the alternating sum on that spectral sequence is unchanged from its second page to its abutment. Ordinary Riemann–Roch, and the trivial Todd class of an abelian variety, give

$$\chi(P, \mathfrak{E}) = \int_{Y_\kappa} \text{ch}(P^\vee) \eta_{\mathfrak{E}}. \quad (10.13)$$

Both sides are additive in triangles in the ordinary perfect source, so the identity also holds for folds of ordinary perfect complexes, using a bounded vector bundle resolution.

**Proposition 10.7** (Nonpolarization detection). *The class  $\zeta_s$  satisfies*

$$\zeta_s \notin \text{span}_{\mathbb{Q}_\ell} \{1, l, l^2, \dots, l^8\}. \quad (10.14)$$

*Proof.* Let  $P_k = L'_k$  and retain  $P_C$  for the ordinary line corresponding to  $C$ . Lemma 7.4 and Proposition 10.2 identify

$$\chi(P_k, \mathfrak{E}) = \nu g_k \cdot \alpha, \quad \chi(P_C, \mathfrak{E}) = \nu c \cdot \alpha.$$

Here  $\alpha = \alpha_{\text{ex}} + g_1 + g_2 + g_3$ ,  $g_k \cdot \alpha_{\text{ex}} = 0$ , and  $c \cdot \alpha_{\text{ex}} \neq 0$ .

Ordinary graph Euler pairings also give

$$\chi(P_k, P_j) = g_k \cdot g_j, \quad \chi(P_C, P_j) = c \cdot g_j. \quad (10.15)$$

In the available strict-arrow direction these follow from full faithfulness and the ordinary intersection-sign formula. Reversing the direction preserves Euler pairings by Serre duality in the even dimension eight, just as it preserves the middle intersection form. For equal labels both self-Euler numbers are zero.

Suppose  $\zeta_s$  belongs to the span in (10.14). Equations (10.7), (10.12), (10.13), and (10.15) imply, for every  $k \geq 1$ ,

$$\begin{aligned} \int_{\mathcal{X}_s} e^{kl} \zeta_s &= \chi(P_k, \mathfrak{E}) = \nu \sum_{j=1}^3 \chi(P_k, P_j) \\ &= \nu \int_{\mathcal{X}_s} e^{kl} \sum_{j=1}^3 e^{-jl}. \end{aligned} \quad (10.16)$$

These evaluations determine every coefficient of a polynomial in  $l$ . Explicitly, if  $z = \sum_{a=0}^8 z_a l^a$ , then

$$\int_{\mathcal{X}_s} e^{kl} z = \left( \int_{\mathcal{X}_s} l^8 \right) \sum_{a=0}^8 \frac{z_a}{(8-a)!} k^{8-a}.$$

The leading intersection factor is nonzero because  $l$  is ample. Equality for all positive integers  $k$  in (10.16) therefore forces

$$\zeta_s = \nu \sum_{j=1}^3 e^{-jl}, \quad \eta_{\mathfrak{E}} = \nu \sum_{j=1}^3 \text{ch}(P_j).$$

Testing the second equality by  $\text{ch}(P_C^\vee)$  in (10.13) would give

$$\chi(P_C, \mathfrak{E}) = \nu \sum_{j=1}^3 \chi(P_C, P_j).$$

The two sides differ, by the Euler and intersection identities, by  $\nu c \cdot \alpha_{\text{ex}} \neq 0$ . This contradiction proves (10.14).  $\square$

## 11 Lifting the alternating Chern character

We now transfer the class detected at the ordinary fiber to the geometric generic fiber. The individual cohomology modules need not be flat over  $R$ . The essential observation is that their torsion contributions cancel after alternating over the four residue degrees.

Retain  $R = \kappa[[t]]$ , the graded theta ring  $\mathcal{S}$ , and the smooth projective scheme  $\mathcal{X} = \text{Proj } \mathcal{S}$  of Proposition 8.7. Denote its special and geometric generic fibers by  $\mathcal{X}_s$  and  $\mathcal{X}_{\bar{\eta}}$ , and put

$$l_s = c_1(\mathcal{O}_{\mathcal{X}_s}(1)), \quad l_{\bar{\eta}} = c_1(\mathcal{O}_{\mathcal{X}_{\bar{\eta}}}(1)).$$

For a module  $M$ , write  $M[t] = \ker(t : M \rightarrow M)$ .

**Lemma 11.1.** *For each  $r \in \mathbb{Z}/4$ , the graded module  $V^r$  is finitely generated over  $\mathcal{S}$ . Its associated sheaf  $\mathcal{V}^r = \widehat{V}^r$  is coherent and perfect on  $\mathcal{X}$ .*

*Proof.* Each slope value is represented by a period-four complex of finite free  $R$ -modules. Applying its cohomology sequence to multiplication by  $t$  and summing over slopes gives short exact sequences

$$0 \longrightarrow V^r/tV^r \longrightarrow V_s^r \longrightarrow V^{r+1}[t] \longrightarrow 0. \quad (11.1)$$

These are graded  $\mathcal{S}$ -linear sequences, where  $t$  acts trivially on every term: each section arrow acts by a chain map on the lifted value complexes, and the cohomology sequence is natural for those maps. The higher module identities ensure that the induced actions are the section-ring actions already defined in Section 8.

By Proposition 10.6,  $V_s^r$  is a finitely generated module over the Noetherian ring  $\mathcal{S}_s = \mathcal{S}/t\mathcal{S}$ . Its submodule  $V^r/tV^r$  is therefore finitely generated. Choose finitely many homogeneous generators and lift them to  $V^r$ . Let  $N^r \subset V^r$  be their  $\mathcal{S}$ -span. For each graded degree  $j$ , the module  $(V^r)_j$  is finite over  $R$ , since it is the cohomology of a finite free complex over the Noetherian ring  $R$ . Thus  $(V^r/N^r)_j$  is finite over  $R$ , and the choice of lifts gives

$$(V^r/N^r)_j = t(V^r/N^r)_j.$$

Nakayama's Lemma implies  $(V^r/N^r)_j = 0$  for every  $j$ . Hence  $N^r = V^r$ . This is a degreewise argument and does not require the direct sum of all graded pieces to be a finite  $R$ -module.

Sheafification now gives a coherent  $\mathcal{V}^r$  on  $\mathcal{X}$ . The scheme  $\mathcal{X}$  is regular and Noetherian of finite dimension: it is smooth of relative dimension eight over the regular discrete valuation ring  $R$ . A coherent sheaf on such a scheme has locally a finite resolution by finite rank vector bundles. Consequently  $\mathcal{V}^r$  is perfect.  $\square$

**Proposition 11.2** (Cancellation under derived restriction). *For the special-fiber inclusion  $i_s : \mathcal{X}_s \hookrightarrow \mathcal{X}$ ,*

$$\sum_{r \in \mathbb{Z}/4} (-1)^r [\mathbf{L}i_s^* \mathcal{V}^r] = \sum_{r \in \mathbb{Z}/4} (-1)^r [\widetilde{V}_s^r] \quad \text{in } K_0(\mathcal{X}_s). \quad (11.2)$$

*Proof.* Flatness of  $\mathcal{X}$  over  $R$  makes  $t$  a nonzero divisor on its structure sheaf. The two-term resolution of  $\mathcal{O}_{\mathcal{X}_s}$  is therefore

$$[\mathcal{O}_{\mathcal{X}} \xrightarrow{t} \mathcal{O}_{\mathcal{X}}],$$

placed in degrees  $-1, 0$ . Tensoring with  $\mathcal{V}^r$  gives

$$[\mathbf{L}i_s^* \mathcal{V}^r] = [\mathcal{V}^r/t\mathcal{V}^r] - [\mathcal{V}^r[t]]. \quad (11.3)$$

Sheafification of graded modules on  $\text{Proj}$  is exact: on a standard affine chart it is localization followed by taking the degree-zero summand, both exact operations. Thus the sheafification of (11.1) identifies

$$[\widetilde{V}_s^r] = [\mathcal{V}^r/t\mathcal{V}^r] + [\mathcal{V}^{r+1}[t]].$$

Multiplying by  $(-1)^r$  and summing over  $\mathbb{Z}/4$ , the last sum becomes  $-\sum_r (-1)^r [\mathcal{V}^r[t]]$  after reindexing. This is precisely the alternating sum of (11.3). The even period is what makes the reindexing compatible with the signs.  $\square$

**Proposition 11.3.** *Fix a prime  $\ell$ . The rational algebraic Chern character*

$$\zeta_{\bar{\eta}} = \sum_{r \in \mathbb{Z}/4} (-1)^r \text{ch}(\mathcal{V}^r|_{\mathcal{X}_{\bar{\eta}}})$$

*in  $\bigoplus_{a=0}^8 H_{\text{ét}}^{2a}(\mathcal{X}_{\bar{\eta}}, \mathbb{Q}_{\ell}(a))$  does not belong to  $\text{span}_{\mathbb{Q}_{\ell}}\{1, l_{\bar{\eta}}, \dots, l_{\bar{\eta}}^8\}$ .*

*Proof.* The ring  $R$  is complete with algebraically closed residue field, hence strictly henselian. Smooth proper base change for a smooth proper scheme over this trait identifies the geometric generic and special  $\ell$ -adic cohomology groups, compatibly with cup products and the restrictions of Chern classes of perfect complexes [20]. Every prime is invertible in  $R$ , which has characteristic zero. These identifications carry  $l_{\bar{\eta}}$  to  $l_s$ . By Proposition 11.2, they carry  $\zeta_{\bar{\eta}}$  to

$$\zeta_s = \sum_{r \in \mathbb{Z}/4} (-1)^r \text{ch}(\widetilde{V}_s^r).$$

The latter class is outside the span of the powers of  $l_s$  by Proposition 10.7. The same is therefore true on the geometric generic fiber.  $\square$

**Proposition 11.4.** *For every field embedding realizing the generic theta algebra at a complex period  $\Pi$  as in Proposition 8.4, the abelian variety  $A_{\Pi}$  has a rational algebraic total cycle class*

$$\zeta_{\Pi} \in \bigoplus_{a=0}^8 H^{2a}(A_{\Pi}, \mathbb{Q})$$

outside  $\text{span}_{\mathbb{Q}}\{1, \theta, \dots, \theta^8\}$ .

*Proof.* Base extension along the embedding identifies the projective spectrum of the generic section ring with  $A_{\Pi}$  and identifies its grading line with the ample line of class  $h_0\theta$ . Extend the embedding of the fraction field to its algebraic closure inside  $\mathbb{C}$ . The sheaves  $\mathcal{V}^r$  restrict to algebraic perfect complexes on this complex fiber. Their alternating Chern character is consequently the cycle class of an element of  $\bigoplus_a \text{CH}^a(A_{\Pi}) \otimes \mathbb{Q}$ .

Invariance of  $\ell$ -adic cohomology under extension of algebraically closed fields, followed by the Betti-étale comparison isomorphism, identifies this character with the extension of  $\zeta_{\bar{\eta}}$  [20]. A rational Betti expression as a linear combination of  $1, \theta, \dots, \theta^8$  would therefore give a  $\mathbb{Q}_{\ell}$ -linear expression for  $\zeta_{\bar{\eta}}$  in powers of  $l_{\bar{\eta}} = h_0\theta$ . Proposition 11.3 excludes this. All objects used in this last step are algebraic sheaves or perfect complexes on projective schemes.  $\square$

## 12 A general period and all special fibers

The construction has produced an algebraic class outside the span of polarization powers whenever the generic theta algebra is realized at a suitable complex period. Two steps remain. We choose such a period where this detects a nonzero Weil component, and then use a projective family of subschemes to carry one fixed Weil class to every period. The exceptional sets used for this choice must be fixed before the matrix direction  $M$  is selected.

### 12.1 Countably many projective parameter spaces

Let  $\pi : \mathcal{A} \rightarrow \mathcal{U}$  be the holomorphic family in (2.2). Its underlying topological family is the marked product  $T \times \mathcal{U}$ . The  $h_0$ th power of the theta line has a holomorphic basis of theta sections over  $\mathcal{U}$ ; their multiplier formulas and normal convergence are those used in Section 8. Since  $h_0 \geq 3$ , this power is very ample on every abelian fiber [22]. These sections give a closed holomorphic embedding

$$\mathcal{A} \hookrightarrow \mathbb{P}^N \times \mathcal{U}, \quad N = h_0^8 - 1. \quad (12.1)$$

Indeed the map is proper and fiberwise an embedding; its differential is injective on vertical tangents, and its base map is the identity.

**Lemma 12.1** (Hilbert parameter spaces). *There is a countable collection of connected reduced complex analytic spaces  $\mathcal{H}$ , each proper over  $\mathcal{U}$ , with a flat family of subschemes*

$$\mathcal{Z}_{\mathcal{H}} \subset \mathcal{A} \times_{\mathcal{U}} \mathcal{H},$$

*such that every closed subscheme of every fiber  $A_{\Pi}$  occurs in at least one family. Each image  $\mathcal{H} \rightarrow \mathcal{U}$  is closed analytic. Consequently the union  $\mathcal{E}_{\text{Hilb}}$  of those images which are proper subsets is a countable union of proper closed analytic subsets of  $\mathcal{U}$ .*

*Proof.* For each Hilbert polynomial  $P$ , take the projective Hilbert scheme  $\text{Hilb}_P(\mathbb{P}^N)$  and its universal flat family [16]. Write  $H_P$  for its analytification. We first construct the reduced closed analytic incidence locus

$$\mathcal{H}_P = \{(\Pi, z) \in \mathcal{U} \times H_P : Z_z \subset A_{\Pi} \text{ scheme-theoretically}\}.$$

Here the scheme structure of  $Z_z$  is retained.

Fix a small neighborhood  $V$  of a period. The ideal of  $\mathcal{A}$  in  $\mathbb{P}^N \times V$  is flat over  $V$ , since both the ambient structure sheaf and its quotient are flat. For a sufficiently large twist  $a$ , relative generation and proper analytic cohomology and base change give finitely many homogeneous polynomial families  $s_1, \dots, s_r$  of degree  $a$  generating that ideal after shrinking  $V$  [15]. One can see the last assertion directly from the evaluation map: it is surjective on the central projective fiber, and the support of its coherent cokernel has proper closed image, which is removed by shrinking.

On the fixed projective Hilbert scheme choose  $e \geq 0$  large enough that the universal family's sheaf of degree- $a + e$  sections pushes forward to a vector bundle  $E_P$  and commutes with base change. One  $e$  works over the entire Hilbert scheme by relative Serre vanishing and its projectivity. Restrict each product  $s_i x^{\beta}$ , with  $|\beta| = e$ , to the pulled-back universal family over  $V \times H_P$ . These restrictions are holomorphic sections of the pulled-back bundle  $E_P$ . Their common zero locus has exactly the required incidence points. For the nontrivial implication, on the projective chart  $x_j \neq 0$  the tested product  $s_i x_j^e$  vanishes, so  $s_i$  vanishes in the structure sheaf of the subscheme. The  $s_i$  generate the ideal of the abelian family, which proves scheme-theoretic containment.

This also proves containment over the entire zero locus: base change identifies the zero testing sections with their evaluations on its universal family, and the same chart argument annihilates the pulled-back ideal. Taking reduced structures, the local zero loci agree on overlaps and give  $\mathcal{H}_P$ . Pullback of the universal family is still flat and is a subscheme of the pulled-back abelian family. Reduction of the parameter space does not require any fiber subscheme to be reduced.

The projection  $\mathcal{H}_P \rightarrow \mathcal{U}$  is proper because  $H_P$  is compact. Complex analytic spaces are locally path connected, so each connected component of  $\mathcal{H}_P$  is open and closed, in particular a closed analytic subspace. There are at most countably many components: the space is second countable, and each nonempty open component contains a member of a countable basis. The components remain proper over  $\mathcal{U}$ , and Remmert's proper mapping theorem makes their images closed analytic [15]. Finally, there are countably many Hilbert polynomials. These components are the asserted spaces.  $\square$

## 12.2 Constancy across singular parameters

The parameter spaces just constructed need not be smooth. We prove constancy of their cohomology classes in a form which includes such parameters and nonreduced specialized subschemes.

**Lemma 12.2.** *For every connected parameter space  $\mathcal{H}$  in Lemma 12.1, the marked rational cohomology class*

$$\mathrm{ch}(\mathcal{O}_{Z_b}) \in H^{\mathrm{even}}(T, \mathbb{Q})$$

*is independent of  $b \in \mathcal{H}$ . In particular its degree-eight component  $\mathrm{ch}_4(\mathcal{O}_{Z_b})$  is constant. On each fiber it is a rational algebraic cycle class.*

*Proof.* Put  $X = \mathcal{A} \times_{\mathcal{U}} \mathcal{H}$  and  $\mathcal{F}_0 = \mathcal{O}_{\mathcal{X}_{\mathcal{H}}}$ , considered as a coherent sheaf on  $X$ . It is flat over  $\mathcal{H}$ . Near any chosen  $b \in \mathcal{H}$ , construct eight exact sequences

$$0 \longrightarrow \mathcal{F}_{j+1} \longrightarrow E_j \longrightarrow \mathcal{F}_j \longrightarrow 0, \quad 0 \leq j \leq 7, \quad (12.2)$$

where  $E_j$  is a vector bundle and every  $\mathcal{F}_j$  is flat over the parameter neighborhood. To obtain a sequence, choose a sufficiently positive twist of  $\mathcal{F}_j$  on the central projective fiber so that it is generated by its sections and has no higher cohomology. Proper analytic cohomology and base change give a vector bundle of sections on a smaller neighborhood. Its evaluation, tensored by the opposite twist, is a surjection near the whole central fiber, and hence over a smaller base neighborhood by properness. Since  $X$  is flat over the base,  $E_j$  is base-flat; the kernel in (12.2) is base-flat as well. In particular these sequences restrict exactly to each fiber.

The smooth fiber  $X_b$  has dimension eight. Its local rings are regular of dimension at most eight, so the eighth syzygy  $\mathcal{F}_8|_{X_b}$  is locally free. This property lifts to a neighborhood of the fiber even though  $\mathcal{H}$  may be singular. Here is the local argument. At a point  $x \in X_b$  put

$$A = \mathcal{O}_{\mathcal{H},b}, \quad S = \mathcal{O}_{X,x}, \quad M = (\mathcal{F}_8)_x.$$

The module  $M$  is finite over the Noetherian local ring  $S$ , is flat over  $A$ , and  $M/\mathfrak{m}_A M$  is free over  $S/\mathfrak{m}_A S$ . Lift a fiber basis to a map  $S^r \rightarrow M$ . Nakayama makes it surjective. If  $K'$  is its kernel, flatness of  $M$  over  $A$  makes reduction exact on the left. The reduced map is an isomorphism, hence  $K'/\mathfrak{m}_A K' = 0$ , and another application of Nakayama gives  $K' = 0$  [26]. Thus  $M$  is free.

Local freeness is open. Properness of  $X \rightarrow \mathcal{H}$  now allows one more shrink of the parameter neighborhood, removing the image of the nonfree locus of  $\mathcal{F}_8$ . We have obtained a finite vector-bundle resolution on the whole inverse image of that neighborhood:

$$0 \longrightarrow \mathcal{F}_8 \longrightarrow E_7 \longrightarrow \cdots \longrightarrow E_0 \longrightarrow \mathcal{F}_0 \longrightarrow 0. \quad (12.3)$$

Only the fibers were required to be regular; no regularity of the total space over a singular parameter was used.

All the bundles in (12.3) are complex topological vector bundles on the marked product  $T \times |\mathcal{H}|$ . Along a path in the chosen parameter neighborhood, their restrictions to its two endpoints have equal Chern characters in  $H^*(T, \mathbb{Q})$ , by homotopy invariance. Exact restriction of (12.3) identifies the alternating sum with  $\mathrm{ch}(\mathcal{O}_{Z_b})$  on each fiber. The class is therefore locally constant. Local path connectedness and connectedness of  $\mathcal{H}$  make it constant on the whole component. On each smooth projective fiber the same resolution computes the usual algebraic Chern character of the coherent structure sheaf, whose cohomology class is the one just computed.  $\square$

### 12.3 Choosing the matrix direction

Let  $\mathcal{E}_{\mathrm{Hdg}}$  be the exceptional union in Proposition 2.2, and set

$$\mathcal{E} = \mathcal{E}_{\mathrm{Hdg}} \cup \mathcal{E}_{\mathrm{Hilb}}.$$

Each constituent is a proper closed analytic subset of the connected complex manifold  $\mathcal{U}$ , hence has empty interior. Baire's theorem supplies a point  $\Pi_* \in \mathcal{U} \setminus \mathcal{E}$ . Choose, before performing the construction,

$$M = 2\pi i(\Pi_* - i1_8). \quad (12.4)$$

It is symmetric and commutes with  $D$ , exactly as required for the relative two-form  $B = dq^t M \wedge dp$ .

**Proposition 12.3.** *For this choice of  $M$ , there is a period  $\Pi \notin \mathcal{E}$  and a codimension-four integral subvariety  $Z \subset A_\Pi$  such that*

$$[Z] = a\theta^4 + w, \quad a \in \mathbb{Q}, \quad 0 \neq w \in W_K. \quad (12.5)$$

*Proof.* Use the convex two-parameter domain  $\mathcal{P}_M$  of (8.7), with period map

$$(\tau, s) \mapsto \tau 1_8 + sM/(2\pi i).$$

It sends  $(i, 1)$  to  $\Pi_*$ . Hence the inverse image of every constituent of  $\mathcal{E}$  is a proper closed analytic subset of  $\mathcal{P}_M$ . Run the symplectic and algebraic construction for this fixed  $M$ . Proposition 8.4 adds only countably many proper closed analytic exceptional sets for the prescribed theta-coefficient embedding. By Baire, a point avoids both lists.

At its period  $\Pi$ , Proposition 11.4 gives a rational algebraic total cycle class outside the span of the powers of  $\theta$ . Formula (2.3) implies that its codimension-four component has nonzero projection to  $W_K$ . Express a cycle representing this component as a finite rational linear combination of integral codimension-four subvarieties. At least one constituent has a nonzero Weil projection. Every such constituent has rational Hodge class and the period lies outside  $\mathcal{E}_{\text{Hdg}}$ , so its class has precisely the form (12.5).  $\square$

## 12.4 The whole Weil plane on every fiber

**Theorem 12.4.** *For every  $\Pi \in \mathcal{U}$ , all classes in  $W_K(A_\Pi)$  are rational algebraic. The same holds for every split Weil-type polarized abelian eightfold over  $\mathbb{C}$ , for every imaginary quadratic field and every compatible polarization type.*

*Proof.* Let  $Z$  and  $a\theta^4 + w$  be as in Proposition 12.3. Its Hilbert point belongs to one of the connected parameter spaces  $\mathcal{H}$  in Lemma 12.1. Since  $\Pi$  was chosen outside  $\mathcal{E}_{\text{Hilb}}$ , that component's image cannot be a proper subset of  $\mathcal{U}$ . Thus  $\mathcal{H} \rightarrow \mathcal{U}$  is surjective.

For a reduced integral codimension-four subvariety of a smooth variety, the leading Chern-character formula is

$$\text{ch}_4(\mathcal{O}_Z) = [Z]$$

[14]. Lemma 12.2 therefore gives the fixed marked class  $a\theta^4 + w$  as  $\text{ch}_4(\mathcal{O}_{Z_b})$  at every point  $b \in \mathcal{H}$ . Over an arbitrary period, choose any point of the surjective parameter space. Its subscheme may be singular, reducible, or nonreduced, but this Chern character is still a rational algebraic cycle class on the smooth abelian fiber. Subtracting the rational algebraic class  $a\theta^4$  proves that the same nonzero rational vector  $w \in W_K$  is algebraic on every fiber.

To obtain a second independent class, choose an integer

$$m > \sqrt{d} \cot(\pi/8).$$

The integral matrix  $m1_8 + D$  commutes with every period in  $\mathcal{U}$ , and hence defines an algebraic endomorphism of every  $A_\Pi$ . On the two determinant lines of  $W_K \otimes \mathbb{C}$ , its pullback has eigenvalues

$$\lambda_+ = (m + i\sqrt{d})^8, \quad \lambda_- = (m - i\sqrt{d})^8.$$

They are nonreal conjugates: the argument of  $m + i\sqrt{d}$  lies strictly between 0 and  $\pi/8$ . A nonzero rational  $w$  has nonzero components on both conjugate determinant lines. If its pullback were a rational multiple  $rw$ , those two components would give  $\lambda_+ = r = \lambda_-$ , a contradiction. Thus  $w$  and its pullback are independent rational algebraic classes and span  $W_K$ .

The integer  $d$  was arbitrary. Finally Proposition 2.1 transports the conclusion to every tuple in the theorem, including every polarization type and every special member with additional endomorphisms.  $\square$

## References

- [1] M. Abouzaid, *The family Floer functor is faithful*, J. Eur. Math. Soc. **19** (2017), no. 7, 2139–2217. doi:10.4171/JEMS/715. Numbering follows arXiv:1408.6794v2.
- [2] M. Abouzaid, *Homological mirror symmetry without correction*, J. Amer. Math. Soc. **34** (2021), 1059–1173. doi:10.1090/jams/973. Numbering follows arXiv:1703.07898v2, titled *Homological mirror symmetry without corrections*.
- [3] M. Akaho and D. Joyce, *Immersed Lagrangian Floer theory*, J. Differential Geom. **86** (2010), no. 3, 381–500. arXiv:0803.0717v1.
- [4] A. Bondal and M. Van den Bergh, *Generators and representability of functors in commutative and noncommutative geometry*, Mosc. Math. J. **3** (2003), no. 1, 1–36. Publisher version; arXiv:math/0204218v2.
- [5] B. Conrad, *Irreducible components of rigid spaces*, Ann. Inst. Fourier **49** (1999), no. 2, 473–541. doi:10.5802/aif.1681.
- [6] T. Ekholm, Y. Eliashberg, E. Murphy, and I. Smith, *Constructing exact Lagrangian immersions with few double points*, Geom. Funct. Anal. **23** (2013), no. 6, 1772–1803. Numbering follows arXiv:1303.0588v2.
- [7] T. Ekholm and I. Smith, *Exact Lagrangian immersions with a single double point*, J. Amer. Math. Soc. **29** (2016), no. 1, 1–59. Publisher version.
- [8] Y. Eliashberg and N. Mishachev, *Introduction to the h-Principle*, Graduate Studies in Mathematics, vol. 48, American Mathematical Society, Providence, RI, 2002. doi:10.1090/gsm/048.
- [9] Y. Eliashberg and E. Murphy, *Lagrangian caps*, preprint (2013), arXiv:1303.0586v1.
- [10] K. Fukaya, Y.-G. Oh, H. Ohta, and K. Ono, *Construction of Kuranishi structures on the moduli spaces of pseudo holomorphic disks: I*, Surveys in Differential Geometry **22** (2018), 133–190. Numbering follows the revised preprint arXiv:1710.01459v2 (2021).
- [11] K. Fukaya, Y.-G. Oh, H. Ohta, and K. Ono, *Construction of Kuranishi structures on the moduli spaces of pseudo-holomorphic disks: II*, preprint (2018), revised 2024, arXiv:1808.06106v2.
- [12] K. Fukaya, Y.-G. Oh, H. Ohta, and K. Ono, *Exponential decay estimates and smoothness of the moduli space of pseudoholomorphic curves*, preprint (2016), revised 2024, arXiv:1603.07026v2.

- [13] K. Fukaya, Y.-G. Oh, H. Ohta, and K. Ono, *Corrigendum of “Construction of Kuranishi structures on the moduli spaces of pseudo holomorphic disks I, Surveys in Differential Geometry XXII (2018), 133–190”*, preprint (2024), [arXiv:2403.19683v1](#).
- [14] W. Fulton, *Intersection Theory*, second edition, *Ergebnisse der Mathematik und ihrer Grenzgebiete, 3. Folge, vol. 2*, Springer, New York, 1998. [Publisher version](#).
- [15] H. Grauert and R. Remmert, *Coherent Analytic Sheaves*, *Grundlehren der mathematischen Wissenschaften*, vol. 265, Springer, Berlin–Heidelberg, 1984. [Publisher version](#).
- [16] A. Grothendieck, *Techniques de construction et théorèmes d’existence en géométrie algébrique IV: les schémas de Hilbert*, *Séminaire Bourbaki*, vol. 6 (1961), exposé 221, 249–276. [Numdam edition](#).
- [17] P. Hall, *Finiteness conditions for soluble groups*, *Proc. London Math. Soc.* (3) **4** (1954), 419–436. [doi:10.1112/plms/s3-4.1.419](#).
- [18] E. Markman, *Cycles on abelian  $2n$ -folds of Weil type from secant sheaves on abelian  $n$ -folds*, preprint (2025), [arXiv:2502.03415v2](#).
- [19] E. Markman, *Secant sheaves and Weil classes on abelian varieties*, preprint (2025), revised 11 February 2026, [arXiv:2509.23403v2](#).
- [20] J. S. Milne, *Lectures on Étale Cohomology*, version 2.21, 22 March 2013, especially Sections 17, 20–21, and 23. [Author’s course notes](#).
- [21] K. Nomizu, *On the cohomology of compact homogeneous spaces of nilpotent Lie groups*, *Ann. of Math.* (2) **59** (1954), 531–538. [doi:10.2307/1969716](#).
- [22] G. Pareschi and M. Popa, *Regularity on abelian varieties I*, *J. Amer. Math. Soc.* **16** (2003), 285–302. [Publisher version](#).
- [23] D. Rathel-Fournier, *On cobordism groups of Lagrangian immersions*, preprint (2024), [arXiv:2409.14651v1](#).
- [24] P. Seidel, *Fukaya Categories and Picard–Lefschetz Theory*, *Zurich Lectures in Advanced Mathematics*, European Mathematical Society, Zürich, 2008. [doi:10.4171/063](#).
- [25] The Stacks Project Authors, *Cohomology of twists of coherent sheaves*, Section 30.14, especially Lemma 30.14.2 and Proposition 30.14.4. [Tag 01YR](#), accessed 18 September 2026.
- [26] The Stacks Project Authors, *Flatness and freeness*, Lemma 10.99.4. [Tag 00MH](#), accessed 18 September 2026.
- [27] B. Toën and M. Vaquié, *Moduli of objects in dg-categories*, *Ann. Sci. École Norm. Sup.* (4) **40** (2007), no. 3, 387–444. [Author-hosted published version](#).
- [28] A. Weil, *Abelian varieties and the Hodge ring* (1977), in *Œuvres scientifiques / Collected Papers*, vol. III, Springer, 1980, 421–429.
- [29] Z. Xie, G. Yu, and R. Zeidler, *On the range of the relative higher index and the higher rho-invariant for positive scalar curvature*, *Adv. Math.* **390** (2021), article 107897. [Publisher version](#); [arXiv:1712.03722v3](#).